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Work Package 2

PROJECT #3

Maintenance Management Platform

P3_E2

Requirement's Analysis of the Bike4All Maintenance platform

Executive Summary

This report outlines the functional and non-functional specifications of the Maintenance bike4All platform, detailing requirements analysis, use cases, and cloud management integration. The document is essential in designing the architecture and development of the maintenance management platform for the bike4All urban mobility initiative. The analysis combines business functionalities, structured and unstructured use case descriptions, and cloud requirements mapped using the NIST CCRA model.

Key outcomes include 18 level 0 use cases, 50 level 1 domain application use cases, and 13 level 1 cloud management use cases. These will inform the platform's development in subsequent tasks under the Bike4All Be.Neutral project.

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Table of contributions

Chapter	Specific Content	Contributions		
		UMinho	CCG	CEiIA
1-8	All Chapters		x	

1 Introduction

This delivery presents the functional and non-functional specifications of the Maintenance bike4All platform, contributing to the detailed requirements analysis for the project. The report is a milestone for the realization of the project's objectives, which comprises the architecture proposal and the development of a maintenance management platform for the 2-wheel urban mobility bike4All platform.

The delivery is structured into three interrelated parts, which integrate requirements analysis, use cases of the domain application, and cloud management use cases. The latter are developed using the NIST CCRA reference model, a key methodology in our approach. The requirements analysis is based on previous preliminary studies, which contributed to the business functionalities for the requirements analysis. The use cases are described using structured and unstructured natural language descriptions, along with diagrammatic representations of the use cases and interaction behaviors. To address the challenge of deriving a cloud architecture from the modeled requirements for the domain application, we map the use case diagram to the NIST CCRA to obtain the cloud requirements. The result is to define the requirements in such a way that they are understood by the cloud service consumers (the owner of the product) because it uncovers more information as if using only the use case model of the application domain.

Overall, this work offers insights into the design of the maintenance bike4All platform and will be the foundation of the next detailed designs.

2 Methodological Approach

Domain-Driven Design (DDD) and Event-Driven Architecture (EDA) are two prominent paradigms in software architecture, each with unique approaches and philosophies.

Domain-Driven Design (DDD) is an approach to software development that centers the design and implementation around the core business concepts (the “domain”). It emphasizes creating a model of the domain, encapsulating complex business logic into a rich, expressive, and cohesive model. This model is typically object-oriented and includes entities, value objects, aggregates, and services that reflect the business domain.

On the other hand, Event-Driven Architecture (EDA) is a design paradigm centered around the production, detection, and reaction to events. An event is a change in state that is meaningful within a business context. EDD enables highly decoupled, scalable, and flexible systems.

While DDD and EDA have different focuses, they can complement each other in a system design. For instance, events are a fundamental part of DDD—we have domain events, and when needed, we even use events as the system’s source of truth. It may be tempting to leverage DDD’s events as the basis for using event-driven architecture. In summary, DDD is more about the business logic and rules, while EDA is about how the system reacts to changes or events. Both can be used together to create a system that is both business-centric and reactive. The domain model can emit events when its state changes. These events, which represent something significant happening in the domain, can then be handled by other parts of the system in an asynchronous and decoupled manner. So, DDD and EDD complement each other. DDD provides a rich and expressive model of the business domain, while EDD allows for decoupled, asynchronous, and reactive behaviors based on the state changes in that domain. Combining them can lead to systems that are more scalable, flexible, and better aligned with the business needs.

Following the preliminary studies, as well as the list of Maintenance Platform initiatives for bike sharing, we have identified ten initiatives around Europe. For this review, we’ve catalogued its business functionalities, and selected the most frequent ones together with interoperability concerns, to be considered in the conception of our platform: Manage fleet’s operations, Real time tracking and reporting, Dashboard and statistics visualization and integrate with other platforms. They were detailed in a set of business requirements to contribute later with the business context for our analysis. We start building a context diagram, to identify the boundaries of the work and interactions with adjacent systems, where these interactions are business events to which the work responds. Business events are happenings in the real world that affect the work, and also happen because it is time for the work to do something. Using these DDD events as the source of truth, we leverage these events as the basis for building use cases using event-driven design, which involves identifying the events that occur in the system and defining how the system should respond to those events.

So, for each event, we define a use case that describes how the system should respond to that event, including the actor, the event, and the system’s response. For

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documenting the detail of the Use Cases, which describes the sequence of interactions between the actor and the system, including the steps the actor takes, the system's responses, and any alternative paths, we use the OpenUP supporting requirements specification, which specifying requirements and constraints in accordance with the FURPS+ classification, which are presented in the annexes.

3 Requirements Analysis

This section involves the analysis of the business requirements identified in the preliminary studies.

Table 1 - Business functionalities

Business Functionality	Description	Business Event
fleet management	1. Real-Time Tracking	1.1 Track Bike Location in Real-Time 1.2 Identify Idle Bikes
	2. Inventory Management	2.1 Manage Bike Details 2.2 Track Bike Availability Status 2.3 Update Bike Inventory
	3. Demand Forecasting	3.1 Collect Historical Data 3.2 Train Predictive Model 3.3 Predict Bike Demand 3.4 Optimize Bike Allocation
	4. Usage Analytics	4.1 Optimize Bike Allocation 4.2 Collect Usage Data 4.3 Analyse Peak Usage Times 4.4 Analyse Popular Routes 4.5 Analyse Average Trip Duration
	5. Incident Reporting	5.1 Receive Issue Report 5.2 Process and Assign Issue 5.3 Track and Update Issue Resolution 5.4 Send Resolution Feedback
	6. Security Features	6.1 Activate Remote Lock 6.2 Trigger Alarm System 6.3 Track Bike Safety Status 6.4 Respond to Security Alerts

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Business Functionality		Description	Business Event
	7. User Interface	The system should have an easy-to-use interface for the administrators (to manage the fleet).	7.1 View Fleet Overview 7.2 Assign Maintenance Tasks 7.3 Monitor Bike Status 7.4 Generate Reports
	8. Integration Capabilities	The system should be able to integrate with other systems like payment gateways, GPS tracking systems, customer support systems, etc.	8.1 Integrate with Bike4All Platform 8.2 Integrate with Mobility Devices Management Platform
preventive maintenance	9. Maintenance Scheduling	The system should be able to schedule regular maintenance checks for each bike in the fleet. This could be based on factors like usage, time, distance traveled, etc.	9.1 Define Maintenance Criteria 9.2 Schedule Maintenance Check 9.3 Notify Maintenance Team 9.4 Perform Maintenance Check 9.5 Update Maintenance Records
	10. Maintenance Alerts	The system should send alerts or notifications when a bike is due for maintenance. This can help ensure that no bike misses its maintenance check.	10.1 Generate Maintenance Alerts 10.2 Send Maintenance Notifications 10.3 Acknowledge Maintenance Alerts 10.4 Escalate Unacknowledged Alerts
	11. Maintenance History	The system should keep a record of all past maintenance activities for each bike. This can help in identifying recurring issues, evaluating the effectiveness of previous repairs, and planning future maintenance activities.	11.1 Retrieve Maintenance History 11.2 Evaluate Repair Effectiveness 11.3 Plan Future Maintenance
	12. Issue Tracking	The system should be able to track and categorize issues identified during maintenance checks. It should flag bikes that are due for maintenance or repair. This can help in prioritizing repairs and identifying common problems across the fleet.	12.1 Track Issues Identified During Maintenance 12.2 Flag Bikes Due for Maintenance or Repair 12.3 Prioritize Repairs
	13. Parts Inventory Management	The system should manage an inventory of spare parts and other materials needed for maintenance. It should track the usage of these parts and trigger alerts when stocks are low.	13.1 Manage Spare Parts Inventory 13.2 Track Usage of Spare Parts 13.3 Trigger Low Stock Alerts 13.4 Generate Inventory Reports
	14. Maintenance Reporting	The system should generate reports detailing the maintenance activities, including the number of bikes serviced, common issues identified, parts used,	14.1 Generate Common Issue Identified Report 14.2 Generate Parts Used Report

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Business Functionality	Description	Business Event
	etc. This can provide valuable insights for improving the maintenance process and the overall operation of the bike-sharing service.	14.3 Generate Comprehensive Maintenance Report

Table 2 - Domain Events

Event No.	Business Event	Input and Output (domain events)	Actor
1.1	Track Bike Location in Real-Time	1.1.1 Send location data (in) 1.1.2 Monitor real-time location (out)	Mobility Devices Management Platform, Maintenance Manager/ Maintenance Operator
1.2	Identify Idle Bikes	1.2.1 Inspect and redistribute idle bikes (out)	Maintenance Manager/ Maintenance Operator
2.1	Manage Bike Details	2.1.1 Input or update bike details (model, age, condition)	Maintenance Manager
2.2	Track Bike Availability Status	2.2.1 Send location data (in)	Mobility Devices Management Platform
2.3	Update Bike Inventory	2.3.1 Add or remove bikes from the inventory (in)	Maintenance Manager
3.1	Collect Historical Data	3.1.1 Send real-time GPS device location data (out)	Mobility Devices Management Platform
3.2	Train Predictive Model	3.2.1 Process historical data to identify demand patterns (out) 3.2.2 Trains the predictive model with the processed data (out)	System
3.3	Predict Bike Demand	3.3.1 Review and utilize predictions (out)	Maintenance Manager
3.4	Optimize Bike Allocation	3.4.1 Review and implement allocation plans (out)	Maintenance Manager
4.1	Collect Usage Data	4.1.1 Send real-time location data (in)	Mobility Devices Management Platform
4.2	Analyse Peak Usage Times	4.2.1 Reviews and validates the peak usage times	Maintenance Manager
4.3	Analyse Popular Routes	4.3.1 Reviews and validates the popular routes	Maintenance Manager
4.4	Analyse Average Trip Duration	4.4.1 Reviews and validates the average trip duration	Maintenance Manager
4.5	Generate Usage Insights Reports	4.5.1 Reviews and distributes the reports	Maintenance Manager
5.1	Receive Issue Report	5.1.1 Send issue report (in)	Bike4All Platform
5.2	Process and Assign Issue	5.2.1 Receive assigned issues (in)	Maintenance Manager
5.3	Track and Update Issue Resolution	5.3.1 Update resolution status of issues (in)	Maintenance Manager
5.4	Send Resolution Feedback	5.4.1 Receive feedback on issue resolution (out)	Bike4All Platform

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Event No.	Business Event	Input and Output (domain events)	Actor
6.1	Activate Remote Lock	6.1.1 Receive command to lock a bike (in)	Mobility Devices Management Platform
6.2	Trigger Alarm System	6.2.1 Receive command to activate the alarm (in)	Mobility Devices Management Platform
6.3	Track Bike Safety Status	6.3.1 Send GPS device status data (out) 6.3.2 Send lock status data (out) 6.3.3 Send alarm status data (out)	Mobility Devices Management Platform
6.4	Respond to Security Alerts	6.4.1 Review and respond to security alerts (in) 6.4.2 Act on security alerts for inspection and repair (out)	Maintenance Manager Maintenance Operator
7.1	View Fleet Overview	7.1.1 Use the interface to manage the fleet (out)	Maintenance Manager
7.2	Assign Maintenance Tasks	7.2.1 Receive and performs tasks (out)	Maintenance Operator
7.3	Monitor Bike Status	7.3.1 Monitor bike status (out)	Maintenance Manager
7.4	Generate Reports	7.4.1 Generate and review reports (in)	Maintenance Manager
8.1	Integrate with Bike4All Platform	8.1.1 Manage user issue reports (in) 8.1.2 Respond to inquiries or issues (in)	Bike4All Platform Maintenance Manager
8.2	Integrate with Mobility Devices Management Platform	8.2.1 Access the GPS tracking data (out) 8.2.2 Provide GPS tracking data (in)	Maintenance Manager Mobility Devices Management Platform
9.1	Define Maintenance Criteria	9.1.1 Define and update maintenance criteria (in)	Maintenance Manager
9.2	Schedule Maintenance Check	9.2.1 Periodically analyse and determine bikes meeting maintenance criteria (out) 9.2.2 Schedule maintenance checks (out)	System
9.3	Notify Maintenance Team	9.3.1 Receive notifications (out) 9.3.2 Receive and act on notifications (out)	Maintenance Manager Maintenance Operator
9.4	Perform Maintenance Check	9.4.1 Perform maintenance checks (out)	Maintenance Operator
9.5	Update Maintenance Records	9.5.1 Provide status of the maintenance checks (in)	Maintenance Operator
10.1	Generate Maintenance Alerts	10.1.1 Generate maintenance alerts for bikes due for maintenance	System
10.2	Send Maintenance Notifications	10.2.1 Receive notifications (out) 10.2.2 Receive notifications and act on notifications (out)	Maintenance Operator Maintenance Manager
10.3	Acknowledge Maintenance Alerts	10.3.1 Acknowledge alerts and plan maintenance (in) 10.3.2 Update alert status (out)	Maintenance Operator

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Event No.	Business Event	Input and Output (domain events)	Actor
10.4	Escalate Unacknowledged Alerts	10.4.1 Notify escalated alert (in) 10.4.2 Assign action for escalated alert (in) 10.4.3 Review escalated alerts and update system with action taken (in)	Maintenance Manager Maintenance Manager Maintenance Operator
11.1	Retrieve Maintenance History	11.1.1 Retrieve maintenance history (in)	Maintenance Manager
11.2	Evaluate Repair Effectiveness	11.2.1 Review evaluation results (in)	Maintenance Manager
11.3	Plan Future Maintenance	11.3.1 Review and approve maintenance plans (in)	Maintenance Manager
12.1	Track Issues Identified During Maintenance	12.1.1 Identify and log issues (in)	Maintenance Operator
12.2	Flag Bikes Due for Maintenance or Repair	12.2.1 Flag bikes due for repair (out) 12.2.2 Update the status of flagged bikes (out)	System
12.3	Prioritize Repairs	12.3.1 Review and adjust repair priorities (in)	Maintenance Manager
13.1	Manage Spare Parts Inventory	13.1.1 Manage inventory records (in)	Maintenance Manager
13.2	Track Usage of Spare Parts	13.2.1 Log the usage of spare parts (in)	Maintenance Operator
13.3	Trigger Low Stock Alerts	13.3.1 Receive and act on low stock alerts (out)	Maintenance Manager
13.4	Generate Inventory Reports	13.4.1 Generate and review reports (in)	Maintenance Manager
14.1	Generate Common Issue Identified Report	14.1.1 Review the common issue report (in)	Maintenance Manager
14.2	Generate Parts Used Report	14.2.1 Review the Parts Used Report (in)	Maintenance Manager
14.3	Generate Comprehensive Maintenance Report	14.3.1 Review comprehensive maintenance report (in)	Maintenance Manager

Table 3 - Mapping domain events with use cases

UC No.	UC Name	Actor	Input and Output	UC Description
1.1	Track Bike Location in Real-Time	Mobility Devices Management Platform, Maintenance Manager/Maintenance Operator	1.1.1 Send location data (in) 1.1.2 Monitor real-time location (out)	The system continuously tracks the real-time location of each bike using GPS technology to monitor the usage and position of the bikes.
1.2	Identify Idle Bikes	Maintenance Manager/Maintenance Operator	1.2.1 Inspect and redistribute idle bikes (out)	The system identifies bikes that have been stationary for a prolonged period, indicating potential underutilization or need for maintenance.
2.1	Manage Bike Details	Maintenance Manager	2.1.1 Input or update bike details (model, age, condition)	This use case describes the process by which the system maintains and

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UC No.	UC Name	Actor	Input and Output	UC Description
				updates details about each bike in the inventory.
2.2	Track Bike Availability Status	Mobility Devices Management Platform	2.2.1 Send location data (in)	This use case describes the process by which the system tracks and updates the current availability status of each bike.
2.3	Update Bike Inventory	Maintenance Manager	2.3.1 Add or remove bikes from the inventory (in)	This use case describes the process by which the system updates the inventory to reflect changes in the fleet.
3.1	Collect Historical Data	Mobility Devices Management Platform	3.1.1 Send real-time GPS device location data (out)	This use case describes the process by which the system collects historical data on bike usage, including times and locations.
3.2	Train Predictive Model	System	3.2.1 Process historical data to identify demand patterns (out) 3.2.2 Trains the predictive model with the processed data (out)	The system shall train a predictive model using collected historical data to identify patterns in bike demand.
3.3	Predict Bike Demand	Maintenance Manager	3.3.1 Review and utilize predictions (out)	The system shall use the trained predictive model to generate predictions of future bike demand at different times and locations.
3.4	Optimize Bike Allocation	Maintenance Manager	3.4.1 Review and implement allocation plans (out)	This use case describes the process by which the system optimizes the allocation of bikes based on demand predictions.
4.1	Collect Usage Data	Mobility Devices Management Platform	4.1.1 Send real-time location data (in)	This use case describes the process by which the system collects data on bike usage, including start and end times, routes taken, and trip durations.
4.2	Analyse Peak Usage Times	Maintenance Manager	4.2.1 Reviews and validates the peak usage times	This use case describes the process by which the system analyzes usage data to identify peak usage times.
4.3	Analyse Popular Routes	Maintenance Manager	4.3.1 Reviews and validates the popular routes	This use case describes the process by which the system analyzes usage data to identify popular routes taken by users.
4.4	Analyse Average Trip Duration	Maintenance Manager	4.4.1 Reviews and validates the average trip duration	This use case describes the process by which the system analyzes usage data to calculate average trip durations.

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UC No.	UC Name	Actor	Input and Output	UC Description
4.5	Generate Usage Insights Reports	Maintenance Manager	4.5.1 Reviews and distributes the reports	This use case describes the process by which the system generates reports based on the analysis of usage patterns, including peak times, popular routes, and average trip durations.
5.1	Receive Issue Report	Bike4All Platform	5.1.1 Send issue report (in)	This use case describes the process by which the system receives issues or incidents reported by users related to the bikes.
5.2	Process and Assign Issue	Maintenance Manager	5.2.1 Receive assigned issues (in)	This use case describes the process by which the system processes and assigns received issues for resolution.
5.3	Track and Update Issue Resolution	Maintenance Manager	5.3.1 Update resolution status of issues (in)	This use case describes the process by which the system tracks and updates the resolution status of reported issues.
5.4	Send Resolution Feedback	Bike4All Platform	5.4.1 Receive feedback on issue resolution (out)	This use case describes the process by which the system sends feedback on the resolution of issues to the Bike4All Platform.
6.1	Activate Remote Lock	Mobility Devices Management Platform	6.1.1 Receive command to lock a bike (in)	This use case describes the process by which the system can remotely lock a bike to prevent unauthorized use.
6.2	Trigger Alarm System	Mobility Devices Management Platform	6.2.1 Receive command to activate the alarm (in)	This use case describes the process by which the system can trigger an alarm on a bike to deter theft or misuse.
6.3	Track Bike Safety Status	Mobility Devices Management Platform	6.3.1 Send GPS device status data (out) 6.3.2 Send lock status data (out) 6.3.3 Send alarm status data (out)	This use case describes the process by which the system continuously monitors the safety status of bikes for any signs of unauthorized use or tampering.
6.4	Respond to Security Alerts	Maintenance Manager Maintenance Operator	6.4.1 Review and respond to security alerts (in) 6.4.2 Act on security alerts for inspection and repair (out)	This use case describes the process by which the system responds to security alerts generated by the monitoring system.
7.1	View Fleet Overview	Maintenance Manager	7.1.1 Use the interface to manage the fleet (out)	This use case describes the process by which Maintenance Manager can view an overview of the entire bike fleet, including the status of each bike.
7.2	Assign Maintenance Tasks	Maintenance Operator	7.2.1 Receive and performs tasks (out)	This use case describes the process by which Maintenance Manager can assign maintenance tasks

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UC No.	UC Name	Actor	Input and Output	UC Description
				to the Maintenance Operator.
7.3	Monitor Bike Status	Maintenance Manager	7.3.1 Monitor bike status (out)	This use case describes the process by which Maintenance Manager can monitor the real-time status of bikes in the fleet.
7.4	Generate Reports	Maintenance Manager	7.4.1 Generate and review reports (in)	This use case describes the process by which the Maintenance Manager can generate reports on various aspects of the fleet, such as usage statistics, maintenance history, and performance metrics.
8.1	Integrate with Bike4All Platform	Bike4All Platform Maintenance Manager	8.1.1 Manage user issue reports (in) 8.1.2 Respond to inquiries or issues (in)	This use case describes how the maintenance bike-sharing platform integrates with the Bike4All platform to handle customer support issues.
8.2	Integrate with Mobility Devices Management Platform	Maintenance Manager Mobility Devices Management Platform	8.2.1 Access the GPS tracking data (out) 8.2.2 Provide GPS tracking data (in)	This use case describes how the maintenance bike-sharing platform integrates with the Mobility Devices Management Platform to enable GPS tracking of bikes.
9.1	Define Maintenance Criteria	Maintenance Manager	9.1.1 Define and update maintenance criteria (in)	This use case describes the process by which administrators define the criteria for scheduling regular maintenance checks for bikes, such as usage, time, or distance traveled.
9.2	Schedule Maintenance Check	System	9.2.1 Periodically analyse and determine bikes meeting maintenance criteria (out) 9.2.2 Schedule maintenance checks (out)	This use case describes the process by which the system schedules regular maintenance checks for each bike based on the defined criteria.
9.3	Notify Maintenance Team	Maintenance Manager Maintenance Operator	9.3.1 Receive notifications (out) 9.3.2 Receive and act on notifications (out)	This use case describes the process by which the system notifies the maintenance manager/maintenance operator about scheduled maintenance checks.
9.4	Perform Maintenance Check	Maintenance Operator	9.4.1 Perform maintenance checks (out)	This use case describes the process by which the Maintenance Manager/Maintenance Operator performs the scheduled maintenance checks on bikes.

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UC No.	UC Name	Actor	Input and Output	UC Description
9.5	Update Maintenance Records	Maintenance Operator	9.5.1 Provide status of the maintenance checks (in)	This use case describes the process by which the system updates maintenance records based on the performed maintenance checks.
10.1	Generate Maintenance Alerts	System	10.1.1 Generate maintenance alerts for bikes due for maintenance	This use case describes the process by which the system generates alerts for bikes that are identified as due for maintenance.
10.2	Send Maintenance Notifications	Maintenance Operator Maintenance Manager	10.2.1 Receive notifications (out) 10.2.2 Receive notifications and act on notifications (out)	This use case describes the process by which the system sends notifications to the Maintenance Manager/ Maintenance Operator about bikes due for maintenance.
10.3	Acknowledge Maintenance Alerts	Maintenance Manager Maintenance Operator	10.3.1 Acknowledge alerts and plan maintenance (in) 10.3.2 Update alert status (out)	This use case describes the process by which the maintenance operator acknowledges the receipt of maintenance alerts and plans for the checks.
10.4	Escalate Unacknowledged Alerts	Maintenance Manager Maintenance Manager Maintenance Operator	10.4.1 Notify escalated alert (in) 10.4.2 Assign action for escalated alert (in) 10.4.3 Review escalated alerts and update system with action taken (in)	This use case describes the process by which the system escalates alerts that have not been acknowledged within a certain timeframe.
11.1	Retrieve Maintenance History	Maintenance Manager	11.1.1 Retrieve maintenance history (in)	This use case describes the process by which the Maintenance Manager can retrieve the maintenance history of a specific bike.
11.2	Evaluate Repair Effectiveness	Maintenance Manager	11.2.1 Review evaluation results (in)	This use case describes the process by which the system evaluates the effectiveness of previous repairs based on subsequent maintenance records.
11.3	Plan Future Maintenance	Maintenance Manager	11.3.1 Review and approve maintenance plans (in)	This use case describes the process by which the system uses maintenance records and analysis to plan future maintenance activities.
12.1	Track Issues Identified During Maintenance	Maintenance Operator	12.1.1 Identify and log issues (in)	This use case describes the process by which the system tracks issues identified during maintenance checks on bikes.
12.2	Flag Bikes Due for Maintenance or Repair	System	12.2.1 Flag bikes due for repair (out) 12.2.2 Update the status of flagged bikes (out)	This use case describes the process by which the system flags bikes that are due for maintenance or

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

UC No.	UC Name	Actor	Input and Output	UC Description
				repair based on identified issues.
12.3	Prioritize Repairs	Maintenance Manager	12.3.1 Review and adjust repair priorities (in)	This use case describes the process by which the system prioritizes repairs based on the severity and frequency of identified issues.
13.1	Manage Spare Parts Inventory	Maintenance Manager	13.1.1 Manage inventory records (in)	This use case describes the process by which the system manages an inventory of spare parts and other materials needed for bike maintenance.
13.2	Track Usage of Spare Parts	Maintenance Operator	13.2.1 Log the usage of spare parts (in)	This use case describes the process by which the system tracks the usage of spare parts during maintenance activities.
13.3	Trigger Low Stock Alerts	Maintenance Manager	13.3.1 Receive and act on low stock alerts (out)	This use case describes the process by which the system triggers alerts when the stock of spare parts is low.
13.4	Generate Inventory Reports	Maintenance Manager	13.4.1 Generate and review reports (in)	This use case describes the process by which the system generates reports on inventory levels, usage, and replenishment activities.
14.1	Generate Common Issue Identified Report	Maintenance Manager	14.1.1 Review the common issue report (in)	This use case describes the process by which the system generates a report detailing common issues identified during maintenance.
14.2	Generate Parts Used Report	Maintenance Manager	14.2.1 Review the Parts Used Report (in)	This use case describes the process by which the system generates a report detailing the spare parts used during maintenance activities.
14.3	Generate Comprehensive Maintenance Report	Maintenance Manager	14.3.1 Review comprehensive maintenance report (in)	This use case describes the process by which the system generates a comprehensive report detailing maintenance activities, including the number of bikes serviced, common issues identified, and parts used.

4 Domain Application Use Cases

4.1 Domain Application Actors Description

Table 4 – Domain Application Actors Description

Actor Bike4ALL-M	Definition
Public Cloud Vendor	It represents the Public Cloud Platform organization, i.e., the entity responsible for making the service available to cloud consumers. It creates platform services (PaaS), manages the technical infrastructure to provide the services, provisions the services through SLAs (service level agreements), and protects the security and privacy of the services.
Bike4All-M Platform Owner	It represents the individual or organization that owns the Maintenance system, which has access to virtual computers, network-accessible storage, network infrastructure, components, and other essential computing resources on which to develop, install, run, and monitor the performance of the Maintenance system.
Maintenance Manager	It represents the individual who manages and supervises the maintenance service, from monitoring alerts to issuing work orders. It also has access to historical data, statistics, and real-time information, registers and removes bikes and parts, sets maintenance parameters, and registers the maintenance team.
Maintenance Operator	It represents the people who perform the necessary bike maintenance work and record all the relevant information for each service. They also have access to the history and reports of their work and must update the bikes' status after maintenance.
Bike4All Platform	It represents the Bicycle Mobility Management Platform, where users can hire bicycles. It allows its users to record incidents reported to the Maintenance Platform.
Mobility Devices Management Platform	It represents the service provider that connects the mobility devices of each bicycle in the fleet to the Bike4All and Maintenance Platform and provides data on the respective bicycles (location and status).

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

4.2 Bike4All Maintenance System Use Case Diagram

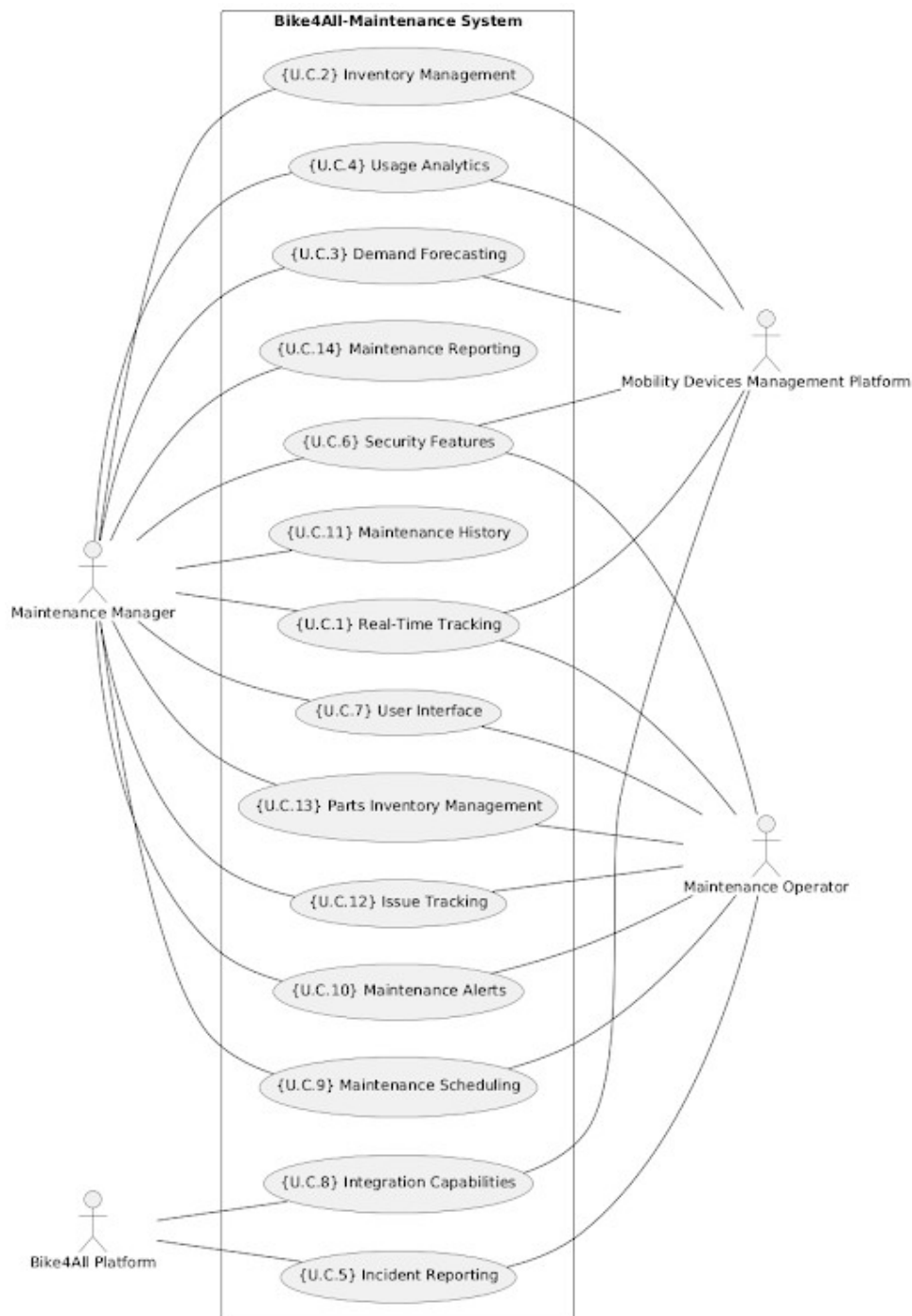


Figure 1 - Use Case Diagram Bike4All-Maintenance System

4.3 Domain Application Use Cases Description

{U.C.1} Real-Time tracking: The system shall track the location of each bike in real-time to monitor usage, identify idle bikes, and ensure optimal distribution across different locations. This functionality supports effective fleet management by providing accurate and current location data. The system must be capable of continuously receiving and processing real-time GPS data from each bike. By monitoring this data, the system can identify bikes that remain stationary for prolonged periods, flagging them as idle for maintenance or redistribution. Additionally, the system will analyze location data to ensure bikes are distributed optimally to meet demand across various locations.

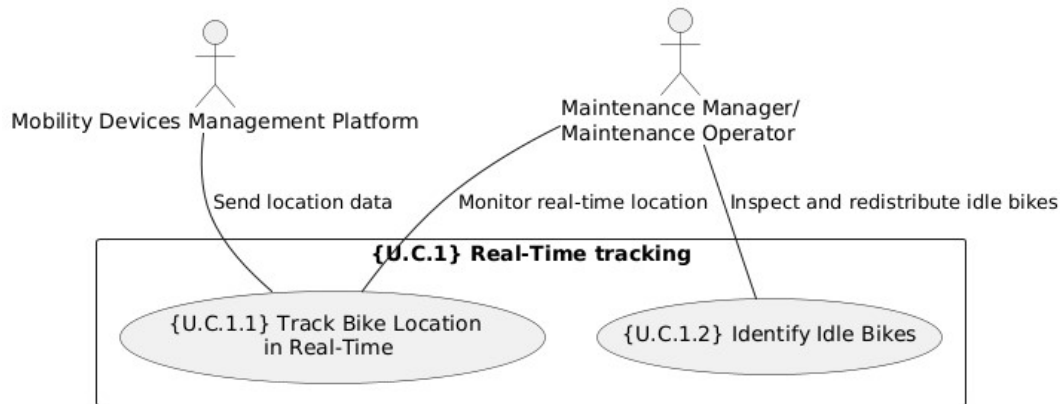


Figure 2 - Use Case diagram for {U.C.1} Real-Time tracking

{U.C.1.1} Track Bike Location in Real-Time: The system shall continuously track the real-time location of each bike using GPS technology to monitor usage and position. This involves the Mobility Devices Management Platform providing real-time GPS data at regular intervals, which the system receives, processes, updates and stores in the database, displaying the location on a monitoring dashboard for the Maintenance Manager and Maintenance Operator.

In the event of GPS Signal Loss, the system will use the last known location and flag the bike for signal recovery monitoring. In the event of Data Transmission Failure, the system will retry multiple times before flagging the bike for manual inspection. The incoming GPS data shall be validated, converting GPS coordinates into a readable format, and updating the database with the latest location data.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

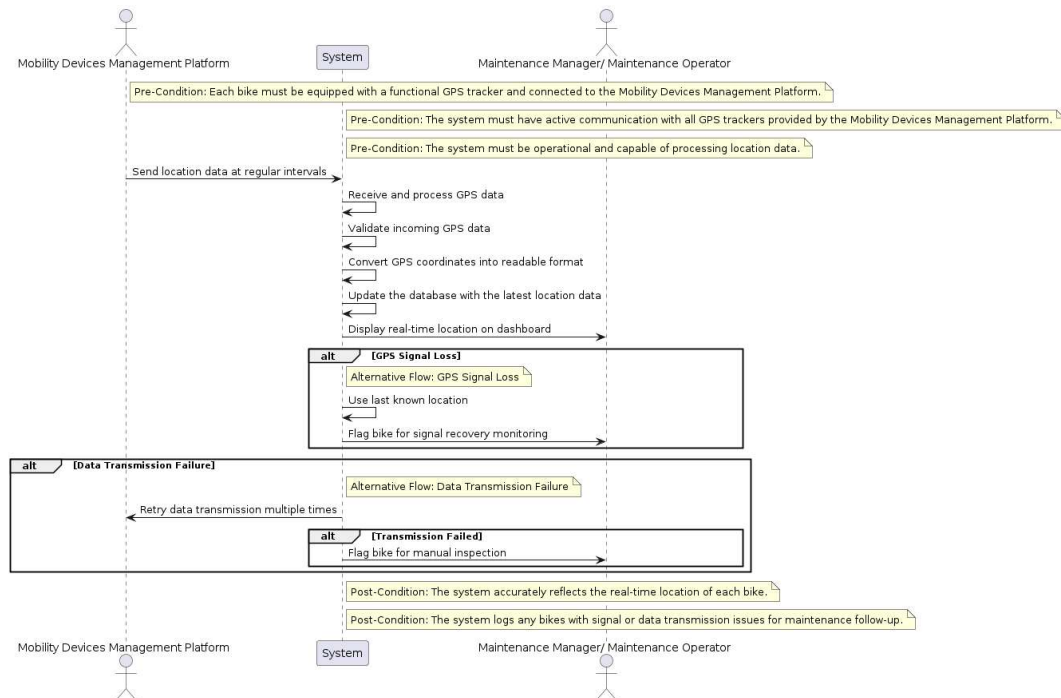


Figure 3 - Sequence diagram, for use case {U.C.1.1} Track Bike Location in Real-Time

{U.C.1.2} Identify Idle Bikes: The system shall identify bikes that have been stationary for a prolonged period, indicating potential underutilization or need for maintenance. This involves continuous monitoring of location data to detect movement, calculate the duration of inactivity and compare it to a predefined threshold, and flagging idle bikes for inspection or redistribution. The system then sends an alert to the Maintenance Manager, who reviews the alert and takes necessary action.

In the event of False Positives, where a bike is flagged as idle but is actually in use, the system allows manual override by the Maintenance Manager or Maintenance Operator. For Multiple Idle Bikes in the same location, the system prioritizes alerts based on the duration of inactivity.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

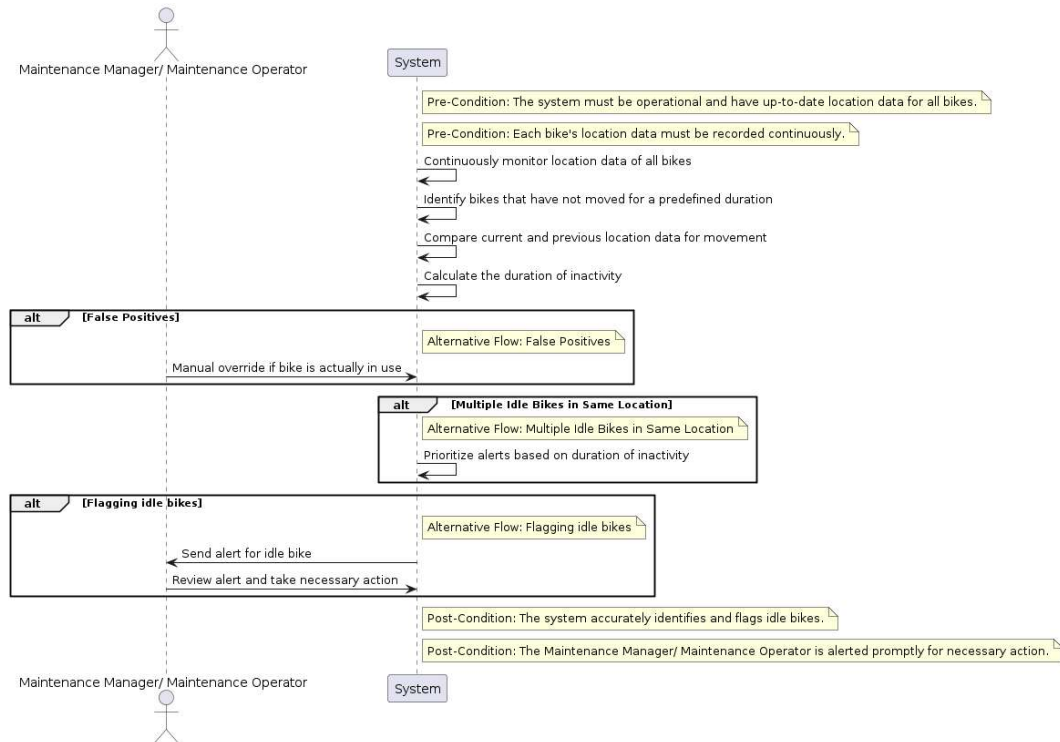


Figure 4 - Sequence diagram for use case {U.C.1.2} Identify Idle Bikes

{U.C.2} Inventory Management: The system shall maintain an up-to-date inventory of all bikes, capturing details such as model, age, condition, and other relevant information. Additionally, the system shall track the total number of bikes, their current status (available, in use, under maintenance), and their locations. This comprehensive inventory management will assist in effectively managing the supply of bikes and making informed decisions regarding the redistribution or procurement of new bikes. The system shall ensure data integrity and accuracy by enabling regular updates and corrections. It will support the Maintenance Manager in monitoring and updating bike details and statuses.

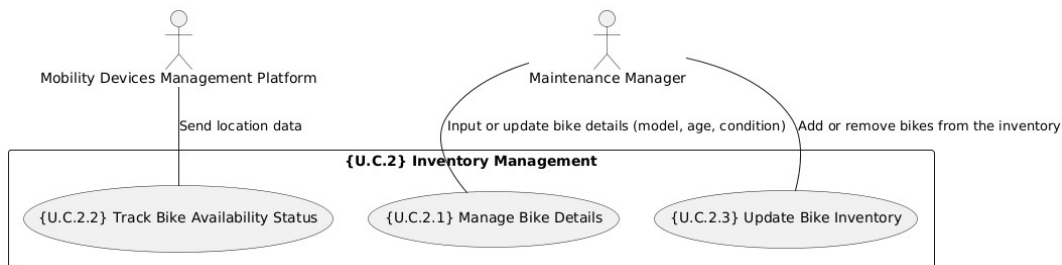


Figure 5 - Use Case diagram for {U.C.2} Inventory Management

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

{U.C.2.1} Manage Bike Details: The system shall maintain and update details about each bike in the inventory to ensure accurate and current records. This involves the Maintenance Manager entering or updating bike details such as model, age, and condition, with the system processing and storing this information.

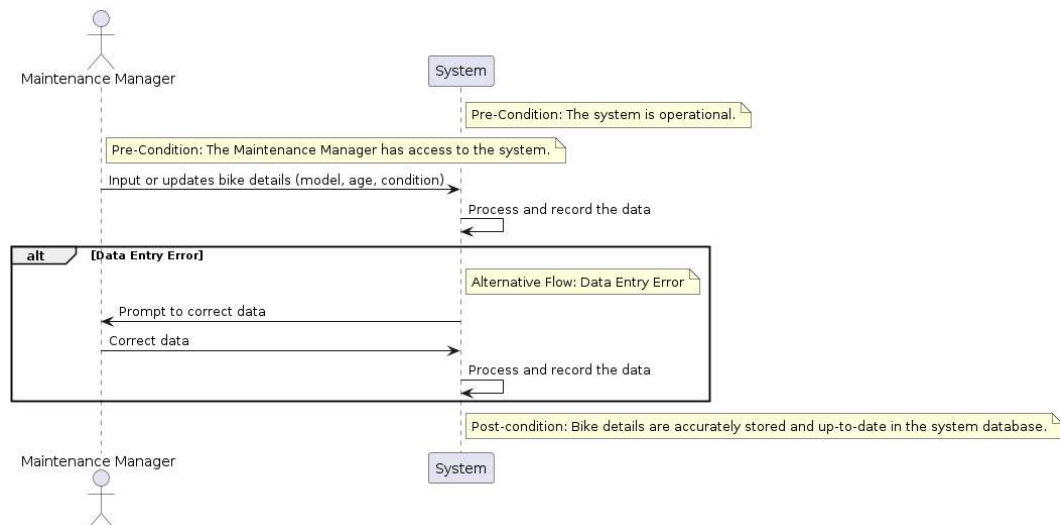


Figure 6 - Sequence diagram for use case {U.C.2.1} Manage Bike Details

{U.C.2.2} Track Bike Availability Status: The system shall track and update the current status of each bike to ensure accurate monitoring and management of the fleet. This process involves the Mobility Devices Management Platform sending real-time location data to the system, which then processes this data to update the bike's status (available, in use, under maintenance). In the event of a GPS Device Failure, where the Mobility Devices Management Platform fails to deliver data from a GPS device, the system shall flag the bike for inspection to ensure timely maintenance and resolution.

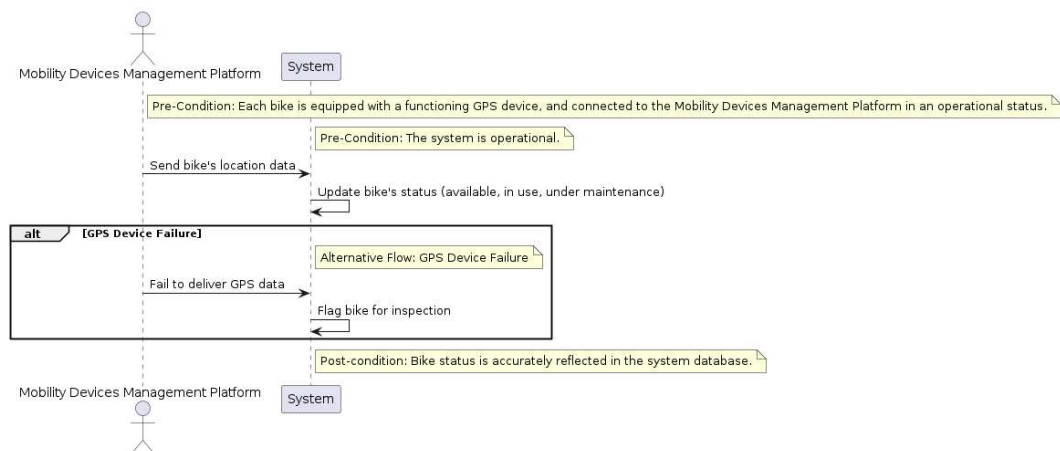


Figure 7 - Sequence diagram for use case {U.C.2.2} Track Bike Availability Status

{U.C.2.3} Update Bike Inventory: The system shall update the bike inventory to reflect changes in the fleet accurately. This involves the Maintenance Manager being able to add or remove bikes from the inventory, with the system processing and storing this information.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

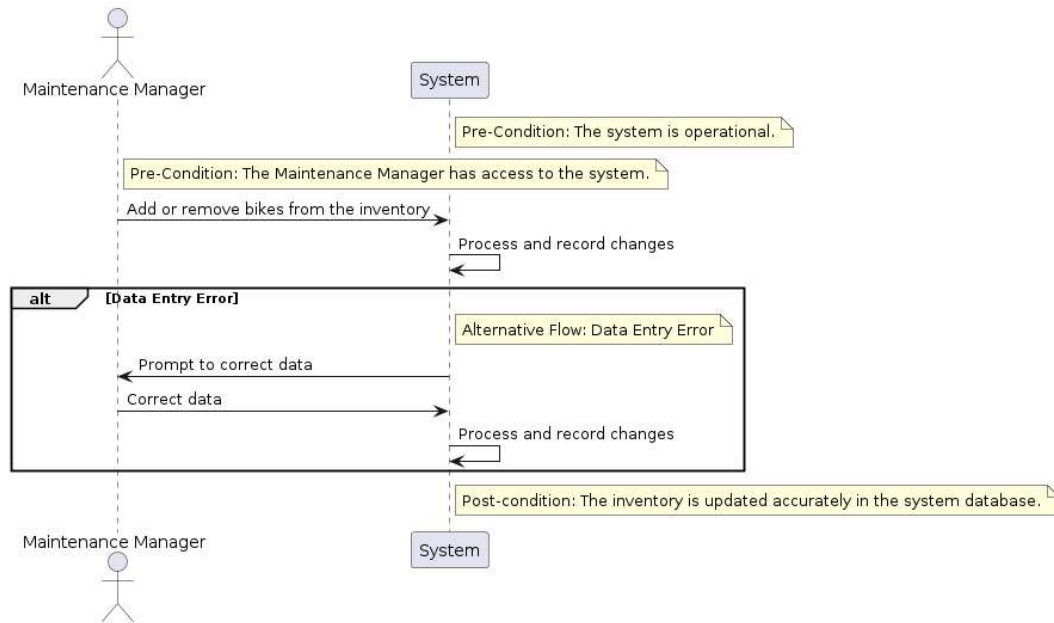


Figure 8 - Sequence diagram for use case {U.C.2.3} Update Bike Inventory

{U.C.3} Demand Forecasting: The system shall predict the demand for bikes at different times and locations to facilitate efficient allocation and redistribution of the bike fleet. It shall analyze historical usage patterns and real-time data to model and forecast future demand accurately. The system shall process data inputs, including time, location, and atmospheric conditions (temperature, precipitation), to generate demand predictions. These predictions shall be used to inform decisions on how to allocate and redistribute bikes across various locations to meet anticipated demand effectively. By predicting demand, the system shall enable optimal management of bike resources, ensuring that bikes are available where and when they are needed most. This functionality shall support strategic planning and operational efficiency, reducing instances of bike shortages or surpluses and improving overall service availability and user satisfaction.

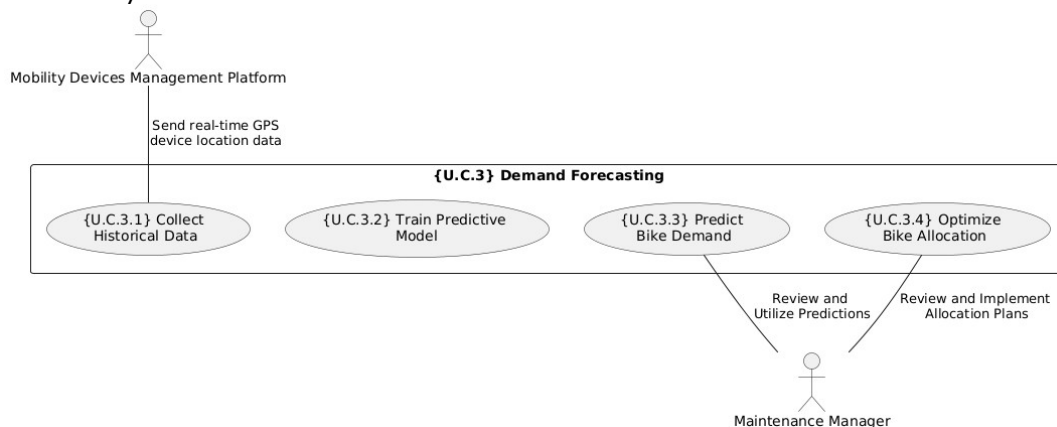


Figure 9 - Use Case diagram for {U.C.3} Demand Forecasting

{U.C.3.1} Collect Historical Data: The system shall collect historical data on bike usage, including times and locations, to support detailed analysis and informed decision-making. The system collects and stores this data in a historical database. The Mobility Devices Management Platform sends real-time GPS location data to the system, which then stores this data. The Mobility Devices Management Platform shall

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send real-time GPS device location data to the system, which stores this data in a historical database. In the event of a GPS device failure, where a GPS device attached to the Mobility Devices Management Platform fails, the system logs the error and continues collecting data from other operational devices.

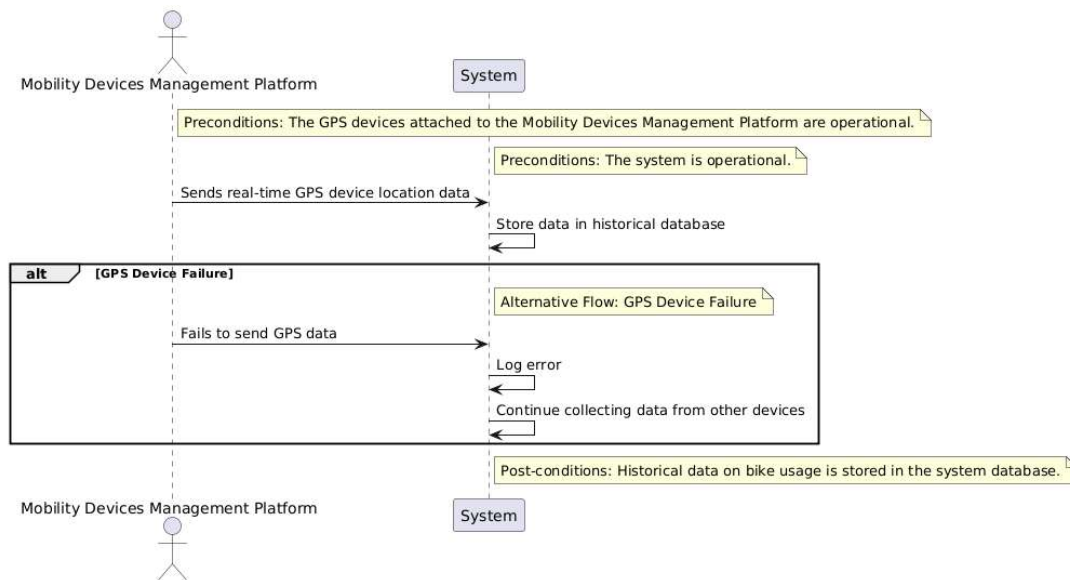


Figure 10 - Sequence diagram for use case {U.C.3.1} Collect Historical Data

{U.C.3.2} Train Predictive Model: The system shall train a predictive model using collected historical data to identify patterns in bike demand. The system is responsible for performing data analysis and model training. The system shall retrieve historical data on bike usage, processing this data to identify demand patterns, and then training the predictive model with the processed data. In the event of insufficient historical data, the system logs the issue and notifies the maintenance manager.

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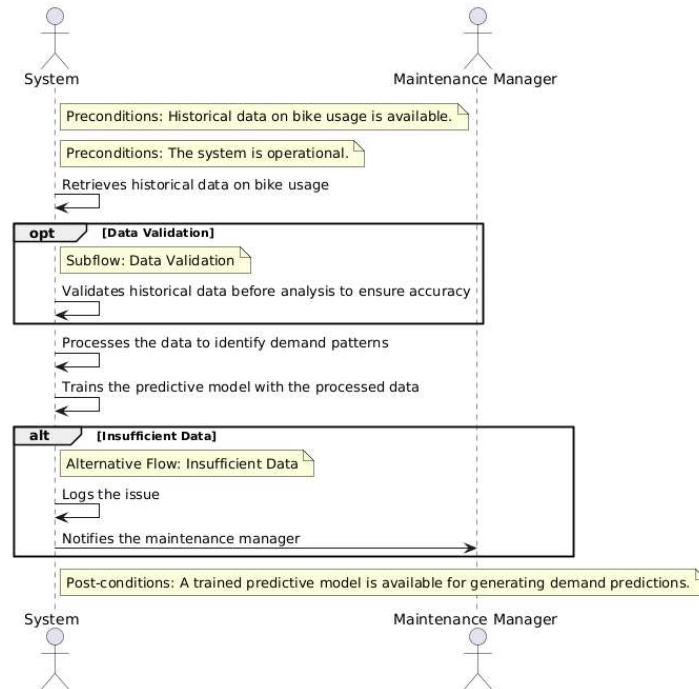


Figure 11 - Sequence diagram for use case {U.C.3.2} Train Predictive Model

{U.C.3.3} Predict Bike Demand: The system shall use the trained predictive model to generate predictions of future bike demand at different times and locations. The system is responsible for generating these demand predictions, while the Maintenance Manager utilizes them for operational planning. The system shall use the predictive model to generate future bike demand predictions and presenting these predictions to the Maintenance Manager, who then reviews and utilizes them for allocation and distribution decisions. In the event of significant prediction errors, the system logs the errors, notifies the Maintenance Manager, and adjusts the predictive model to improve accuracy. In the event of a Prediction Error, where significant errors are found in the predictions, the system shall adjust the prediction model accordingly.

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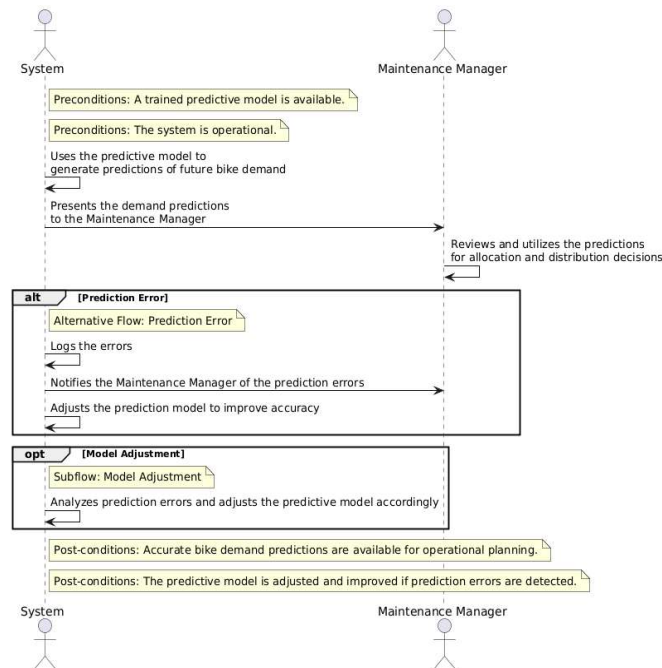


Figure 12 - Sequence diagram for use case {U.C.3.3} Predict Bike Demand

{U.C.3.4} Optimize Bike Allocation: The system shall optimize the allocation of bikes based on demand predictions to ensure efficient distribution and availability. The System shall analyze demand predictions alongside current inventory levels to generate recommendations for optimal bike allocation. These recommendations shall be reviewed and implemented by the Maintenance Manager. The optimal bike allocation should consider two or more goals, including minimizing the lack of bikes and the kilometers needed for the proposed reallocation or the costs of reallocation. In the event of Resource Constraints, where there are limitations such as a shortage of bikes, the system shall adjust its recommendations accordingly. Regular optimization of bike allocation is required to ensure that bikes are distributed efficiently to meet predicted demand, and adjustments may be made based on resource constraints.

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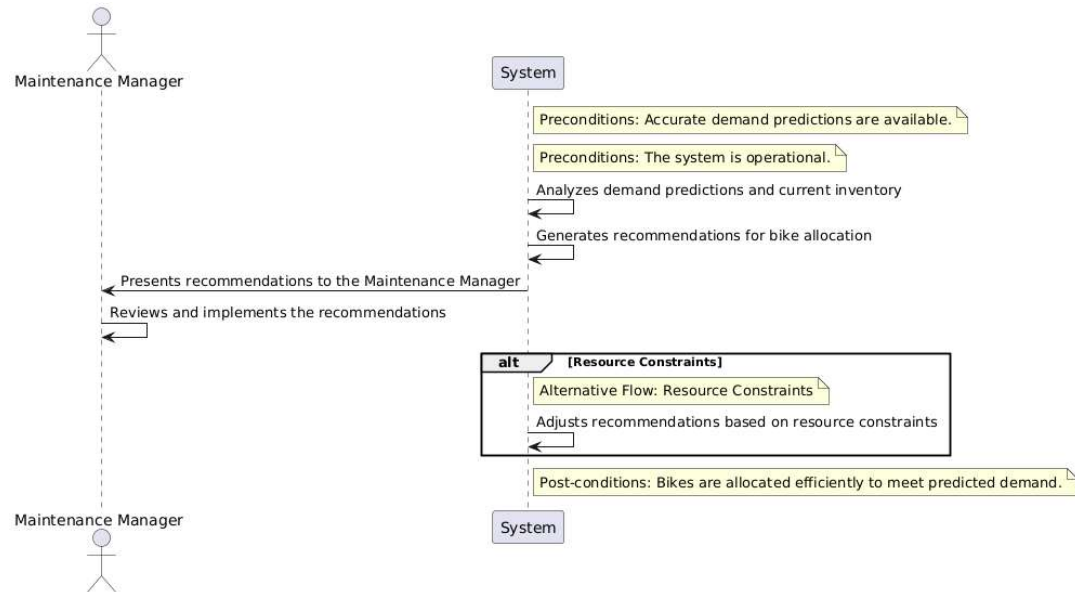


Figure 13 - Sequence diagram for use case {U.C.3.4} Optimize Bike Allocation

{U.C.4} Usage Analytics: The system shall analyze usage patterns, including peak usage times, popular routes, and average trip durations, to provide valuable insights for service improvement and future growth planning. The system shall process and analyze data collected from bike usage to identify these patterns accurately. The analysis of peak usage times shall help determine periods of highest demand, while the identification of popular routes shall reveal the most frequently traveled paths by users. Calculating the average trip duration shall provide insights into typical usage behaviors and preferences.

The system shall compile the results of these analyses into comprehensive reports, which can be reviewed by the Maintenance Manager to inform decisions related to service enhancements, resource allocation, and strategic planning for expansion. The system shall ensure that the analysis and reporting processes are accurate and reliable.

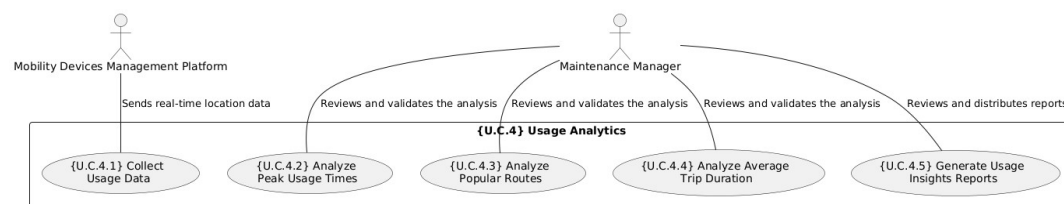


Figure 14 - Use Case diagram for {U.C.4} Usage Analytics

{U.C.4.1} Collect Usage Data: The system shall collect data on bike usage, including start and end times, routes taken, and trip durations, to provide detailed insights into bike utilization. This process involves receiving real-time location and trip data from the Mobility Devices Management Platform and storing it in a dedicated usage database.

In the event of a GPS Device Failure, where the Mobility Devices Management Platform fails to deliver data from a GPS device, the system shall log the error and continue collecting data from other operational devices to ensure continuous data collection.

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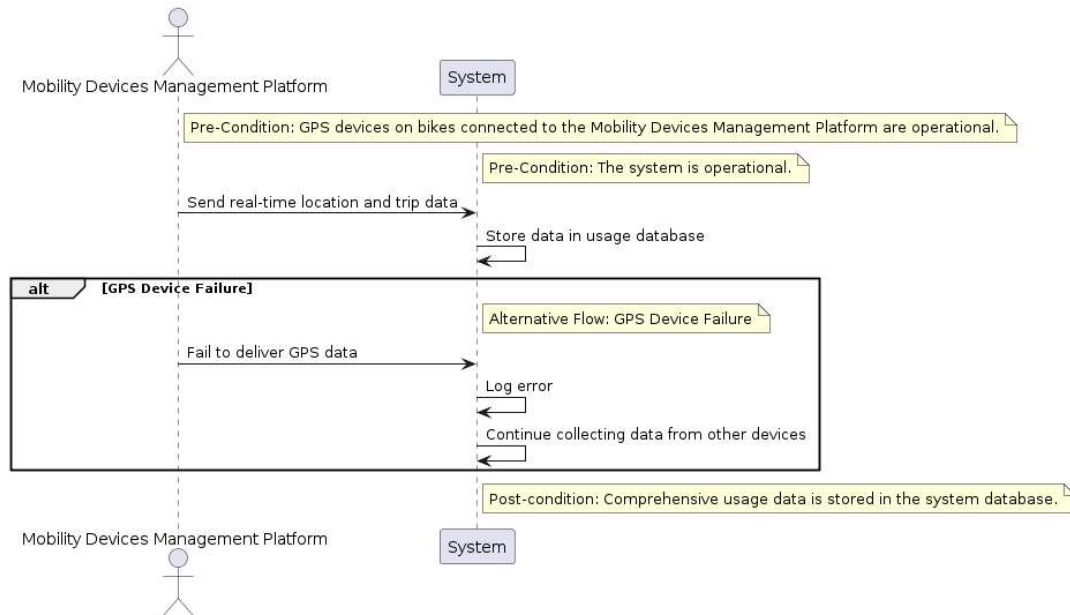


Figure 15 - Sequence diagram for use case {U.C.4.1} Collect Usage Data

{U.C.4.2} Analyze Peak Usage Times: The system shall analyze the available usage data to identify peak usage times, providing insights into the periods of highest bike demand. The Maintenance Manager then reviews and validates the analysis for accuracy.

In the event of a Data Anomaly, where anomalies are detected in the data, the system shall flag these anomalies for further review. Regular analysis is necessary to ensure that peak usage times are identified accurately, and any data anomalies are handled and flagged appropriately.

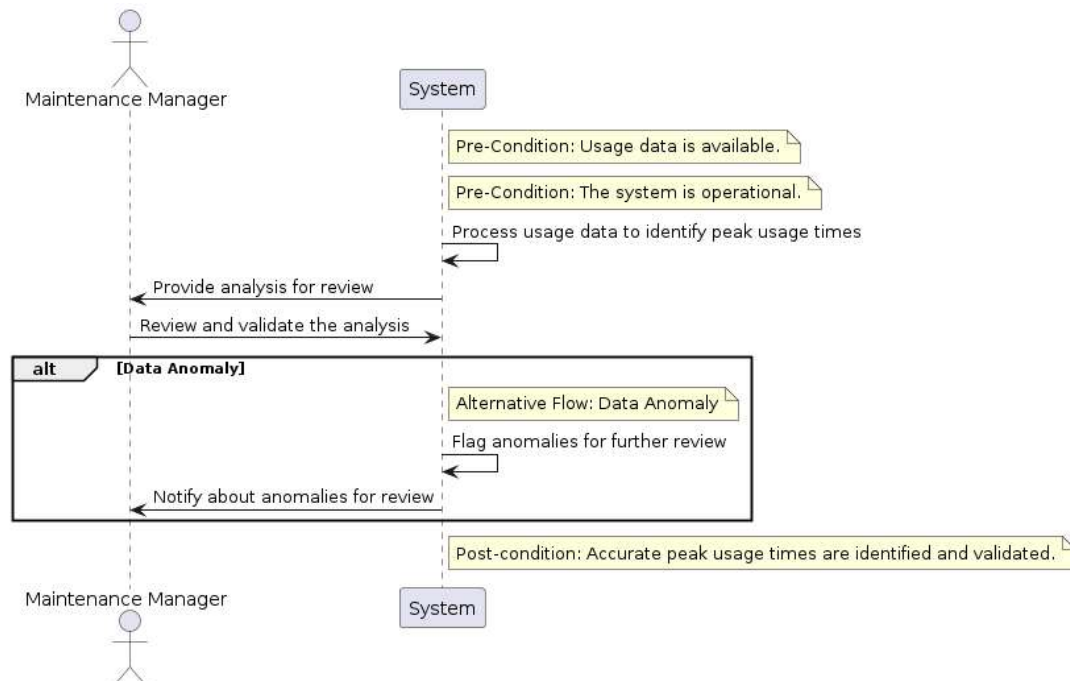


Figure 16 - Sequence diagram for use case {U.C.4.2} Analyze Peak Usage Times

{U.C.4.3} Analyze Popular Routes: The system shall analyze the available usage data to identify popular routes taken by users, providing insights into user preferences and travel patterns. The Maintenance Manager then reviews and validates the analysis to ensure accuracy.

In the event of a Data Anomaly, where anomalies are detected in the data, the system shall flag these anomalies for further review. Regular analysis is required to ensure that popular routes are accurately identified and any data anomalies are handled and flagged appropriately.

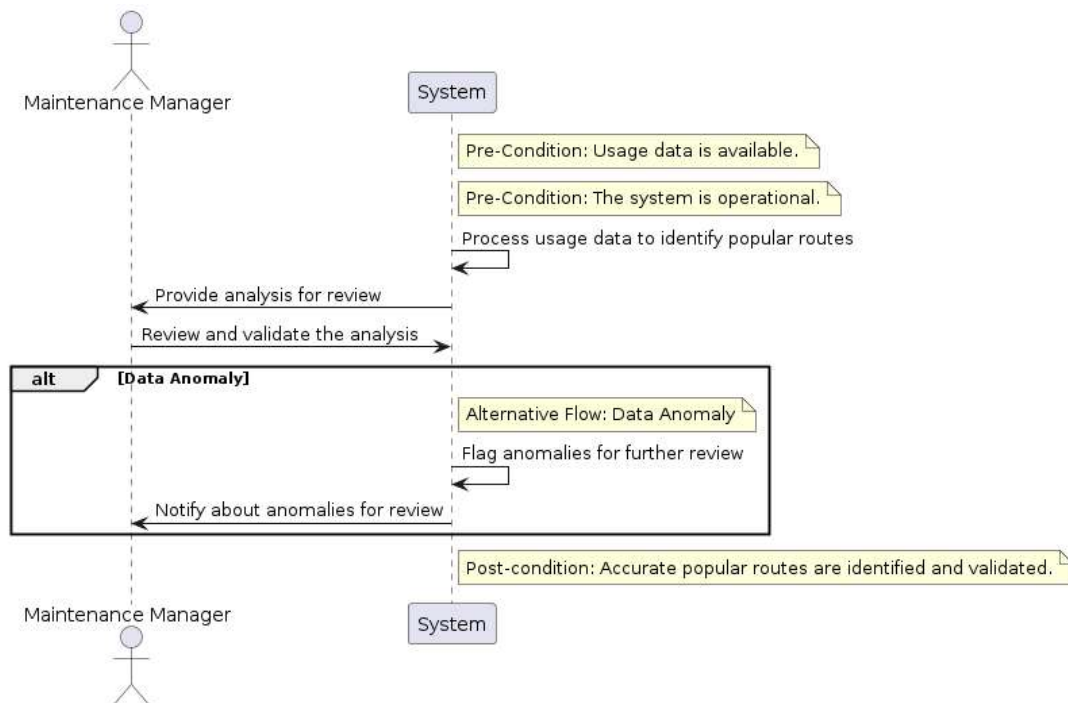


Figure 17 - Sequence diagram for use case {U.C.4.3} Analyze Popular Routes

{U.C.4.4} Analyze Average Trip Duration: The system shall analyze the available usage data to calculate average trip durations, providing insights into the typical length of bike trips. The Maintenance Manager then reviews and validates the analysis for accuracy.

In the event of a Data Anomaly, where anomalies are detected in the data, the system shall flag these anomalies for further review. Regular analysis is required to ensure that average trip durations are accurately calculated and any data anomalies are handled and flagged appropriately.

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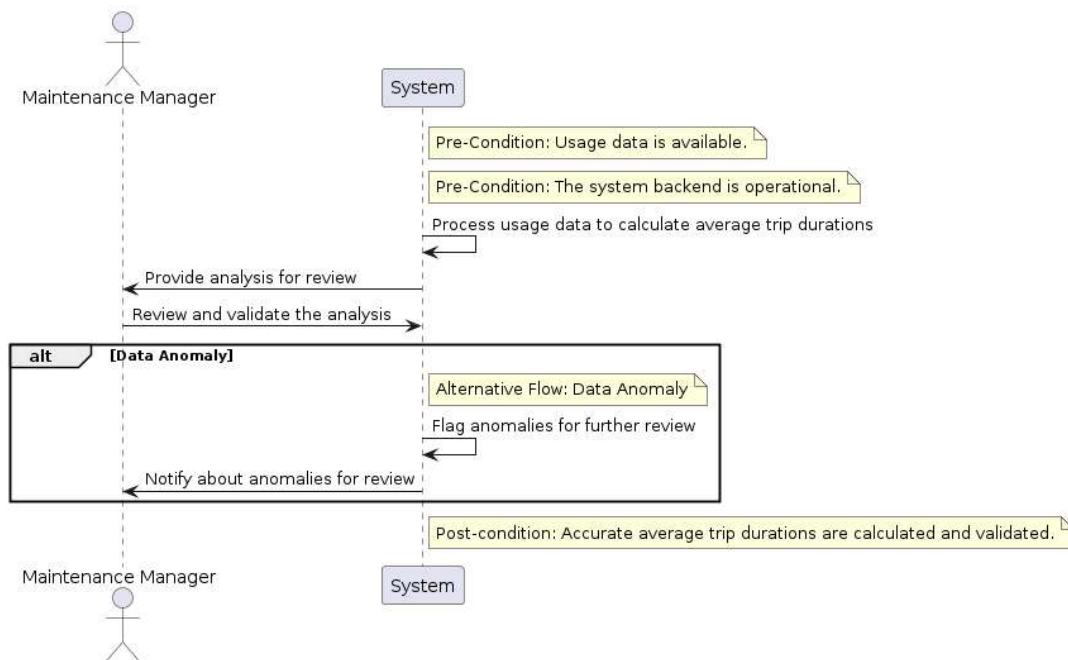


Figure 18 - Sequence diagram for use case {U.C.4.4} Analyze Average Trip Duration

{U.C.4.5} Generate Usage Insights Reports: The system shall generate comprehensive reports based on the results of the analysis of usage patterns, including peak times, popular routes, and average trip durations. The Maintenance Manager shall review the reports for accuracy and act on it.

In the event of a Report Generation Error, where errors occur during the report generation process, the system shall log these errors and retry the process to ensure successful report creation. Regular generation and distribution of usage insights reports are essential for operational planning and strategic decision-making, ensuring that the Maintenance Manager and other stakeholders are well-informed about usage patterns and trends.

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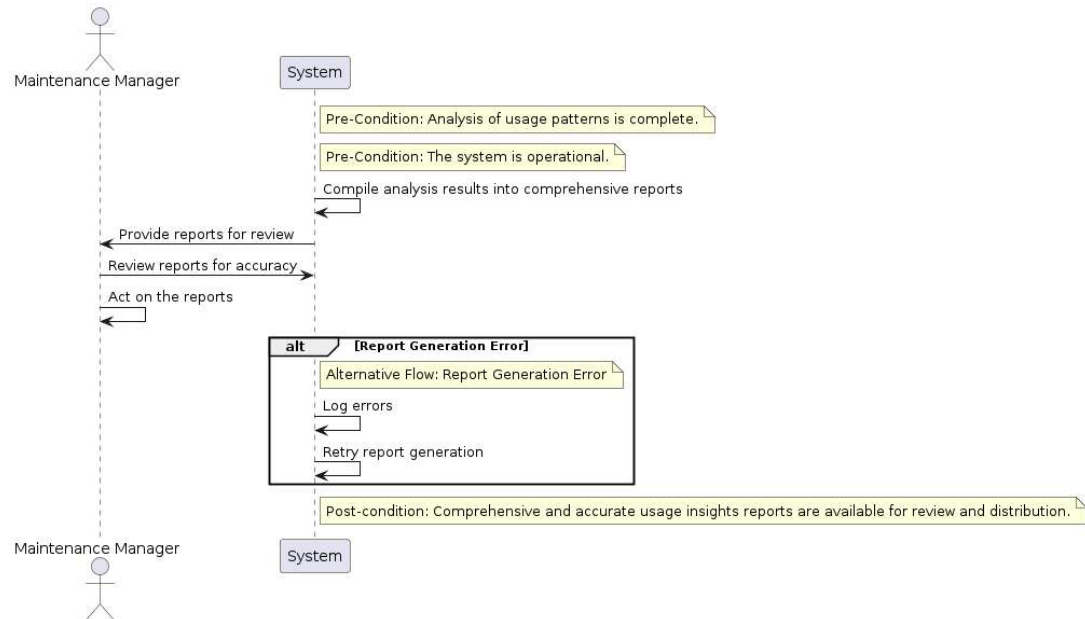


Figure 19 - Sequence diagram for use case {U.C.4.5} Generate Usage Insights Reports

{U.C.5} Incident Reporting: The system shall receive issues or incidents reported by users related to the bikes from the Bike4All Platform and send resolution feedback back to the Bike4All Platform. This functionality is essential for identifying bikes that need immediate maintenance or repair.

The system shall be capable of receiving issue reports from the Bike4All Platform, logging these reports in the system's database, and processing them to assign to maintenance operators. The system shall track the resolution status of each issue, ensuring that updates are recorded accurately as maintenance operators work on resolving the issues.

Once an issue is resolved, the system shall generate and send resolution feedback to the Bike4All Platform. In case of errors during report reception or feedback transmission, the system shall log the errors and retry the process to ensure successful communication.

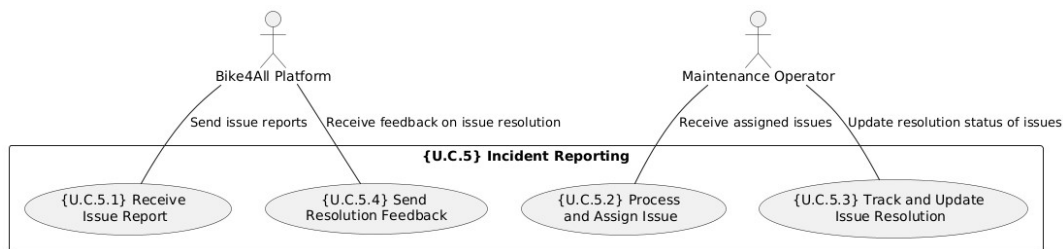


Figure 20 - Use case diagram for {U.C.5} Incident Reporting

{U.C.5.1} Receive Issue Report: The system shall receive issues or incidents reported by users related to the bikes. The Bike4All Platform shall send issue reports to the system, which shall then receive and log these reports in the database.

In the event of a Transmission Error, where there is an error in receiving the report, the system shall log the error and request a retransmission to ensure that all issue reports are captured accurately.

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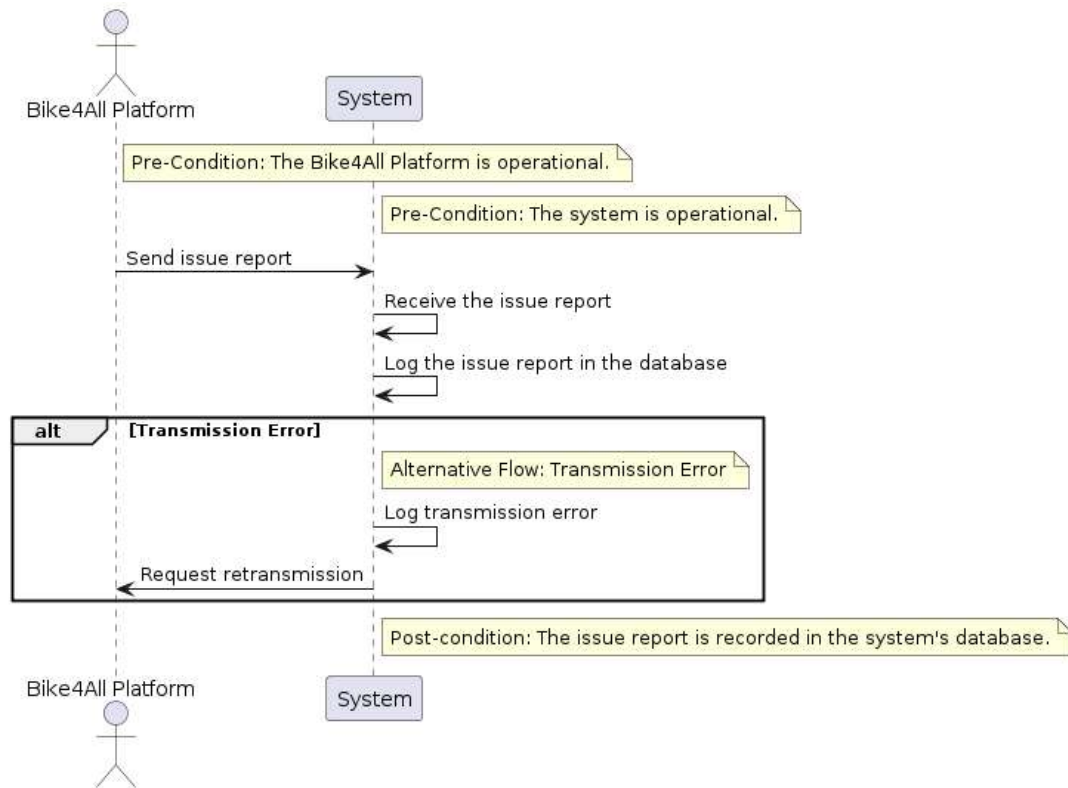


Figure 21 - Sequence diagram for use case {U.C.5.1} Receive Issue Report

{U.C.5.2} Process and Assign Issue: The system shall process and assign received and logged issues for resolution to the appropriate maintenance operator. In the event of an Assignment Error, where there is an error in the assignment process, the system shall log the error and retry the assignment to ensure that all issues are assigned correctly.

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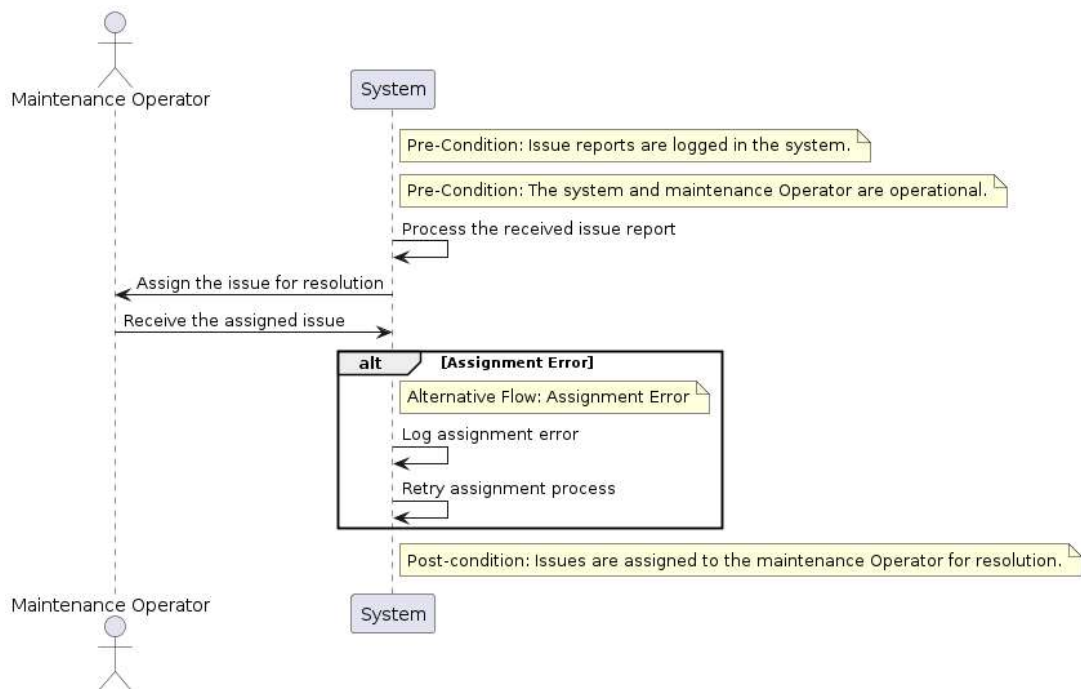


Figure 22 - Sequence diagram for use case {U.C.5.2} Process and Assign Issue

{U.C.5.3} Track and Update Issue Resolution: The system shall track and update the resolution status of reported issues to ensure effective maintenance operations. The Maintenance Operator shall update the status of assigned issues as they work on them, and the System shall track and record these updates accurately.

In the event of an Update Error, where there is an error in updating the status, the system shall log the error and request a status update retry to ensure the information remains accurate.

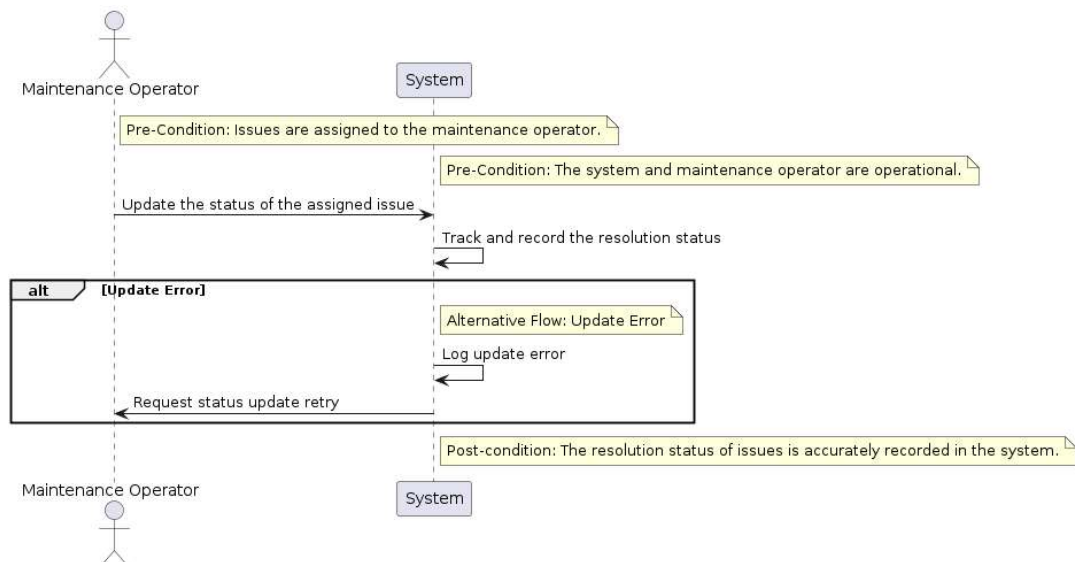


Figure 23 - Sequence diagram for use case {U.C.5.3} Track and Update Issue Resolution

{U.C.5.4} Send Resolution Feedback: The system shall generate and send feedback based on the updated statuses of resolved issues to the Bike4All Platform to ensure effective communication and tracking of maintenance activities. In the event of a Transmission Error, where there is an error in sending the feedback, the system shall log the error and retry the transmission to ensure that feedback is successfully communicated.

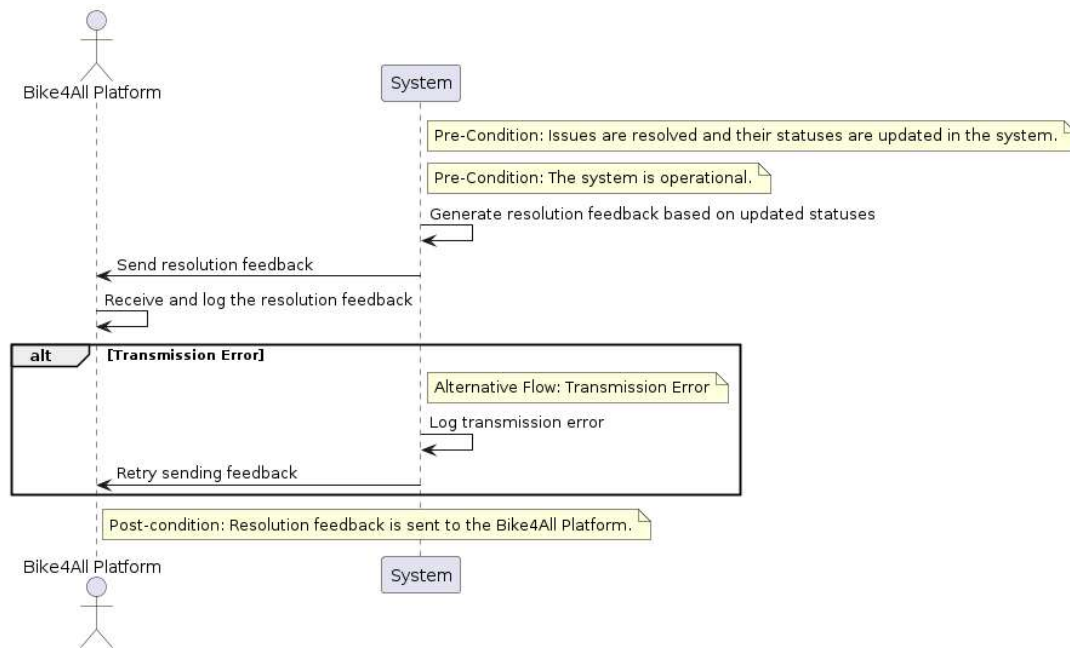


Figure 24 - Sequence diagram for use case {U.C.5.4} Send Resolution Feedback

{U.C.6} Security Features: The system shall incorporate security features to prevent the theft and misuse of bikes, ensuring the protection and optimal use of the bike-sharing fleet. Key security features shall include GPS tracking, remote locking mechanisms, and alarm systems.

GPS tracking shall enable real-time monitoring of bike locations, allowing for the detection of unauthorized movements. Remote locking mechanisms shall allow the system to lock bikes remotely, preventing unauthorized use. Alarm systems shall be able to trigger audible alerts to deter theft or tampering.

The system shall be capable of continuously monitoring the status of these security features and generating alerts in case of any signs of unauthorized use or tampering. The system shall also enable the Maintenance Manager to respond promptly to security alerts, ensuring swift action to secure the bikes.

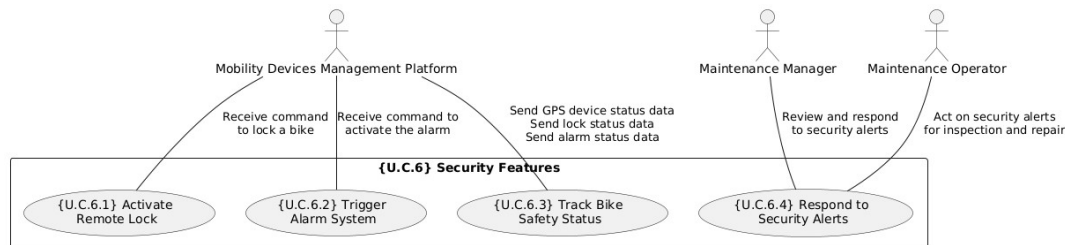


Figure 25 - Use case diagram for {U.C.6} Security Features

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{U.C.6.1} Activate Remote Lock: The system shall send a lock command to a bike to prevent unauthorized use through the Mobility Devices Management Platform, which then locks the bike and sends a confirmation back to the system. In the event of a Lock Mechanism Failure, where the platform fails to lock the bike, the system shall log an error and retry the command to ensure the bike is securely locked.

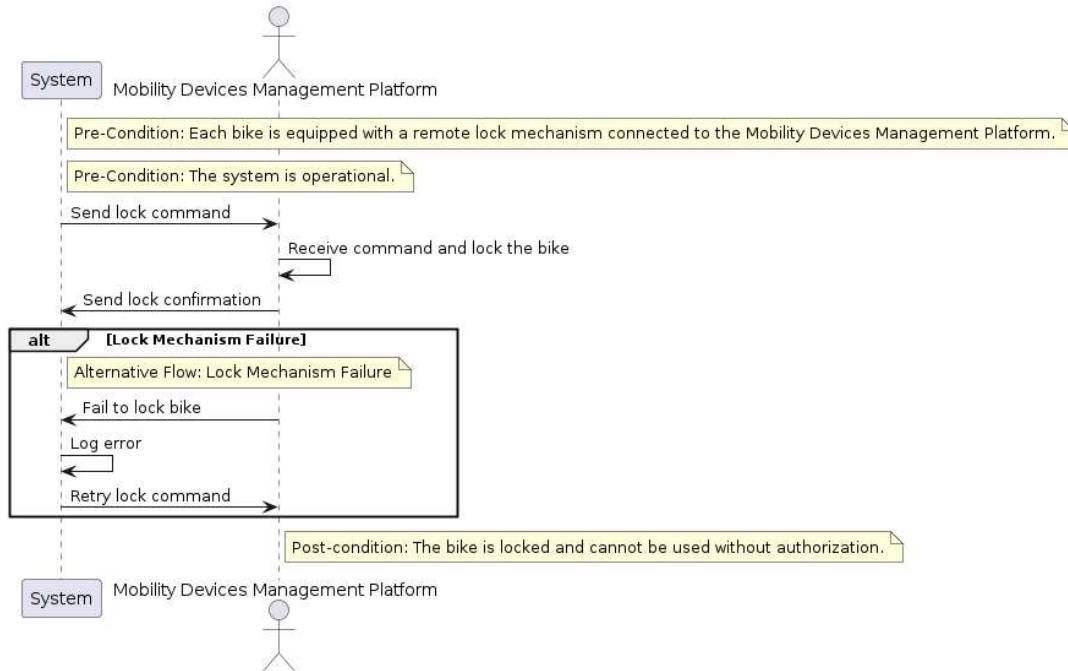


Figure 26 - Sequence diagram for use case {U.C.6.1} Activate Remote Lock

{U.C.6.2} Trigger Alarm System: The system shall trigger an alarm on a bike to deter theft or misuse by sending commands to the Mobility Devices Management Platform, which activates the alarm on the bike, and sends a confirmation back to the system. Each bike must be equipped with an alarm system connected to the Mobility Devices Management Platform. In the event of an Alarm Trigger Failure, where the platform fails to activate the alarm, the system shall log the error and retry the command to ensure successful activation.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

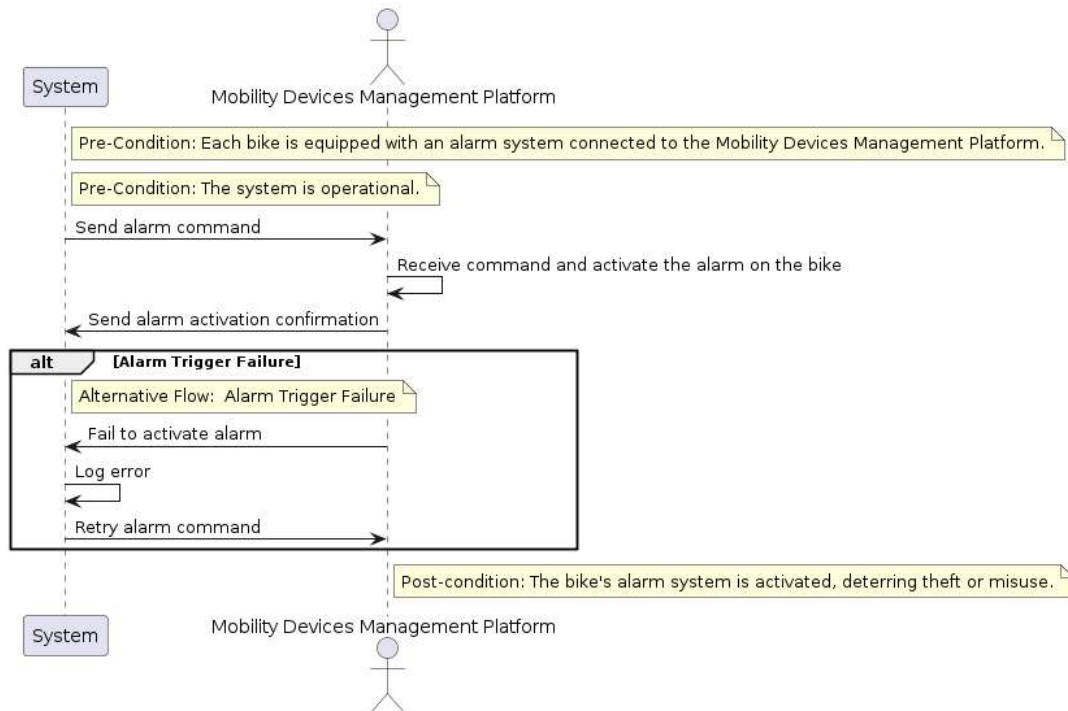


Figure 27 - Sequence diagram for use case {U.C.6.2} Trigger Alarm System

{U.C.6.3} Track Bike Safety Status: The system shall continuously monitor the status of bikes for any signs of unauthorized use or tampering, utilizing data from GPS devices, remote lock mechanisms, and alarm systems. Each bike must be equipped with functioning GPS, lock, and alarm systems, and the system must be operational.

The Mobility Devices Management Platform sends status data about the GPS device, remote lock mechanism, and alarm system of each bike to the system at regular intervals. The system monitors and processes this status data, flagging any signs of unauthorized use or tampering for further inspection.

In the event of a System Error, where the system encounters an error, it shall log the error and continue monitoring the status of other bikes to maintain overall system functionality.

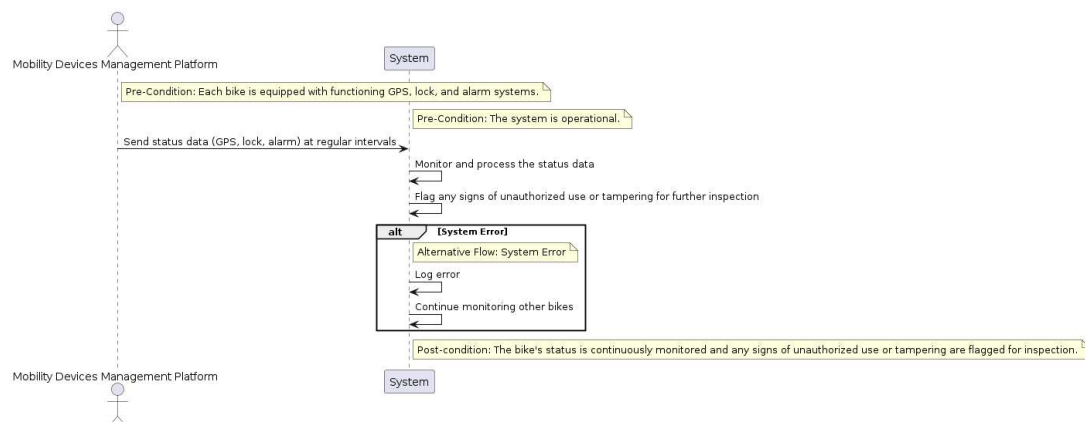


Figure 28 - Sequence diagram for use case {U.C.6.3} Track Bike Safety Status

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{U.C.6.4} Respond to Security Alerts: The system shall respond to security alerts generated by the monitoring system to ensure bike security and prompt maintenance actions. When signs of unauthorized use or tampering are detected, the system shall generate a security alert and notify the Maintenance Manager, who will review the alert and decide on the necessary action, like assigning any required physical inspection or repair tasks to the Maintenance Operator. The Maintenance Operator acts on the alert and updates the system with the resolution status. In the event of a False Alarm, where the alert is found to be unnecessary, the Maintenance Manager shall mark it as resolved without further action.

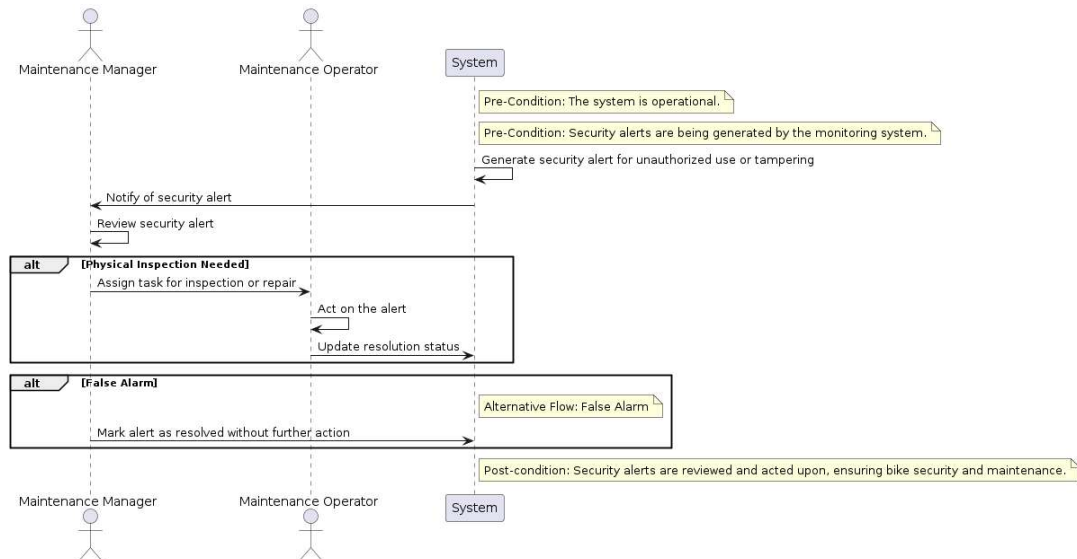


Figure 29 - Sequence diagram for use case {U.C.6.4} Respond to Security Alerts

{U.C.7} User Interface: The system shall provide an easy-to-use interface for the Maintenance Manager to effectively manage the fleet. This interface shall enable the Maintenance Manager to perform various tasks such as viewing fleet overviews, assigning maintenance tasks, monitoring bike status and generating reports. The interface shall be intuitive, allowing the Maintenance Manager to navigate and execute tasks efficiently without extensive training. It should provide clear access to real-time data and status updates, ensuring that the Maintenance Manager can make informed decisions promptly.

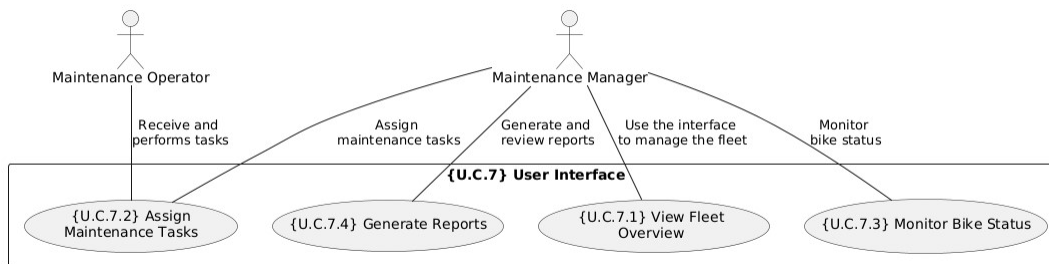


Figure 30 – Use case diagram for {U.C.7} User Interface

{U.C.7.1} View Fleet Overview: The system shall provide a comprehensive overview of the entire bike fleet, including the status of each bike, to the Maintenance Manager. The Maintenance Manager shall be able to use the system interface to

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manage the fleet effectively, showing the status of each bike (available, in use, under maintenance), relying on real-time data and status updates provided by the system. In the event of a Data Retrieval Error, where the system fails to retrieve data, an error message shall be displayed and the system shall retry the data retrieval process.

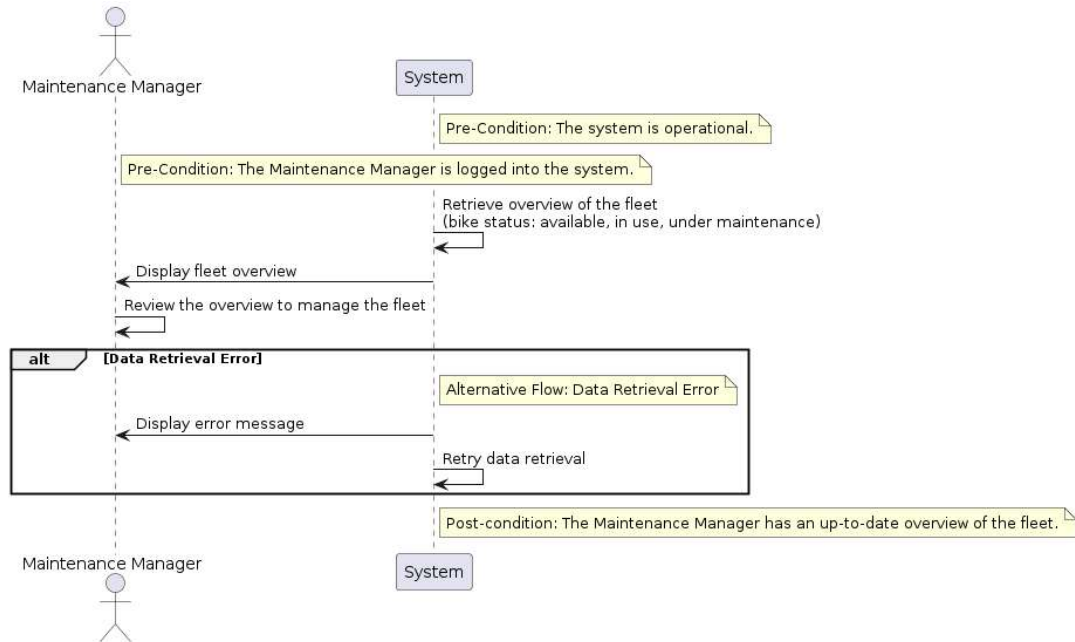


Figure 31 - Sequence diagram for use case {U.C.7.1} View Fleet Overview

{U.C.7.2} Assign Maintenance Tasks: The system shall allow the Maintenance Manager to identify bikes that require maintenance and assign maintenance tasks to the Maintenance Operator efficiently. The system shall then notify the maintenance operator while logging the task assignment. The Maintenance Manager shall identify bikes requiring maintenance from the fleet overview and assign the necessary tasks to the Maintenance Operator.

If a notification fails, the system shall log the error and retry the notification process.

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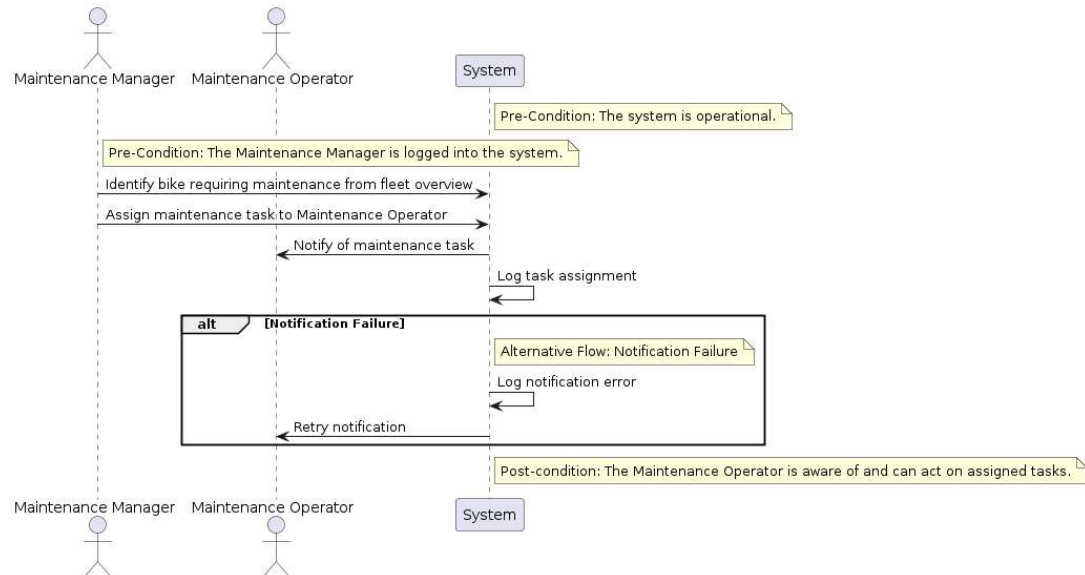


Figure 32 - Sequence diagram for use case {U.C.7.2} Assign Maintenance Tasks

{U.C.7.3} Monitor Bike Status: The system shall allow the Maintenance Manager to monitor the detailed status of each bike of the fleet. The maintenance manager shall select a bike, access the detailed menu of that bike and the system provides real-time updates on the status (location, usage, maintenance...). In the event of an Update Failure, the system shall log the error and retry to update the status of the selected bike.

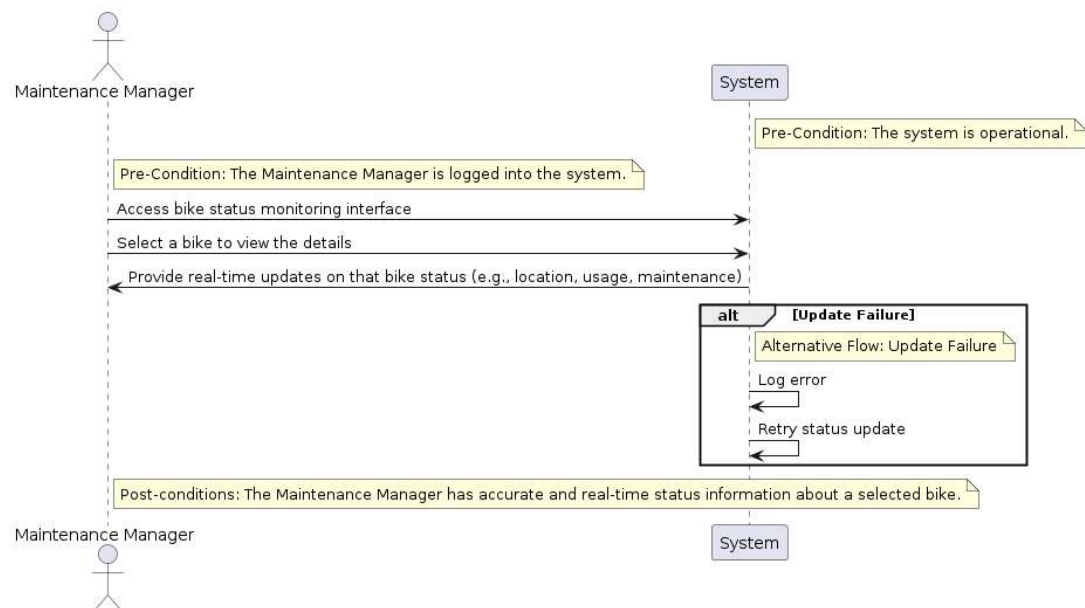


Figure 33 - Sequence diagram for use case {U.C.7.3} Monitor Bike Status

{U.C.7.4} Generate Reports: The system shall enable the Maintenance Manager to generate comprehensive reports on various aspects of the fleet, such as usage statistics, maintenance history, and performance metrics. The Maintenance Manager shall select the type of report to generate, the system compiles the necessary data, and generates the report for review and analysis by the Maintenance Manager.

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If a report generation error occurs, the system shall log the error and retry the report generation process.

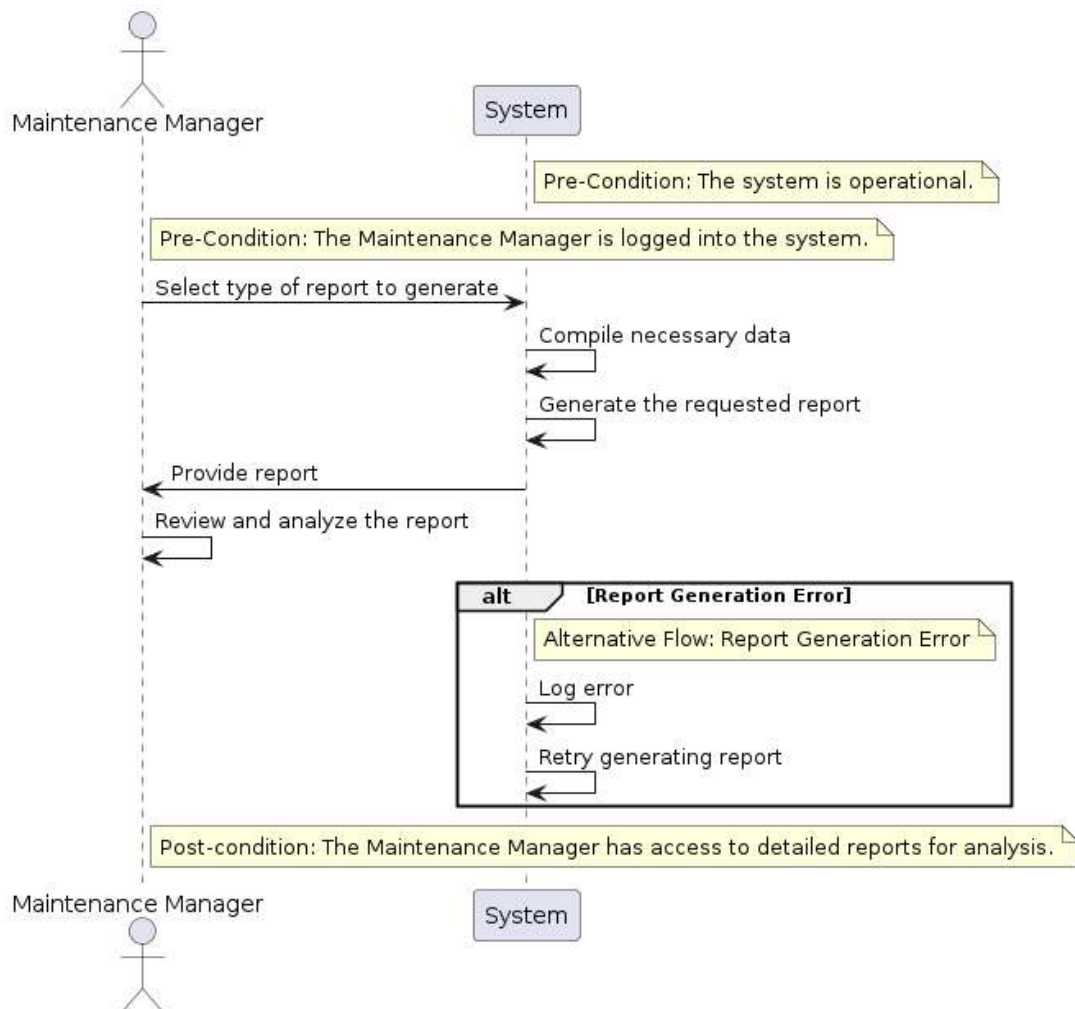


Figure 34 - Sequence diagram for use case {U.C.7.4} Generate Reports

{U.C.8} Integration Capabilities: The system shall be able to integrate with external systems such as the Bike4All platform and the Mobility Devices Management Platform. The integration with the Bike4All platform is necessary for handling customer issues reports and sending the maintenance works feedback. This involves the system managing data synchronization, receiving customer issues reports the Bike4All platform, processing them, and enabling maintenance operators or managers to respond through the system. The integration ensures that the issues are addressed promptly and efficiently.

Additionally, the system shall integrate with the Mobility Devices Management Platform to enable GPS tracking of bikes. This requires the system to communicate with the Mobility Devices Management Platform to request and receive real-time GPS tracking data. The system processes this data to update bike statuses and locations in its database. Maintenance operators and managers use this data to locate and manage bikes effectively. The system shall handle data synchronization, ensure continuous and accurate reception of GPS data, and address any unavailability of data gracefully.

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These integrations are essential for providing seamless customer support and accurate bike tracking, enhancing the overall efficiency and reliability of the maintenance bike-sharing platform.

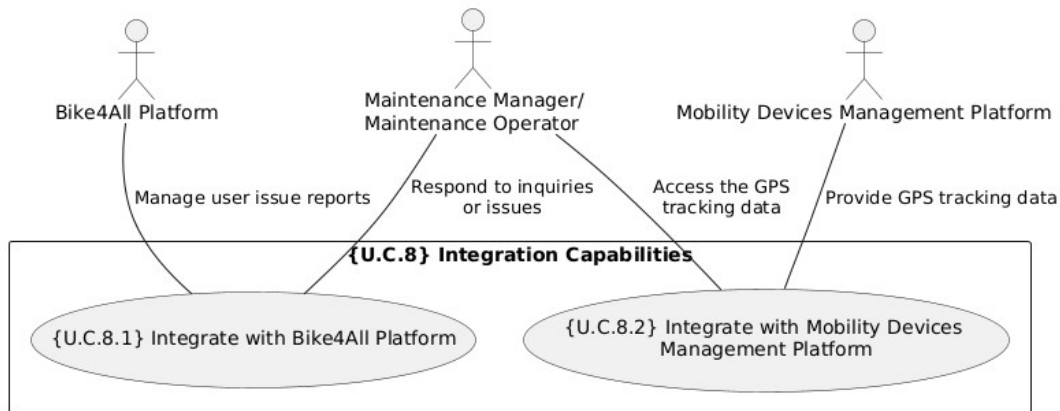


Figure 35 - Use case diagram for {U.C.8} Integration Capabilities

{U.C.8.1} Integrate with Bike4All Platform: The system shall integrate with the Bike4All Platform to manage customer issues reporting. The integration process involves the system handling data synchronization and ensuring communication with the Bike4All Platform. Maintenance operators or managers will respond to these issues through the system.

The issue reports submitted by the users of the Bike4All Platform are forwarded to the system, which receives and processes it. The system notifies the maintenance operator and maintenance manager of new issues, who then responds through the system. The system sends the response back to the Bike4All Platform with the response or resolution.

In case the system is unavailable, the system must log the issue and notify the Bike4All Platform of the delay.

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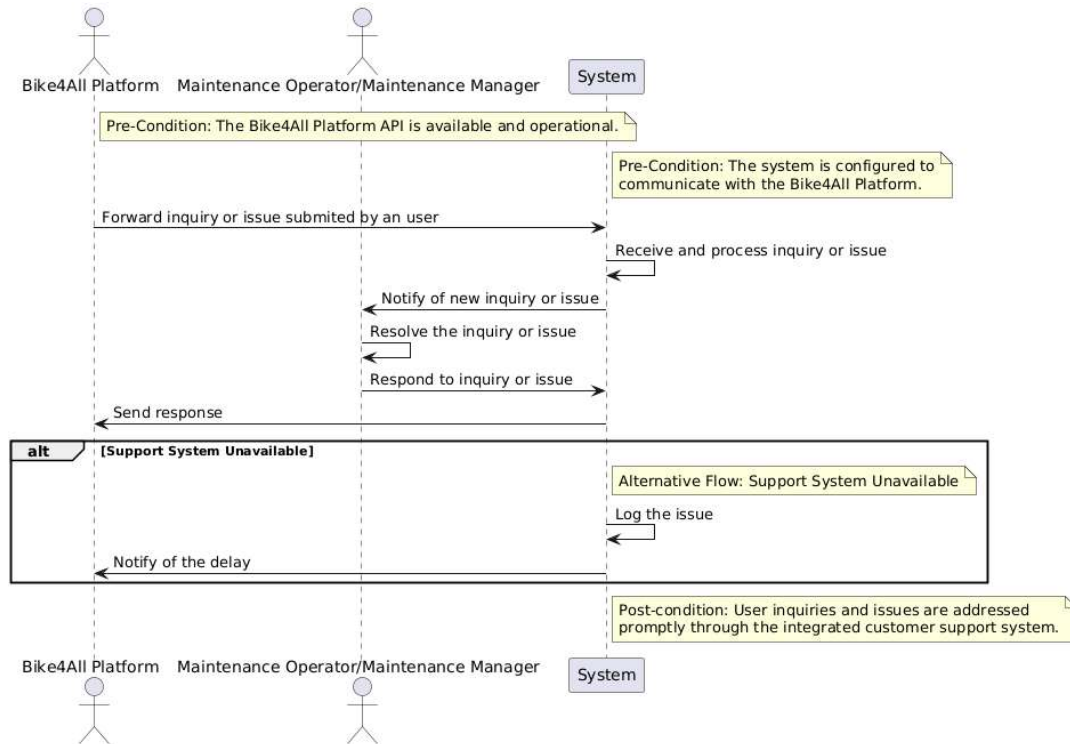


Figure 36 - Sequence diagram for use case {U.C.8.1} Integrate with Bike4All Platform

{U.C.8.2} Integrate with Mobility Devices Management Platform: The system shall integrate with the external Mobility Devices Management Platform, equipped with a GPS tracking system to monitor bike locations in real time. This integration ensures continuous and accurate tracking of bike locations to support fleet management.

The system receives, processes and updates in the database the real-time location data from the Mobility Devices Management Platform.

In the event of a Data Transmission Error, where there is an error in data transmission, the system shall log the error and attempt to re-establish the connection to ensure data continuity.

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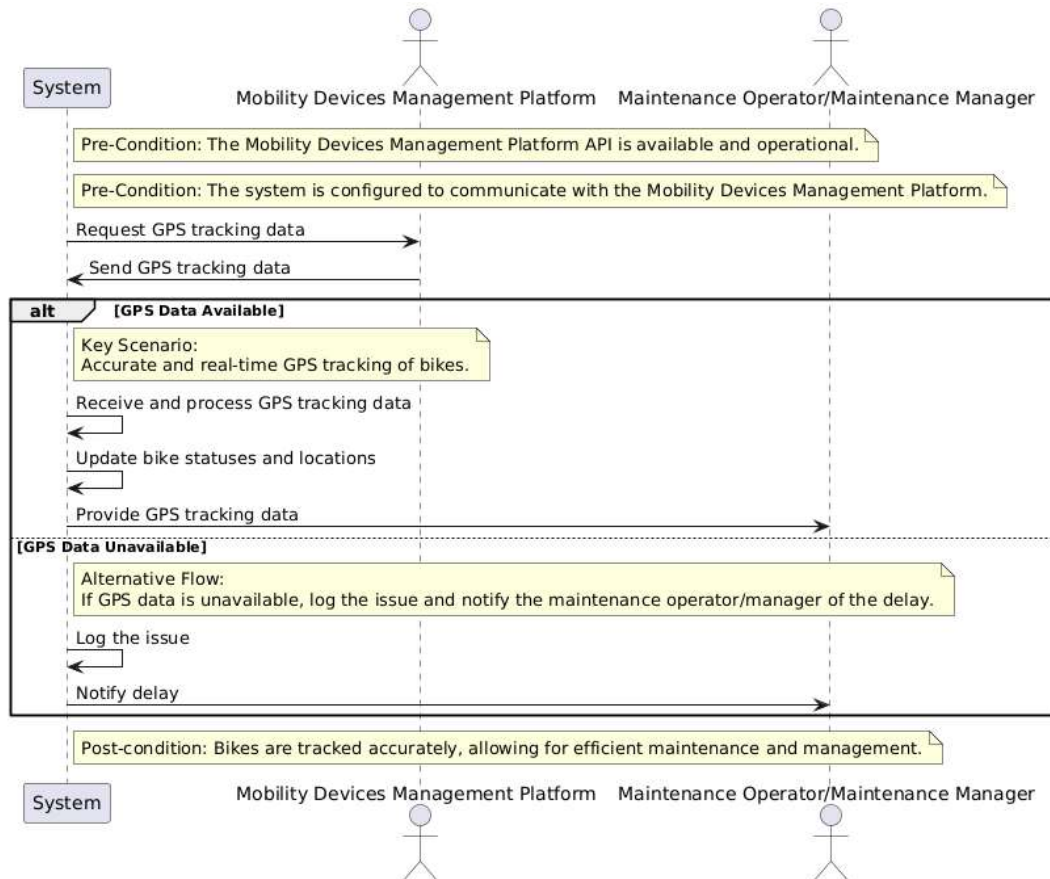


Figure 37 - Sequence diagram for use case {U.C.8.2} Integrate with Mobility Devices Management Platform

{U.C.9} Maintenance Scheduling: The system shall schedule regular maintenance checks for each bike in the fleet based on predefined criteria such as usage, time, and distance traveled. The system will analyze each bike's data periodically to determine which bikes meet the criteria for maintenance checks. Upon identifying the bikes that require maintenance, the system will schedule the maintenance checks and log the schedule in its database. This scheduling process ensures that all bikes receive timely maintenance based on their usage patterns, operational time, and distance covered.

The system shall handle any errors in data analysis and scheduling, ensuring continuous and accurate scheduling of maintenance activities.

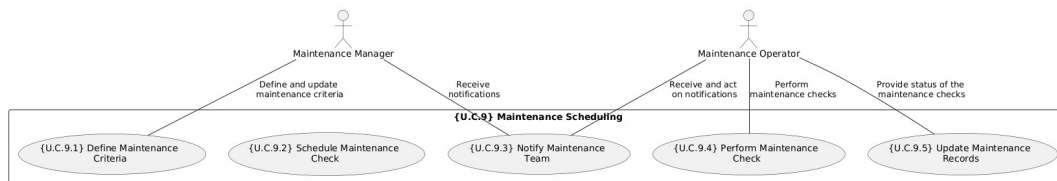


Figure 38 – Use case diagram for {U.C.9} Maintenance Scheduling

{U.C.9.1} Define Maintenance Criteria: The system shall allow the Maintenance Manager to define and adjust criteria for scheduling regular maintenance checks for bikes. These criteria encompass factors such as usage hours, time intervals or distance traveled. The system then saves and applies these criteria to the entire fleet.

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In cases where the entered criteria are invalid, the system will prompt the Maintenance Manager to correct the entries.

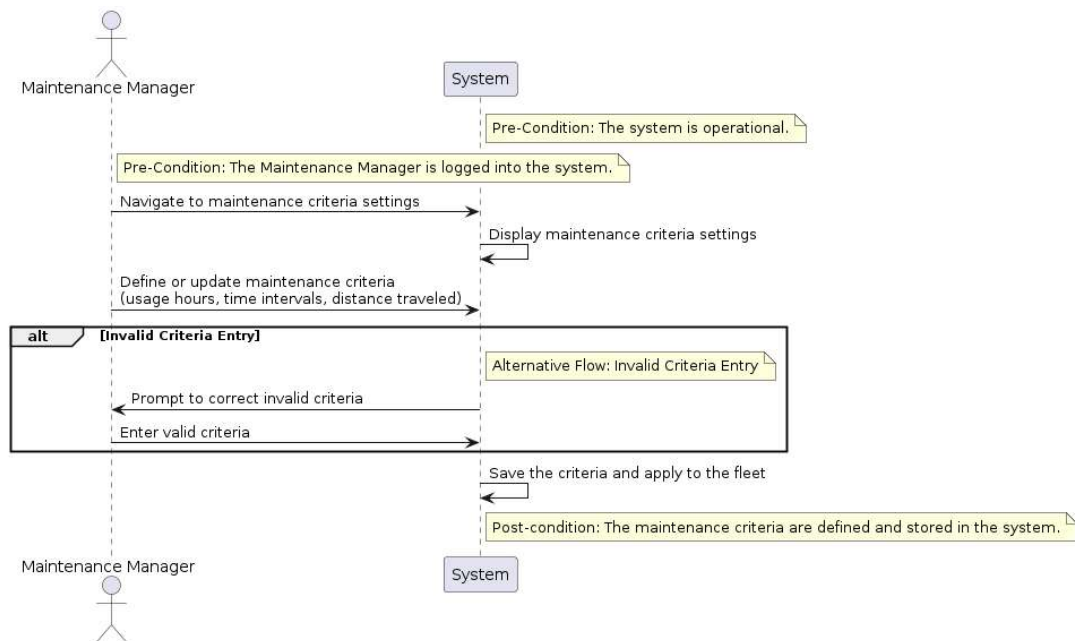


Figure 39 - Sequence diagram for use case {U.C.9.1} Define Maintenance Criteria

{U.C.9.2} Schedule Maintenance Check: The system shall be capable of scheduling regular maintenance checks for each bike based on the criteria predefined by the Maintenance Manager. The Maintenance Manager oversees this process while the system analyzes relevant data and schedules the checks.

The system shall periodically analyze each bike's data, such as usage, time, and distance traveled, determine which bikes meet the criteria for a maintenance check and schedule these checks accordingly, logging the schedule for tracking and management purposes.

In cases where there is an error in data analysis, the system must log the error and retry the analysis.

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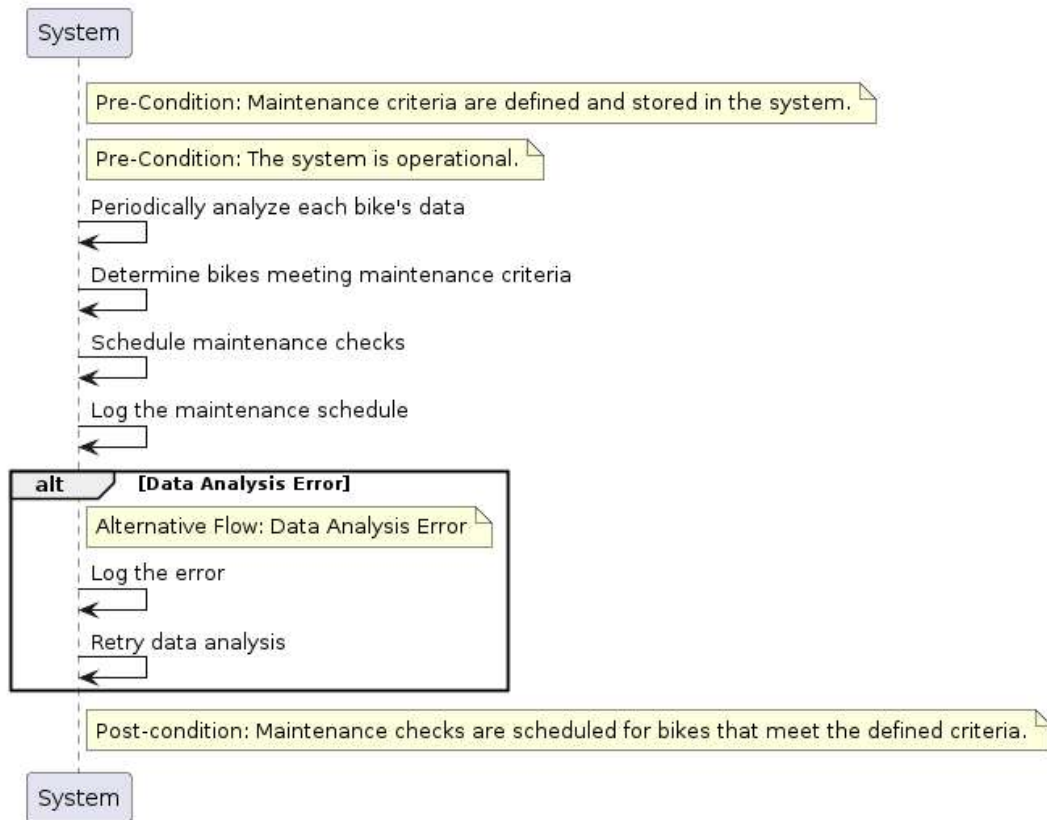


Figure 40 – Sequence diagram for use case {U.C.9.2} Schedule Maintenance Check

{U.C.9.3} Notify Maintenance Team: The system shall generate notifications for the scheduled maintenance checks and send them to the maintenance manager and maintenance operator. Upon receiving the notifications, the maintenance manager and maintenance operator review them to prepare for the maintenance tasks. In cases where a notification fails to send, the system must log the error and retry sending the notification.

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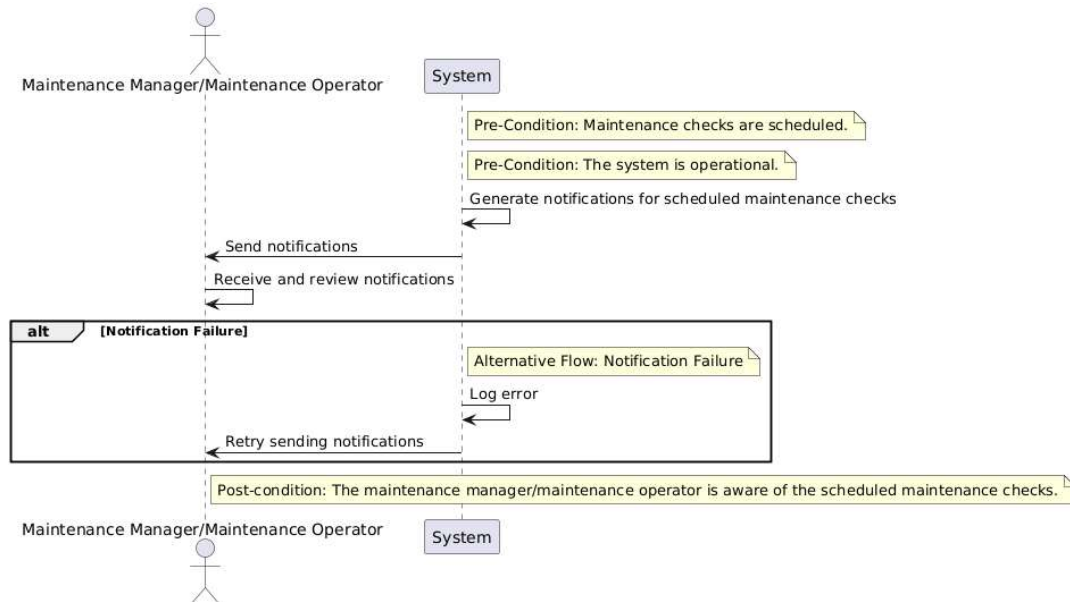


Figure 41 - Sequence diagram for use case {U.C.9.3} Notify Maintenance Team

{U.C.9.4} Perform Maintenance Check: The system shall support the process by which the Maintenance Operator performs scheduled maintenance checks on bikes. The Maintenance Operator reviews notifications, performs the maintenance checks, and logs all the tasks performed, status and results in the system. In scenarios where the Maintenance Operator is unable to perform a maintenance check, such as the bike being unavailable, the system shall log the issue and reschedule the maintenance.

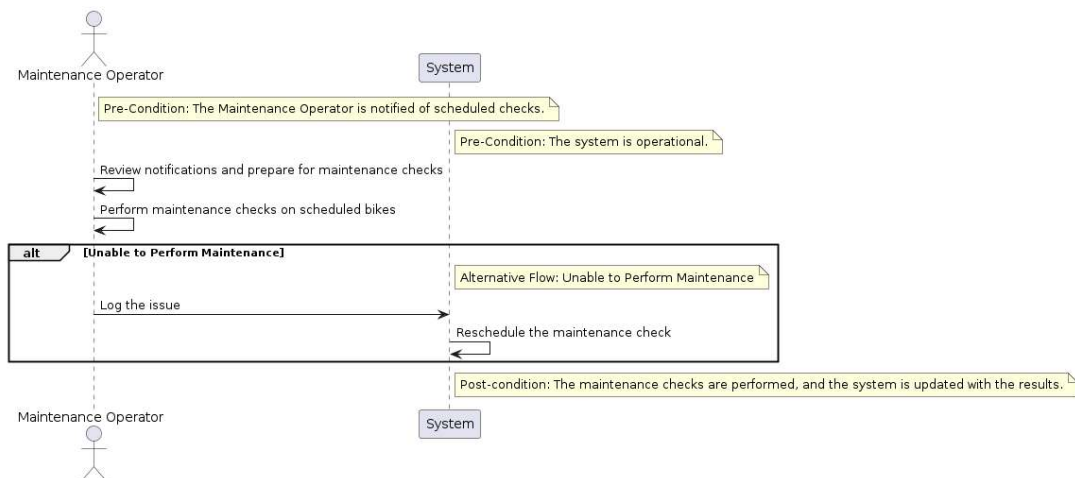


Figure 42 - Sequence diagram for use case {U.C.9.4} Perform Maintenance Check

{U.C.9.5} Update Maintenance Records: The system shall update maintenance records based on the results of performed maintenance checks. The Maintenance Operator provides the status and results of these checks, which the system then updates and stores in the database, and confirms the update to the Maintenance Operator. These records capture detailed information such as maintenance dates, actions taken, and parts replaced, ensuring an up-to-date history for each bike. In cases where there is an error in updating the records, the system must log the error and retry the update.

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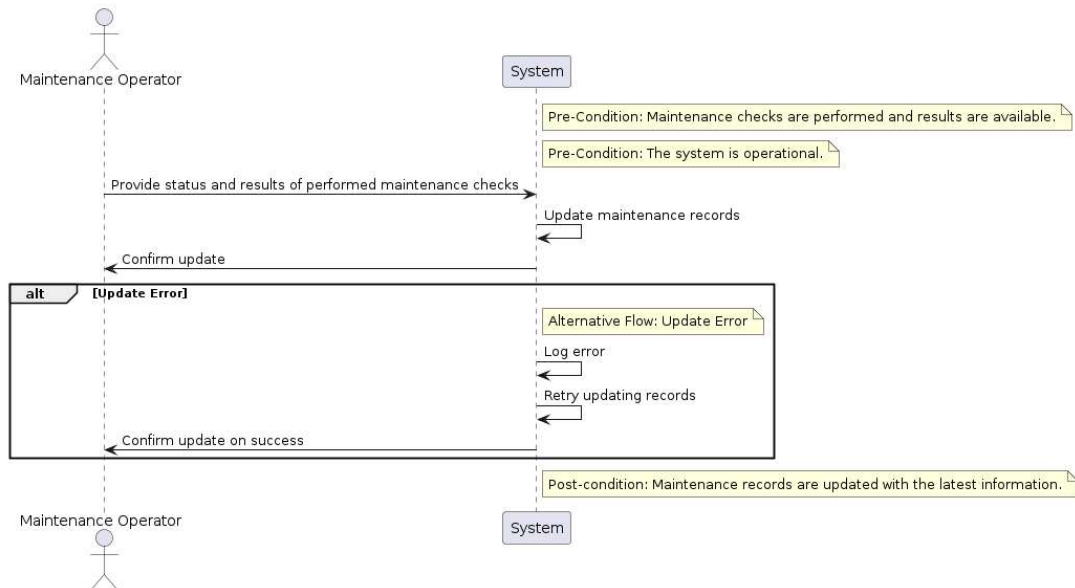


Figure 43 - Sequence diagram for use case {U.C.9.5} Update Maintenance Records

{U.C.10} Maintenance Alerts: The system shall generate alerts and send notifications when a bike is due for maintenance to ensure that no bike misses its scheduled maintenance check. The system should continuously monitor bike maintenance schedules to determine when a bike is due for maintenance. Upon identifying a bike that requires maintenance on a past schedule, the system should generate an alert and send a notification to the maintenance operator or maintenance manager. The notification should include details such as the bike's identification, the type of maintenance required, and the due date for the maintenance check. The system shall handle and log any errors in generating alerts or sending notifications, retrying the process if necessary. Additionally, the system shall provide a user-friendly interface for the maintenance manager or maintenance operators to manage and acknowledge received notifications. Ensuring timely alerts and notifications helps maintain the optimal performance and safety of the bike fleet.

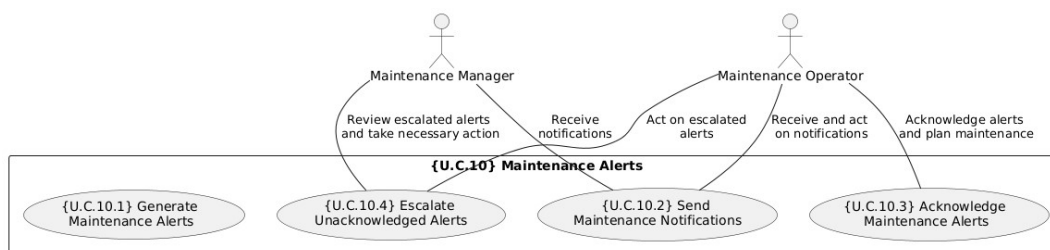


Figure 44 – Use case diagram for {U.C.10} Maintenance Alerts

{U.C.10.1} Generate Maintenance Alerts: The system shall generate alerts for bikes that are identified as due for maintenance and log these alerts in the database. These alerts are based on the maintenance schedules and are generated for bikes that have been flagged in the system as needing maintenance. In the event of an alert generation error, the system must log the error and retry the alert generation.

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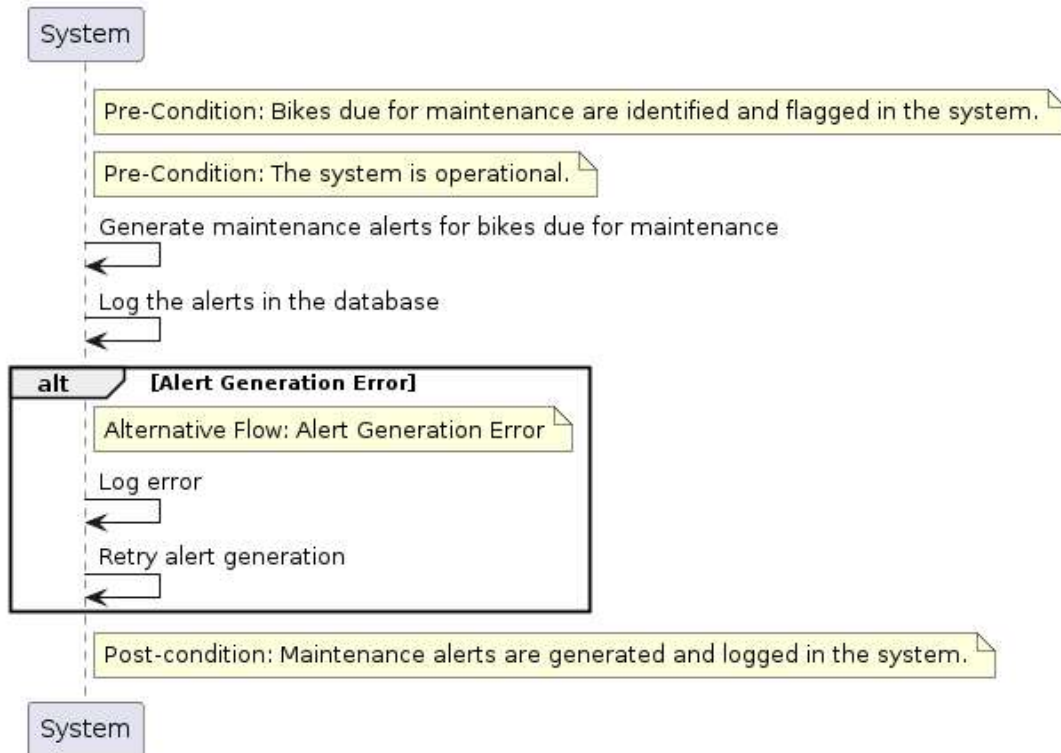


Figure 45 - Sequence diagram for use case {U.C.10.1} Generate Maintenance Alerts

{U.C.10.2} Send Maintenance Notifications: The system shall generate and send notifications to the Maintenance Manager and Maintenance Operator about bikes that are due for maintenance. This process is based on previously generated maintenance alerts. The Maintenance Manager and Maintenance Operator shall receive and review the notifications to take appropriate action. In the event of a notification failure, the system must log the error and retry sending the notification.

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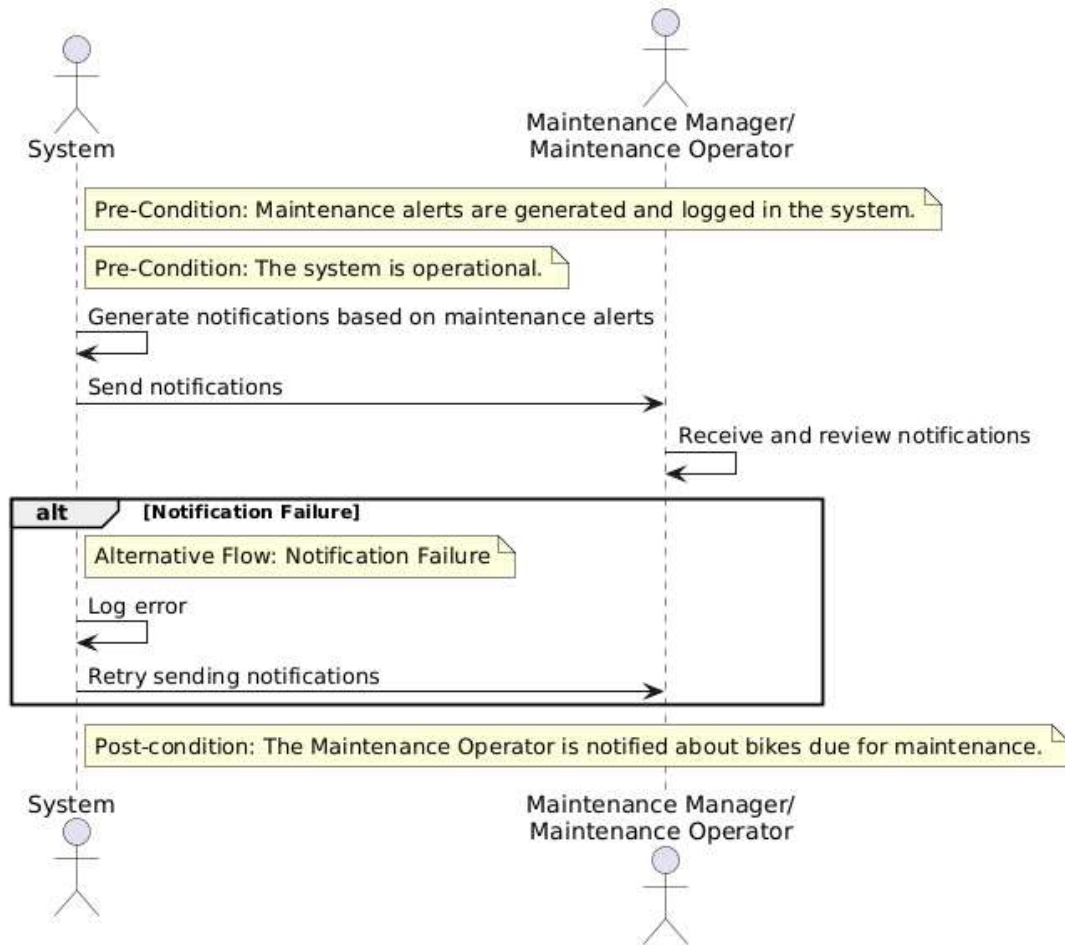


Figure 46 - Sequence diagram for use case {U.C.10.2} Send Maintenance Notifications

{U.C.10.3} Acknowledge Maintenance Alerts: The system shall support the process by which the maintenance operator acknowledges the receipt of maintenance alerts and plans for the necessary checks. The maintenance operator will review the notifications, acknowledge the alerts in the system for the system to log these acknowledgments and update the alert status, and plan the maintenance activities. In cases where the acknowledgment fails, the system must log the error and retry the acknowledgment process.

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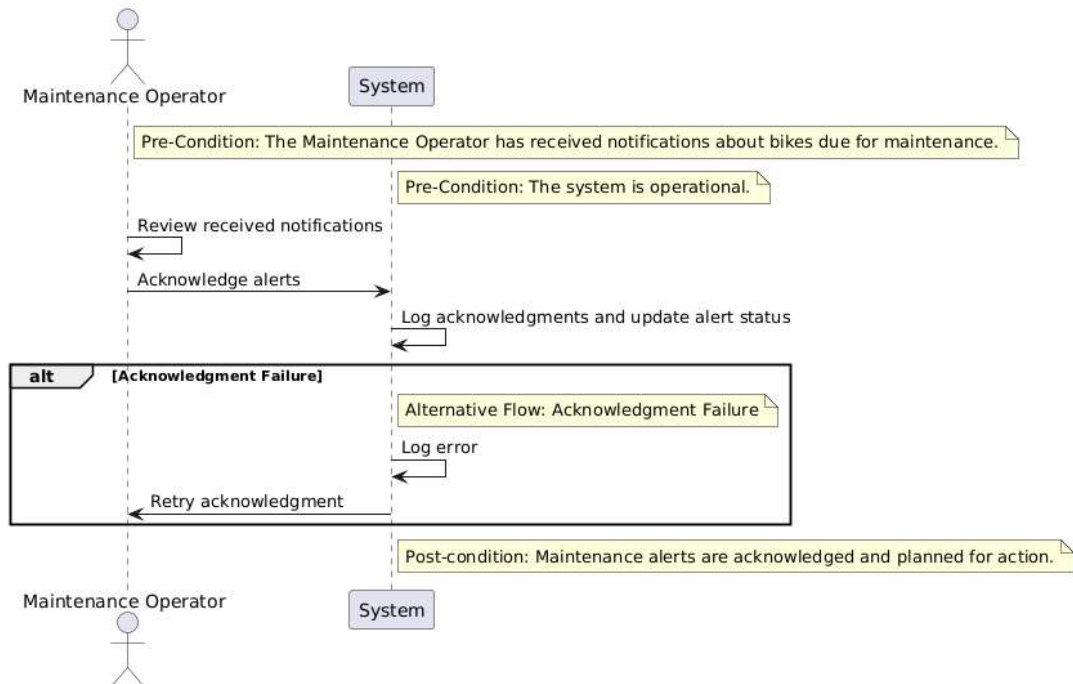


Figure 47 - Sequence diagram for use case {U.C.10.3} Acknowledge Maintenance Alerts

{U.C.10.4} Escalate Unacknowledged Alerts: The system shall monitor the acknowledgment status of maintenance alerts and escalate those that have not been acknowledged within a specified timeframe. This process involves notifying the Maintenance Manager of the escalated alerts to ensure prompt action. The Maintenance Manager then reviews the escalated alert and takes necessary action. In cases where there is an error in the escalation process, the system must log the error and retry the escalation.

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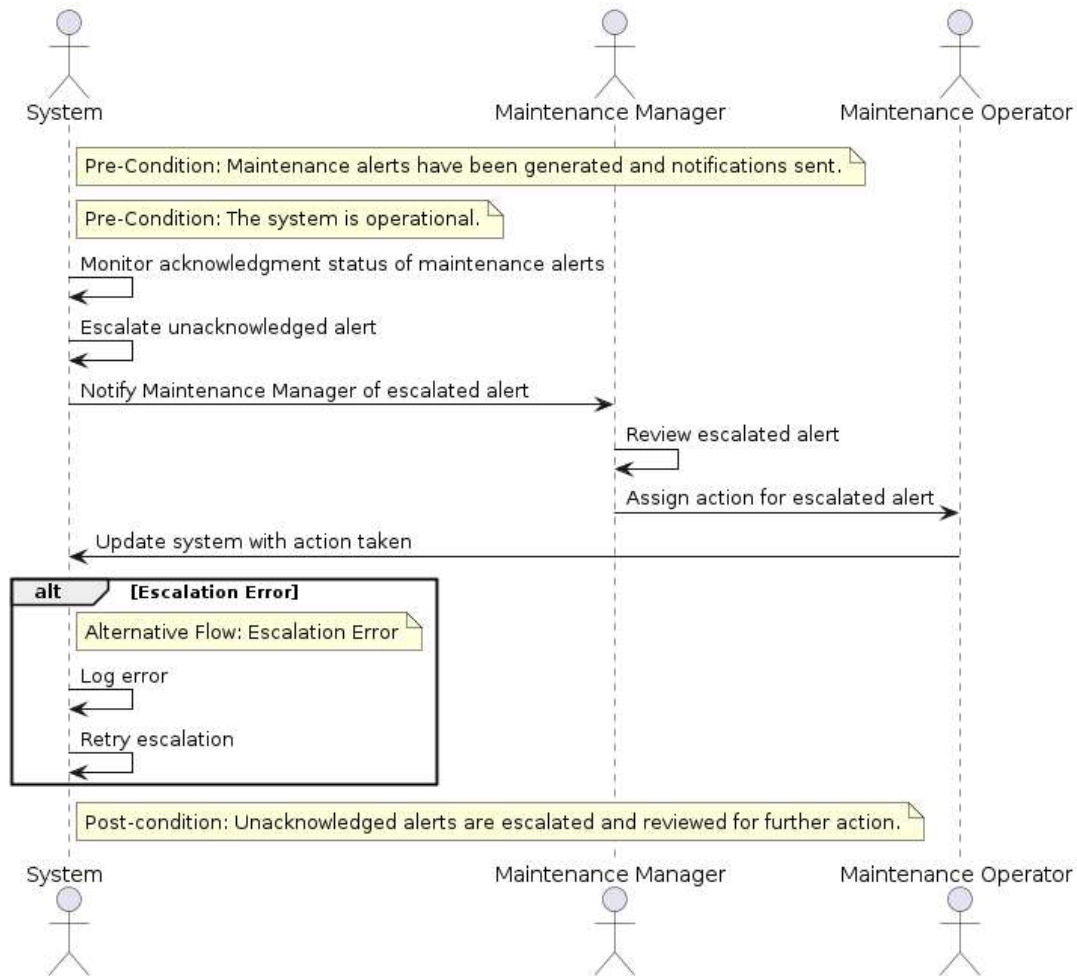


Figure 48 - Sequence diagram for use case {U.C.10.4} Escalate Unacknowledged Alerts

{U.C.11} Maintenance History: The system shall keep a detailed record of all past maintenance activities for each bike. These records shall include information on the type of maintenance performed, the date, the parts used, and the maintenance operator involved. Maintaining comprehensive maintenance records helps evaluate the effectiveness of previous repairs, and plan future maintenance activities. By leveraging this historical maintenance data, the system ensures that maintenance management is proactive, data-driven, and capable of addressing both current and potential issues with the bike fleet.

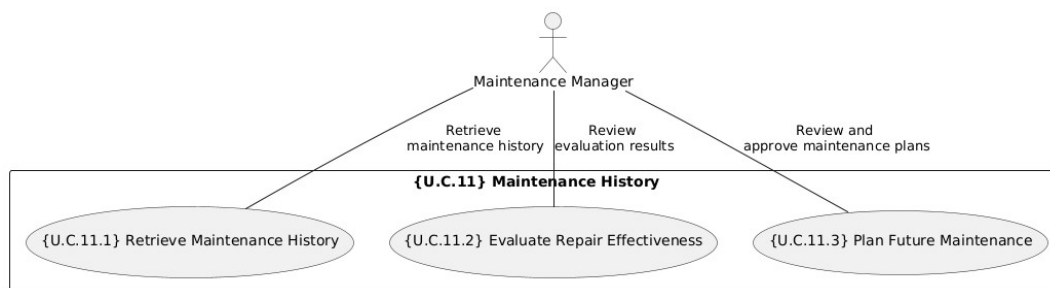


Figure 49 - Use case diagram for {U.C.11} Maintenance History

{U.C.11.1} Retrieve Maintenance History: The system shall allow the Maintenance Manager to retrieve the maintenance history of a specific bike. This involves accessing stored maintenance records and presenting them to the Maintenance Manager.

In the event of a data retrieval error, the system must log the error and retry retrieving the data.

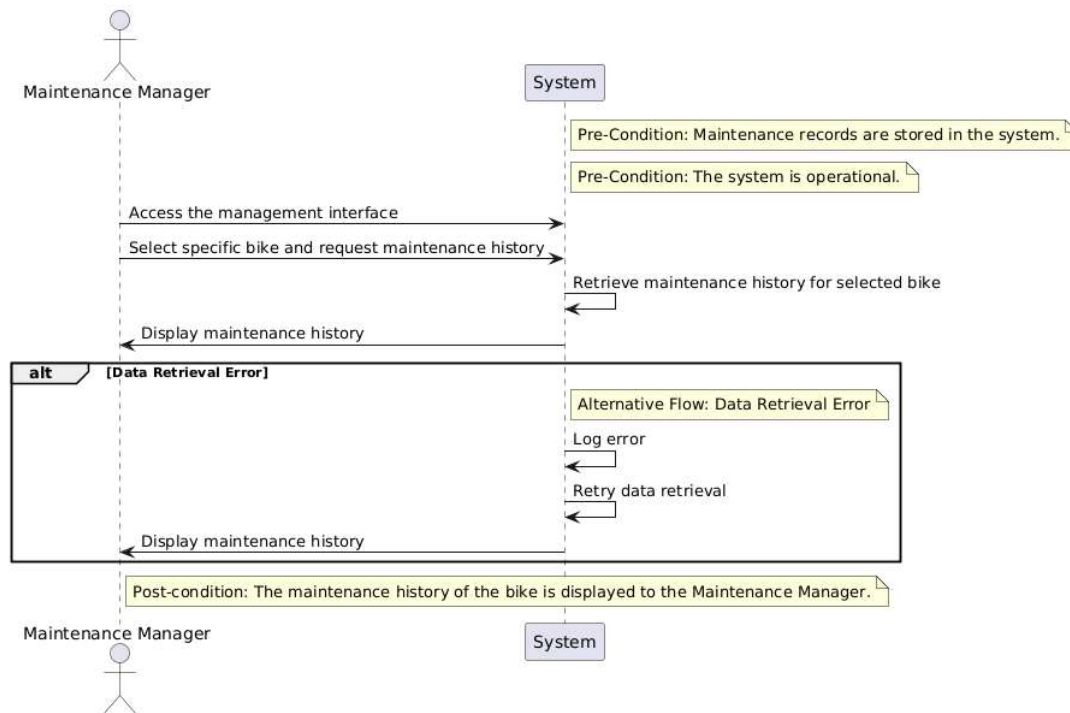


Figure 50 - Sequence diagram for use case {U.C.11.1} Retrieve Maintenance History

{U.C.11.2} Evaluate Repair Effectiveness: The system shall evaluate the effectiveness of previous repairs based on subsequent maintenance records. This involves the system analyzing stored maintenance records to determine how well past repairs have addressed issues. The system then generates a report detailing the findings on repair effectiveness. The Maintenance Manager reviews this report and takes any necessary action based on the evaluation results.

In the event of an evaluation error, the system must log the error and retry the evaluation process.

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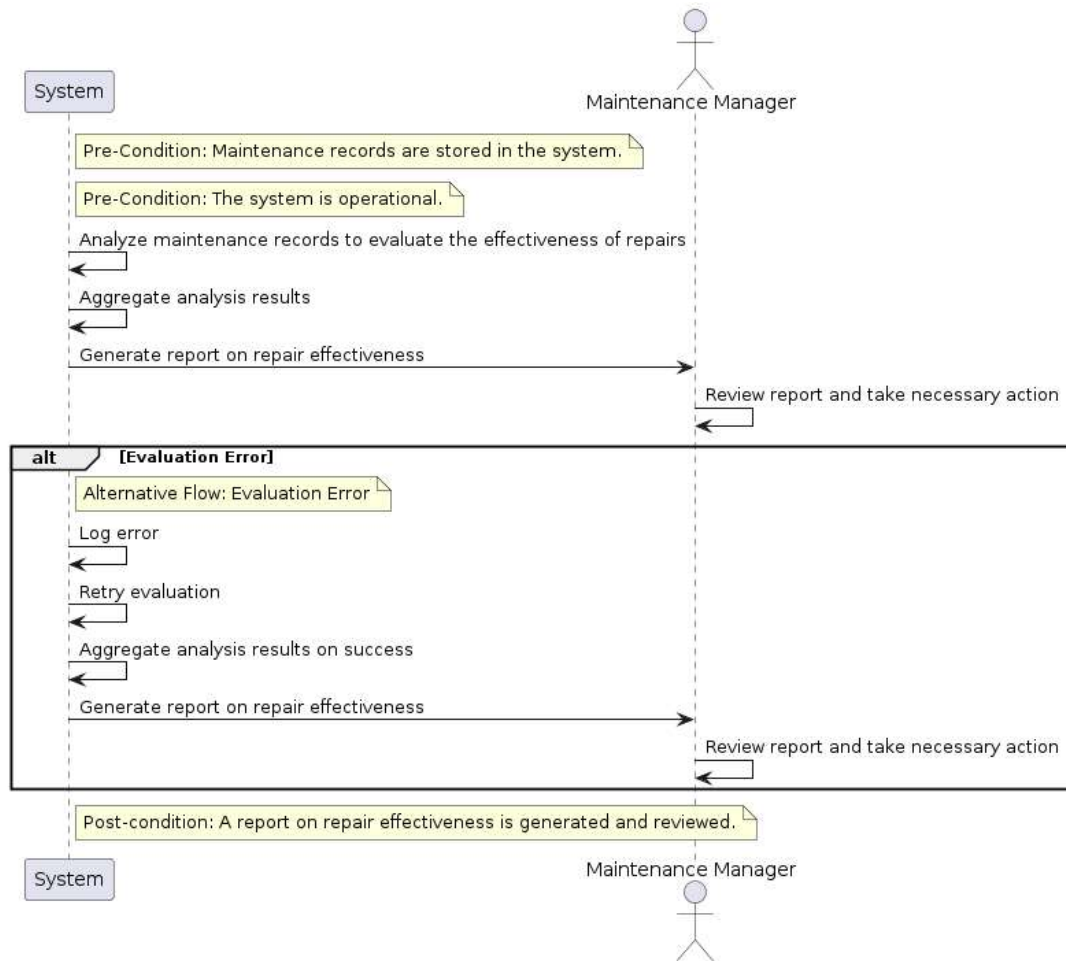


Figure 51 - Sequence diagram for use case {U.C.11.2} Evaluate Repair Effectiveness

{U.C.11.3} Plan Future Maintenance: The system shall analyze the maintenance records and reports to develop a plan for future maintenance activities. This involves analyzing available data to generate a maintenance plan, which the Maintenance Manager reviews and approves.

In the event of a planning error, the system must log the error and retry the planning process.

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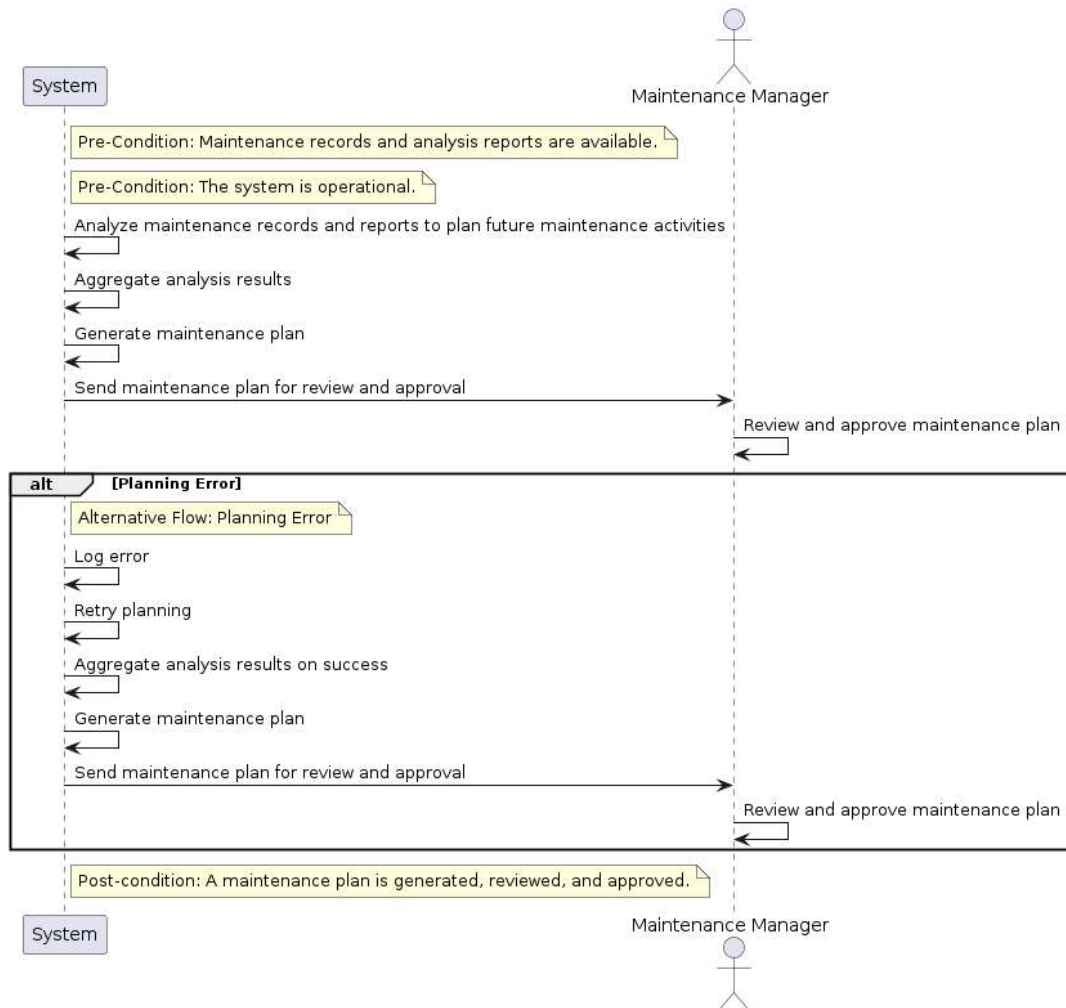


Figure 52 - Sequence diagram for use case {U.C.11.3} Plan Future Maintenance

{U.C.12} Issue Tracking: The system shall track and categorize issues identified during maintenance checks. It should analyze these issues to flag bikes that are due for maintenance or repair. The system shall then prioritize these repairs based on severity and frequency, helping to manage maintenance tasks efficiently. Additionally, the system shall identify common problems across the fleet by analyzing the categorized issues, enabling the Maintenance Manager to address widespread issues effectively. These functionalities ensure that maintenance activities are well-organized, prioritized, and targeted towards both individual and fleet-wide problems.

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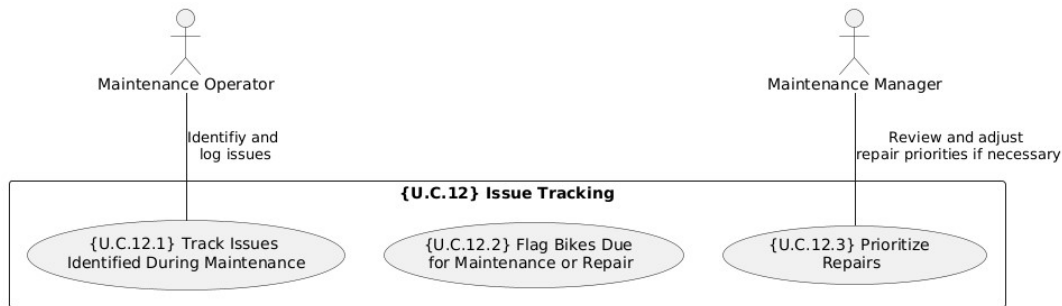


Figure 53 – Use case diagram for {U.C.12} Issue Tracking

{U.C.12.1} Track Issues Identified During Maintenance: The system shall track issues identified during the performing of maintenance checks on bikes and categorize these issues. This involves the Maintenance Operator logging identified issues into the system, which then analyzes the logged issues to assign them to predefined categories such as mechanical, electrical, or wear and tear, and stores it in the database.

In the event of a logging error, the system must log the error and prompt the Maintenance Operator to retry logging the issue.

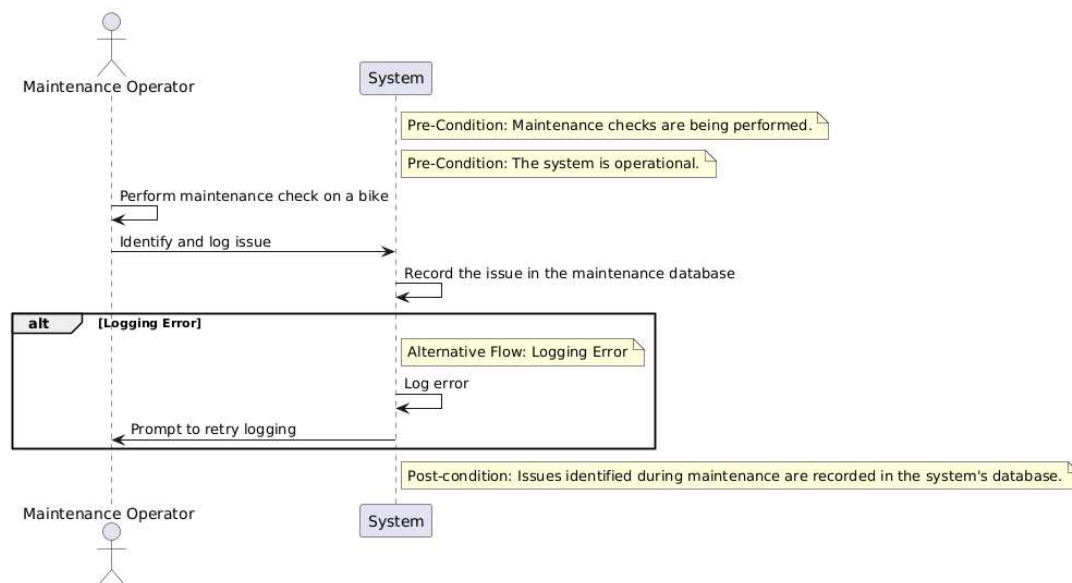


Figure 54 - Sequence diagram for use case {U.C.12.1} Track Issues Identified During Maintenance

{U.C.12.2} Flag Bikes Due for Maintenance or Repair: The system shall flag bikes that are due for maintenance or repair based on identified and categorized issues and update the status of these flagged bikes in the database. This involves reviewing logged issues and maintenance criteria to determine which bikes need attention.

In the event of a flagging error, the system must log the error and retry the flagging process.

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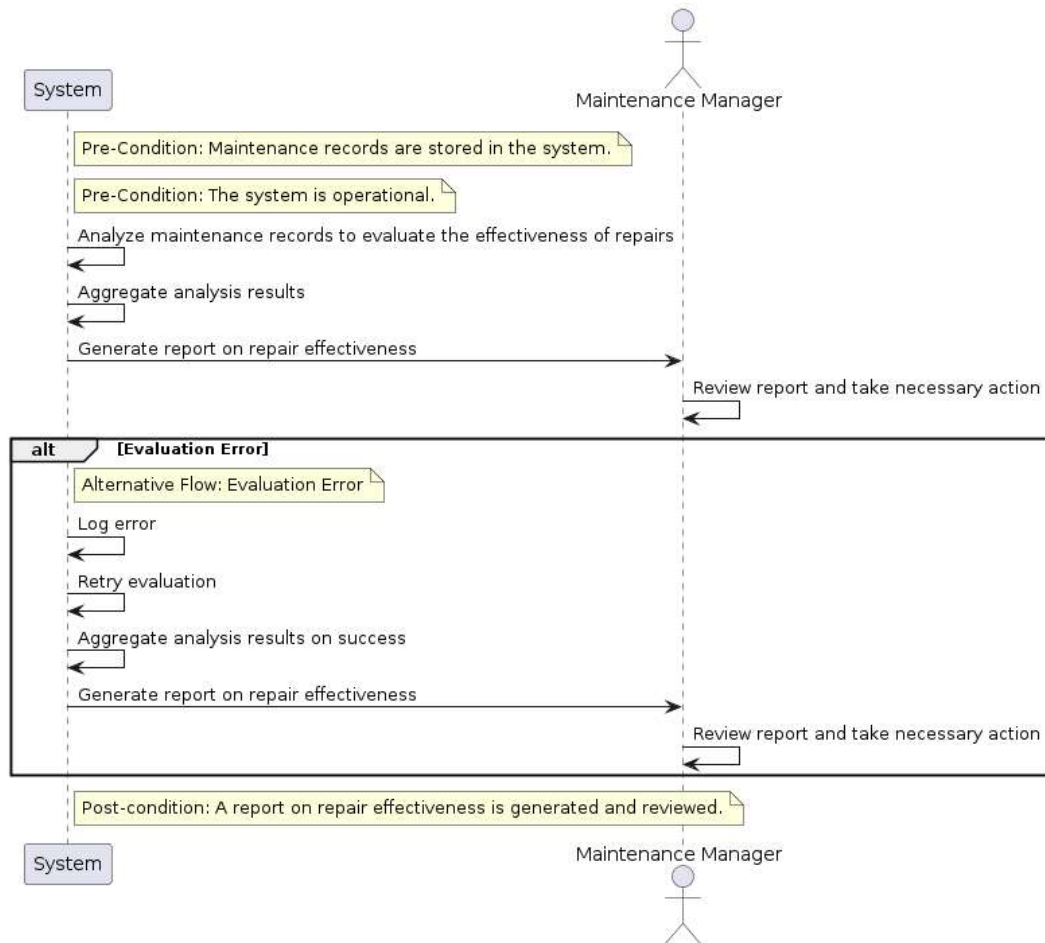


Figure 55 - Sequence diagram for use case {U.C.12.2} Flag Bikes Due for Maintenance or Repair

{U.C.12.3} Prioritize Repairs: The system shall prioritize repairs based on the severity and frequency of identified issues. This involves analyzing the flagged bikes and their issues to determine the order of repairs. The Maintenance Manager can review and adjust these priorities if necessary. The system updates the priority list in the database.

In the event of a prioritization error, the system must log the error and retry the prioritization process.

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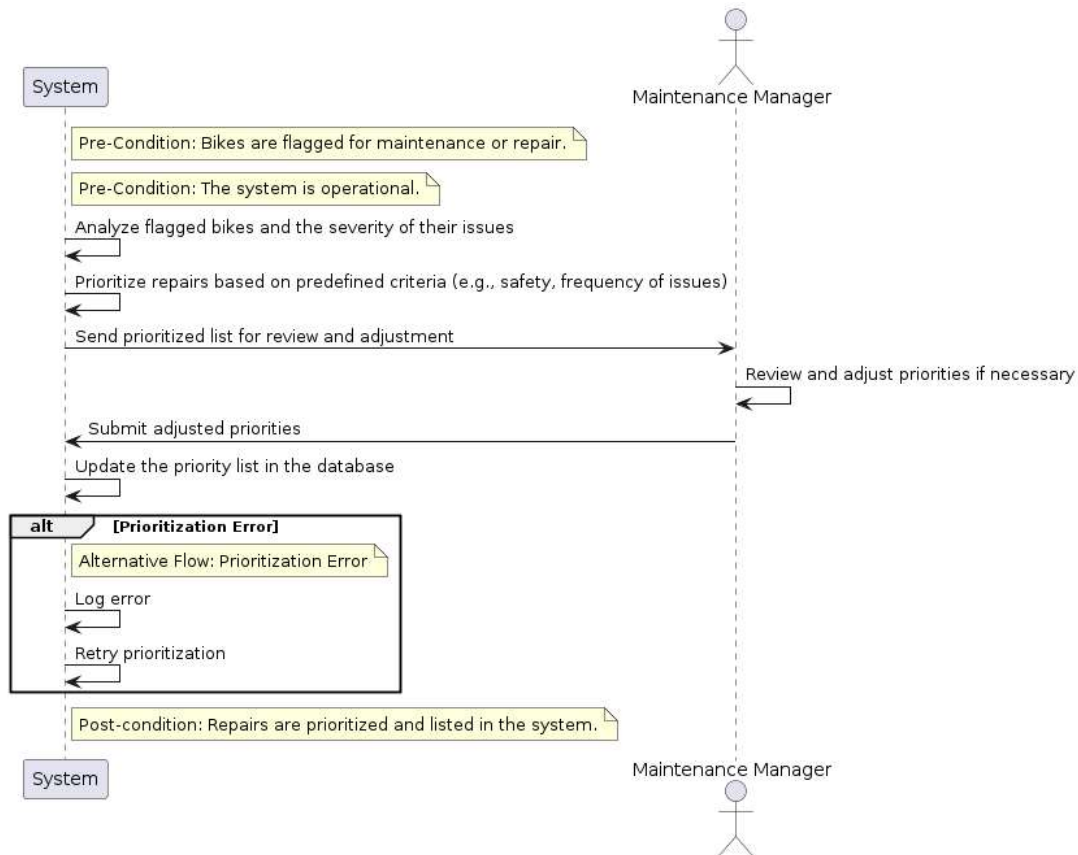


Figure 56 - Sequence diagram for use case {U.C.12.3} Prioritize Repairs

{U.C.13} Parts Inventory Management: The system shall manage an inventory of spare parts and other materials required for bike maintenance. This includes tracking the usage of these parts during maintenance activities. The system shall also monitor the inventory levels and trigger alerts when stock levels fall below a predefined threshold, ensuring that necessary materials are always available for maintenance tasks. This functionality helps maintain an adequate supply of spare parts, preventing delays in maintenance due to stock shortages and ensuring the efficient operation of the maintenance process.

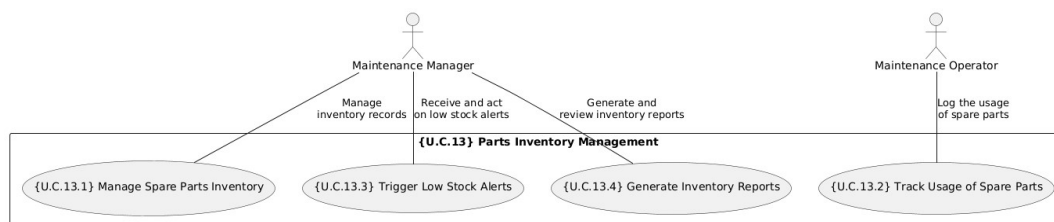


Figure 57 - Use case diagram for {U.C.13} Parts Inventory Management

{U.C.13.1} Manage Spare Parts Inventory: The system shall manage an inventory of spare parts and other materials needed for bike maintenance. This process involves the Maintenance Manager or Maintenance Operator accessing the inventory management interface to add, update, or remove spare parts and materials. The system processes the updated inventory data and stores the updates in the inventory database.

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In cases of data entry errors, the system must prompt the Maintenance Manager or Maintenance Operator to correct the errors.

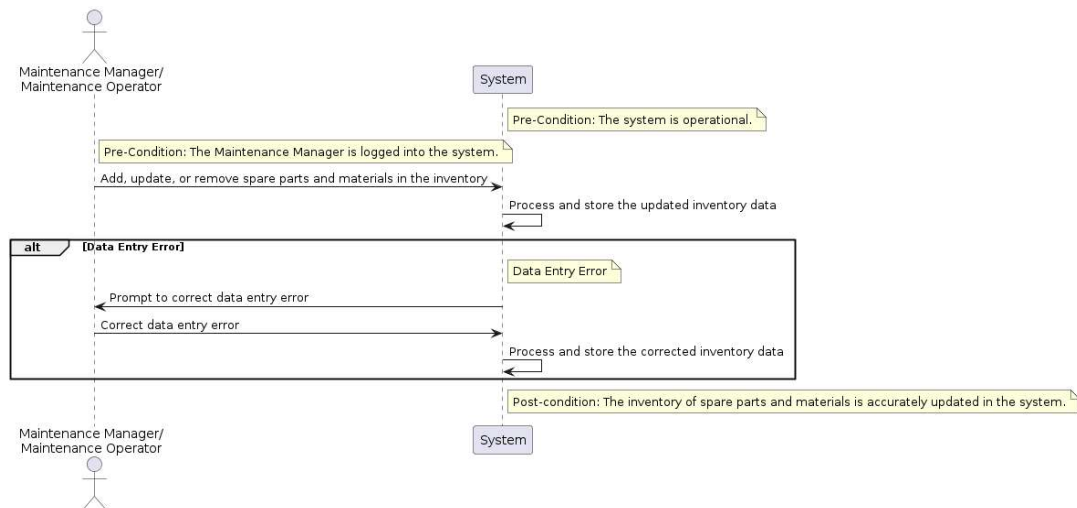


Figure 58 - Sequence diagram for use case {U.C.13.1} Manage Spare Parts Inventory

{U.C.13.2} Track Usage of Spare Parts: The system shall track the usage of spare parts during maintenance activities. This involves the Maintenance Operator logging the usage of spare parts during maintenance activities, which the system then processes and stores the updates of the count of the used spare parts in the inventory database.

In cases of logging errors, the system must log the error and prompt the team to retry logging the usage.

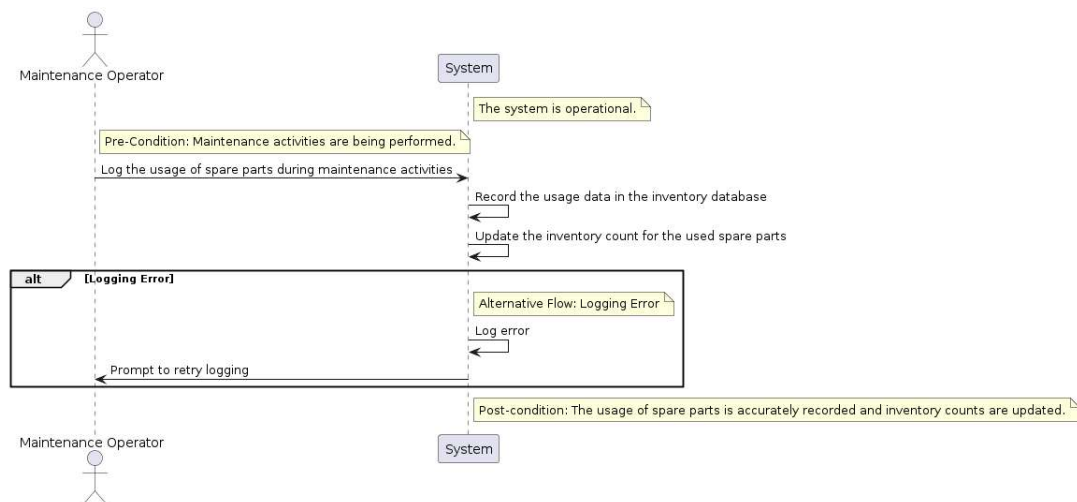


Figure 59 - Sequence diagram for use case {U.C.13.2} Track Usage of Spare Parts

{U.C.13.3} Trigger Low Stock Alerts: The system shall trigger alerts when the stock of spare parts is low to ensure timely replenishment. This involves the system continuously monitoring inventory levels of spare parts and triggering low stock alerts when stock falls below a predefined threshold. The Maintenance Manager receives these alerts and takes necessary action.

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In cases where there is an alert failure, the system must log the error and retry triggering the alert.

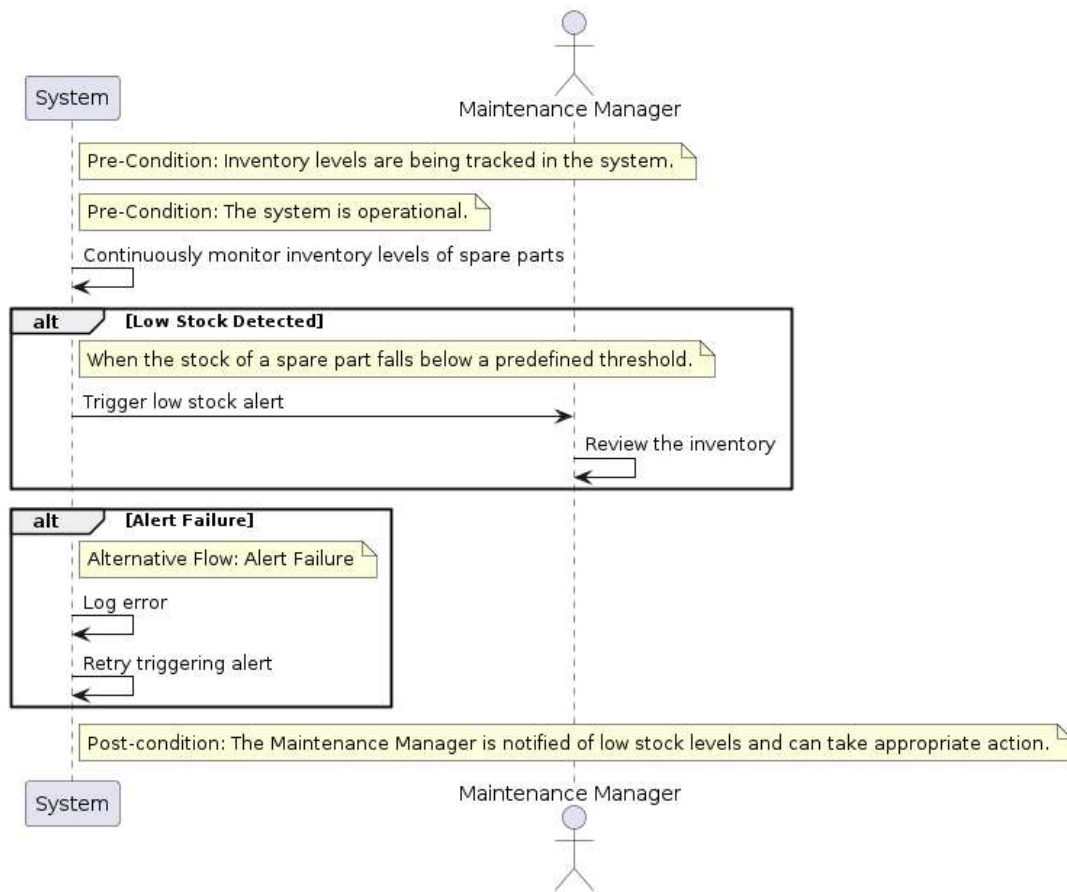


Figure 60 - Sequence diagram for use case {U.C.13.3} Trigger Low Stock Alerts

{U.C.13.4} Generate Inventory Reports: The system shall generate reports on inventory levels and usage of spare parts to assist in effective inventory management. This process involves the Maintenance Manager selecting the type of inventory report to generate and the system compiling the necessary data and generating the report. The Maintenance Manager reviews the generated report and takes any necessary actions based on the insights provided. In cases of report generation errors, the system shall log the error and retry the report generation process.

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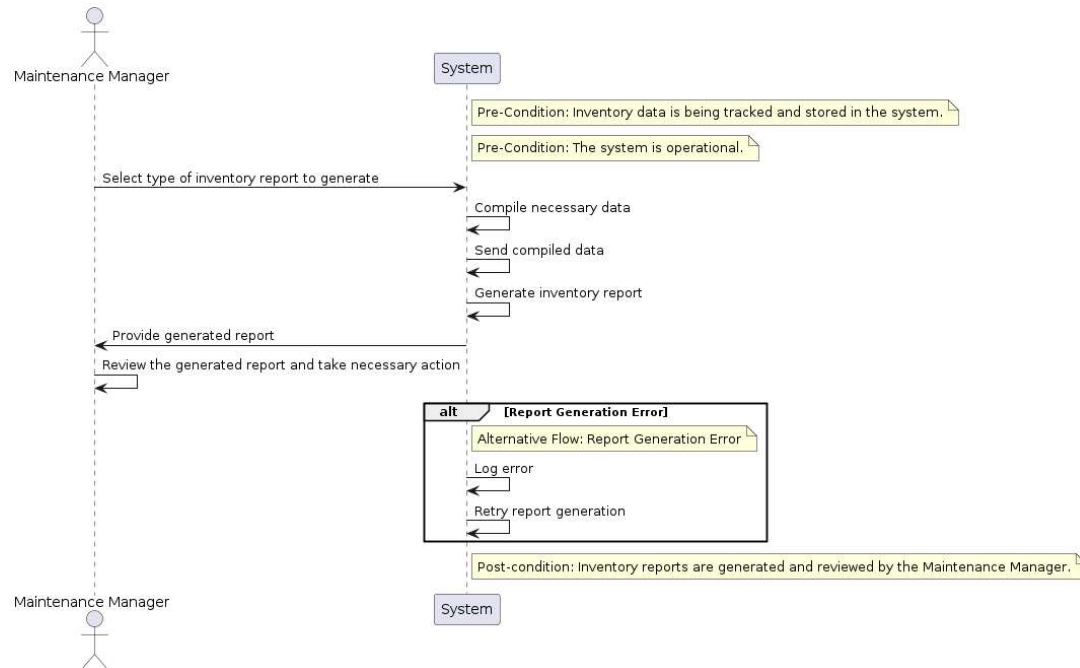


Figure 61 - Sequence diagram for use case {U.C.13.4} Generate Inventory Reports

{U.C.14} Maintenance Reporting: The system shall generate detailed reports on maintenance activities to provide valuable insights for improving the maintenance process and the overall operation of the bike-sharing service. These reports should include the common issues identified, parts used, and other relevant data. The system shall track and store all maintenance activity data accurately to compile these reports. By analyzing the generated reports, the maintenance manager can identify trends, address recurring problems, optimize resource use, and enhance maintenance strategies. This functionality is essential for ensuring efficient and effective maintenance operations, ultimately contributing to the smooth operation of the bike-sharing service.

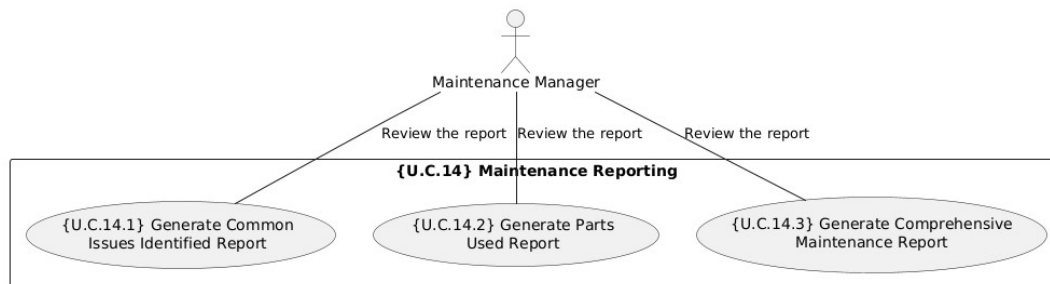


Figure 62 - Use case diagram for {U.C.14} Maintenance Reporting

{U.C.14.1} Generate Common Issues Identified Report: The system shall generate a report detailing common issues identified during maintenance activities. This process involves the Maintenance Manager selecting a reporting period, after which the system retrieves and analyzes maintenance activity data for that period to identify common issues. The system then compiles this data and generates the report, which is presented to the Maintenance Manager for review. In case of an error in data analysis, the system shall log the error and retry the process.

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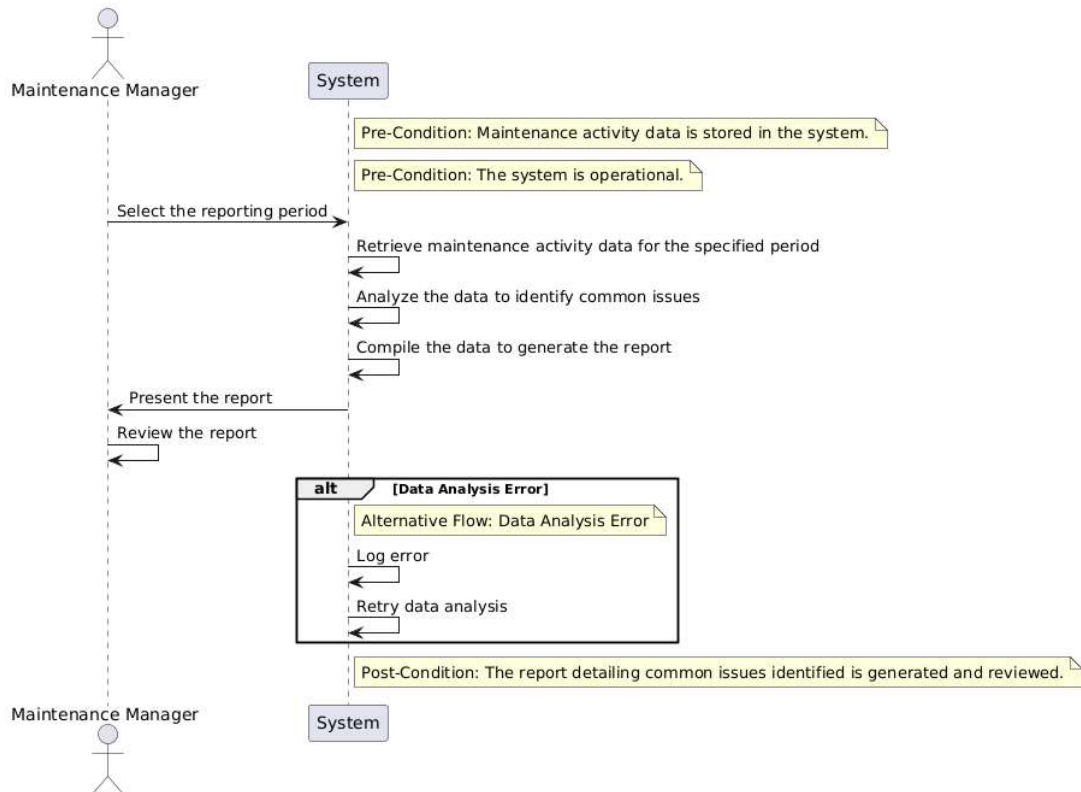


Figure 63 - Sequence diagram for use case {U.C.14.1} Generate Common Issues Identified Report

{U.C.14.2} Generate Parts Used Report: The system shall generate a report detailing the spare parts used during maintenance activities. This process involves the Maintenance Manager selecting a reporting period, after which the system retrieves and analyzes maintenance activity data for that period to identify the parts used. The system then compiles this data and generates the report, which is presented to the Maintenance Manager for review. In case of an error in data analysis, the system shall log the error and retry the process.

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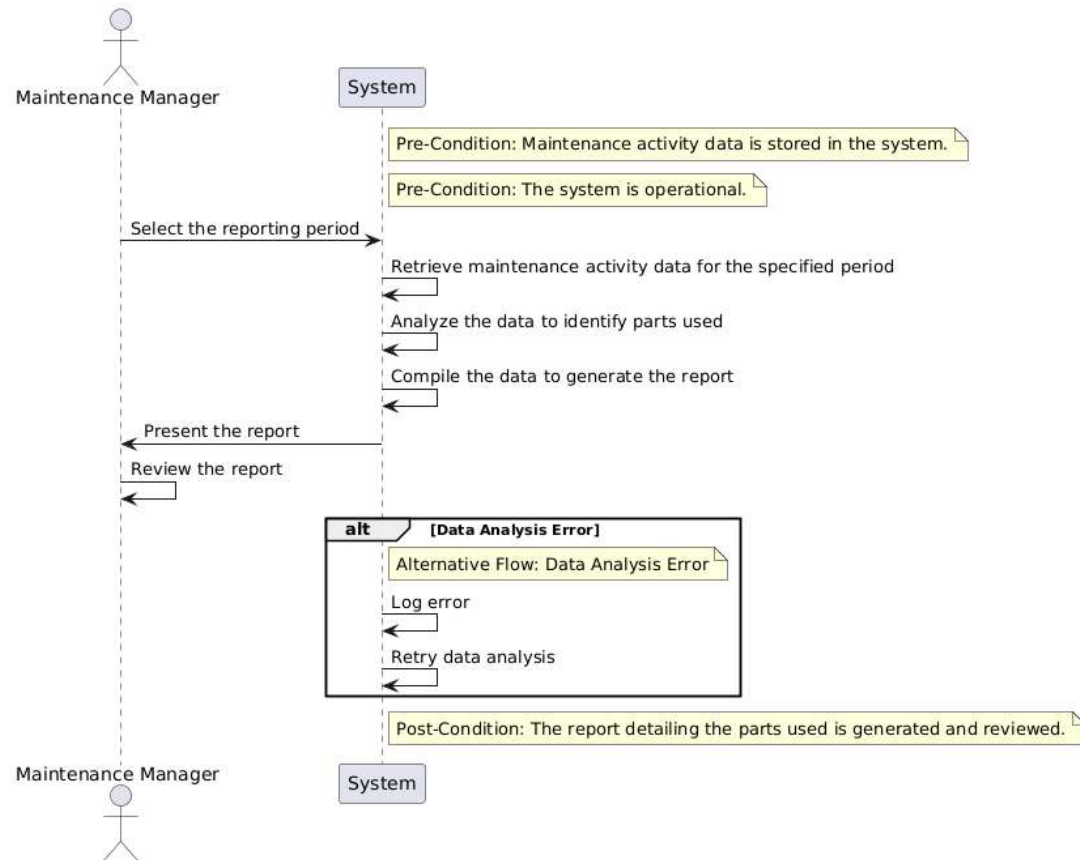


Figure 64 - Sequence diagram for use case {U.C.14.2} Generate Parts Used Report

{U.C.14.3} Generate Comprehensive Maintenance Report: The system shall generate a comprehensive report detailing maintenance activities, including the number of bikes serviced, common issues identified, and parts used. This involves the Maintenance Manager selecting a reporting period, after which the system retrieves maintenance activity data for that period, compiles the data, and generates the comprehensive report. The report is then presented to the Maintenance Manager for review.

In case of an error in report generation, the system shall log the error and retry the process.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

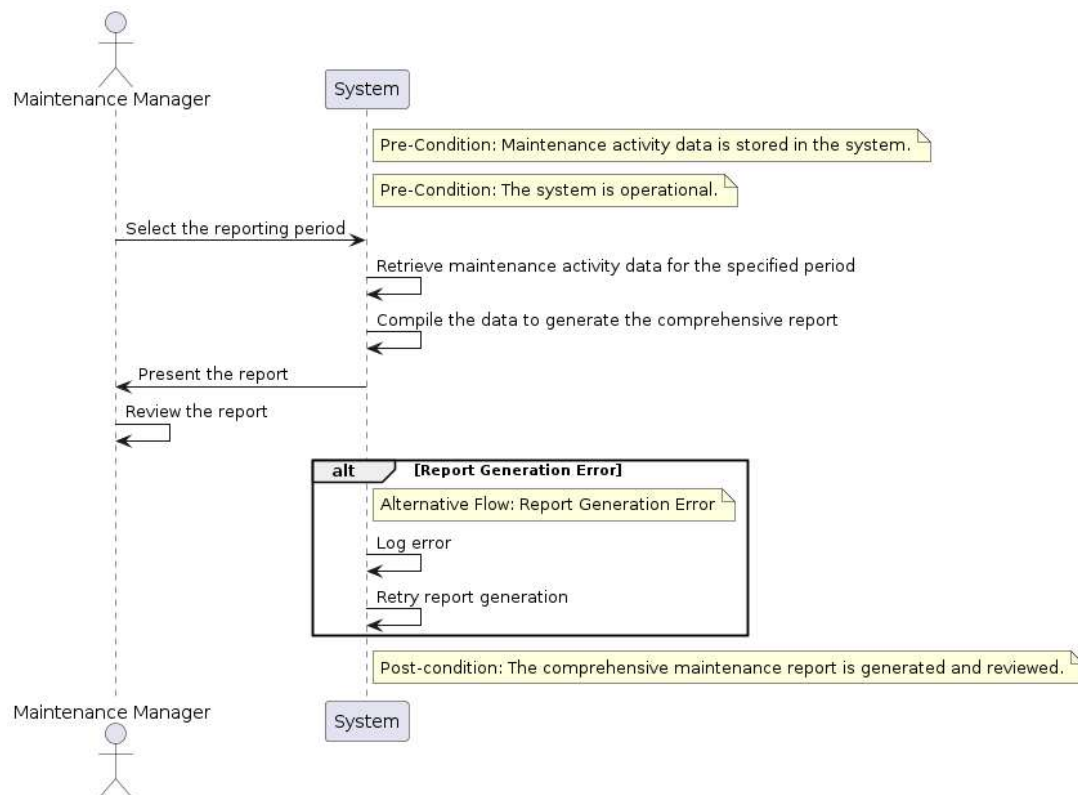


Figure 65 - Sequence diagram for use case {U.C.14.3} Generate Comprehensive Maintenance Report

5 Mapping the Domain Application Use Cases to The Cloud Computing Framework

5.1 Platform Actors Description

Table 5 – Platform Actors Description

Actor NIST	Actor Bike4ALL-M	Definition
Public PaaS Cloud Provider	Public Cloud Vendor	Represents the Public Cloud Platform organization, i.e., the entity responsible for making the service available to cloud consumers. It creates platform services (PaaS), manages the technical infrastructure to provide the services, provisions the services through SLAs (service level agreements), and protects the security and privacy of the services.
Public Cloud PaaS consumer	Platform Administrator	Represents the Administrator of the Public Cloud PaaS service account of the Be.Neutral-bike4all-maintenance organization, which uses and maintains a contractual relationship with the cloud provider. Browse the cloud provider's service catalog, request services, configure contracts, and use the services. It is billed for the service provisioned, which may include the number of users, the type of resources consumed by the platform, or the duration of use of the platform.
	Application Administrator	Represents the administrator of the Be.Neutral-bike4all-maintenance organization who configures and monitors the performance of the applications hosted on the cloud platform.
	Application Developer	Represents the Be.Neutral-bike4all-maintenance organization person who designs and implements software applications.
	Application Tester	Represents the person in the Be.Neutral-bike4all-maintenance organization who tests the applications in the cloud platform.
	Application Deployer	Represents the person in the Be.Neutral-bike4all-maintenance organization who publishes the applications in the cloud platform.
Cloud SaaS consumer	Maintenance Manager	It represents the individual who manages and supervises the maintenance service, from monitoring alerts to issuing work orders. It also has access to historical data, statistics, and real-time information, registers and removes bikes and parts, sets maintenance parameters, and registers the maintenance team.
	Maintenance Operator	It represents the people who perform the necessary bike maintenance work and record all the relevant information for each service. They also have access to the history and reports of their work and must update the bikes' status after maintenance.
	Bike4All Platform	It represents the Bicycle Mobility Management Platform, where users can hire bicycles. It allows its users to record incidents reported to the Maintenance Platform.
	Mobility Devices Management Platform	It represents the service provider that connects the mobility devices of each bicycle in the fleet to the Bike4All and Maintenance Platform and provides data on the respective bicycles (location and status).

5.2 Mapping of NIST layers with domain application use cases

Table 6 - Mapping Use Cases with NIST Layers

NIST Layers	Use Cases
1.Cloud Consumers	not applied
2.Cloud Auditor	not applied
3.Cloud Broker	not applied
3.1. Service Intermediation	not applied
3.2. Service Aggregation	not applied
3.3. Service Arbitrage	not applied
4. Cloud Provider	
4.1. Service Layer	
4.1.1. SaaS	{U.C.1.1} Track Bike Location in Real-Time {U.C.1.2} Identify Idle Bikes {U.C.2.1} Track Bike Details {U.C.2.2} Track Bike Status {U.C.2.3} Update Bike Inventory {U.C.3.1} Collect Historical Data {U.C.3.2} Analyze Demand Patterns {U.C.3.3} Predict Bike Demand {U.C.3.4} Optimize Bike Allocation {U.C.3.5} Monitor and Adjust Predictions {U.C.4.1} Collect Usage Data {U.C.4.2} Analyze Peak Usage Times {U.C.4.3} Analyze Popular Routes {U.C.4.4} Analyze Average Trip Duration {U.C.4.5} Generate Usage Insights Reports {U.C.5.1} Receive Issue Report {U.C.5.2} Process and Assign Issue {U.C.5.3} Track and Update Issue Resolution {U.C.5.4} Send Resolution Feedback {U.C.6.1} Activate Remote Lock {U.C.6.2} Trigger Alarm System {U.C.6.3} Monitor Bike Status {U.C.6.4} Respond to Security Alerts {U.C.7.1} View Fleet Overview {U.C.7.2} Assign Maintenance Tasks {U.C.7.3} Monitor Bike Status {U.C.7.4} Generate Reports {U.C.8.1} Integrate with Bike4All Platform {U.C.8.2} Integrate with Mobility Devices Management Platform {U.C.9.1} Define Maintenance Criteria {U.C.9.2} Schedule Maintenance Check {U.C.9.3} Notify Maintenance Team {U.C.9.4} Perform Maintenance Check {U.C.9.5} Update Maintenance Records {U.C.10.1} Generate Maintenance Alerts {U.C.10.2} Send Maintenance Notifications {U.C.10.3} Acknowledge Maintenance Alerts {U.C.10.4} Escalate Unacknowledged Alerts {U.C.11.1} Retrieve Maintenance History {U.C.11.2} Evaluate Repair Effectiveness {U.C.11.3} Plan Future Maintenance {U.C.12.1} Track Issues Identified During Maintenance

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NIST Layers	Use Cases
	{U.C.12.2} Flag Bikes Due for Maintenance or Repair {U.C.12.3} Prioritize Repairs {U.C.13.1} Manage Spare Parts Inventory {U.C.13.2} Track Usage of Spare Parts {U.C.13.3} Trigger Low Stock Alerts {U.C.13.4} Generate Inventory Reports {U.C.14.1} Generate Common Issues Identified Report {U.C.14.2} Generate Parts Used Report {U.C.14.3} Generate Comprehensive Maintenance Report
4.1.2. PaaS	Responsibility of Cloud Infrastructure
4.1.3. IaaS	Responsibility of Cloud Infrastructure
4.2. Resource Abstraction and Control Layer	Responsibility of Cloud Infrastructure
4.3. Physical Resource Layer	Responsibility of Cloud Infrastructure
4.3.1. Hardware	Responsibility of Cloud Infrastructure
4.3.2. Facility	Responsibility of Cloud Infrastructure
4.4. Cloud Service Management	
4.4.1. Business Support	
4.4.1.1. Customer Management	{U.C.15.1} Account management
4.4.1.2. Contract Management	{U.C.15.2} Contract Management
4.4.1.3. Inventory Management	{U.C.15.3} Service Catalog Management
4.4.1.4. Accounting & Billing	{U.C.15.4} Invoice Management
4.4.1.5. Reporting & Auditing	{U.C.16.2} Platform Monitoring
4.4.1.6. Pricing & Rating	{U.C.15.3} Service Catalogue Management
4.4.2. Provisioning / Configuration	
4.4.2.1. Rapid Provisioning	{U.C.16.1} Application Management
4.4.2.2. Resource Change	{U.C.16.1} Application Management
4.4.2.3. Monitoring & Reporting	{U.C.16.2} Platform Monitoring
4.4.2.4. Metering	{U.C.16.3} Resource Utilization Measurement
4.4.2.5. SLA Management	{U.C.16.4} SLA Management
4.4.3. Portability / Interoperability	
4.4.3.1. Data Portability	
4.4.3.1.1. Copy Data To-From	{U.C.17.1} Data synchronization
4.4.3.1.2. Bulk Data Transfer	{U.C.17.1} Data synchronization
4.4.3.2. Service Interoperability	
4.4.3.2.1. Unified Mgmt Interface	{U.C.18.2} Systems integration
4.4.3.3. System Portability	
4.4.3.3.1. VM Images Migration	Responsibility of Cloud Infrastructure
4.4.3.3.2. App/Sync Migration	Responsibility of Cloud Infrastructure
4.4.3.3. Security	{U.C.18.1} Backup management {U.C.18.2} Configuration of data access control

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

NIST Layers	Use Cases
4.4.3.4. Privacy	{U.C.15.1} Account management {U.C.18.2} Configuration of data access control

5.3 Platform Use Case Diagram

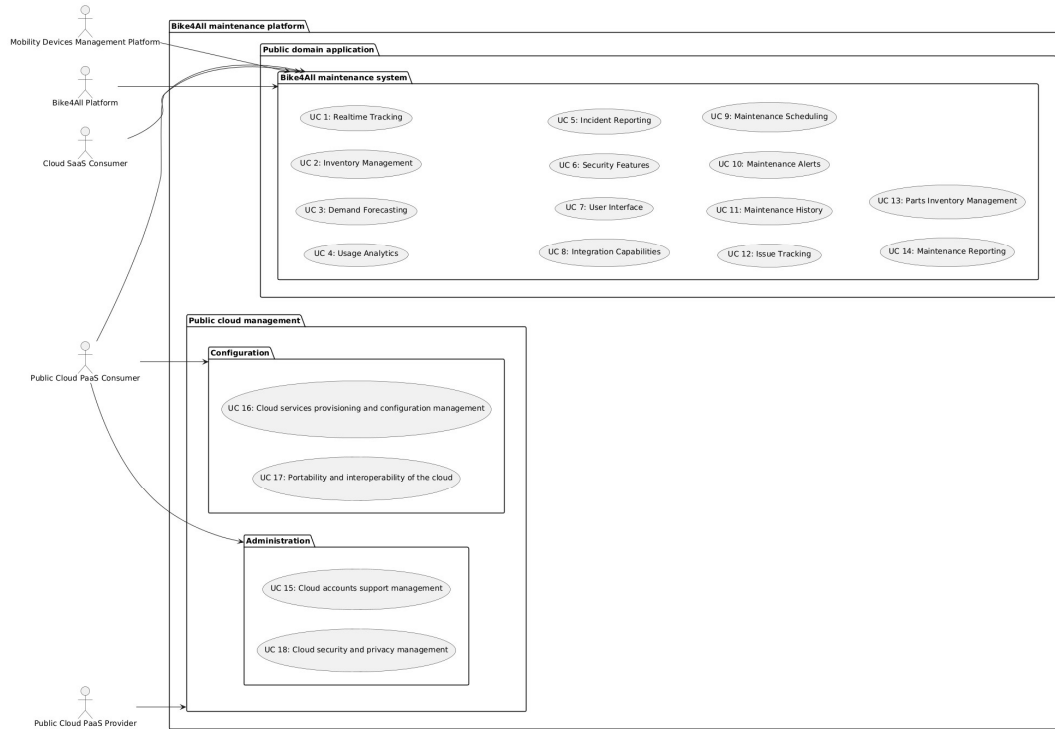


Figure 66 - Bike4All-Maintenance Platform

5.4 Cloud management use cases description

{U.C.15} Cloud account support management: involves the set of services supporting the contract between cloud provider and cloud consumer in the public cloud. It includes the components used to perform these business operations, such as account management, contract management, catalogue management, billing management. These use cases are detailed in Account management, Contract management, Service catalogue management and invoicing management refined use cases.

{U.C.15.1} Account management: The platform administrator shall manage accounts, activate/inactivate/terminate platform user accounts and their profiles. It gives the application developer, tester and deployer access to use the integrated development tools (IDE), software development kits (SDK), and software management and installation tools. Manages the business relationship with the

provider, and contacts to resolve technical support issues or other problems (billing, provisioning of services).

{U.C.15.2} Contract Management: The platform administrator manages contracts, initiates, negotiates and terminates contracts with the provider.

{U.C.15.3} Service Catalogue Management: The platform administrator installs and manages the catalog services, which include the cloud storage environment, compute operations, and all service-related functions that are necessary for the management and operation of the services offered to end customers. The platform administrator has control over the applications and some parameters of the programming environment, but limited or no access to the infrastructure that supports the platform, such as network, servers, operating systems (OS), or data storage.

{U.C.15.4} Invoice Management: The administrator evaluates the cloud services, and determines prices based on the resource usage profiles of the end users of the applications. It also allows you to compare resource prices with other cloud providers.

{U.C.16} Cloud services provisioning and configuration management: includes the set of services for provisioning, monitoring and reporting on cloud services, measuring resource usage and service level management, such as QoS parameters (SLA). These use cases are detailed in Application management, Platform monitoring, Resource utilization measurement and SLA management refined use cases.

{U.C.16.1} Application Management: The platform administrator, application administrator, programmer, tester and deployer of the applications use the integrated development tools (IDE), software development kits (SDK), and software management and installation tools to develop, configure and install the software applications and cloud resources. They also scale applications to optimize their performance (and cost).

{U.C.16.2} Platform Monitoring: The platform administrator and application administrator explore and monitor virtual resources, monitor cloud operations, notifications and events, and generate performance reports.

{U.C.16.3} Resource Utilization Measurement: The platform administrator and application administrator consult the measured values related to the use of cloud services and resources. These values refer to storage space, CPU processing, network bandwidth, active user accounts. Reports can also be generated on these measurements.

{U.C.16.4} SLA Management: The platform administrator and application administrator define the service agreement level (SLA), with the basic scheme of QoS parameters), and perform its monitoring and enforcement.

{U.C.17} Portability and interoperability management of the cloud: includes the mechanisms to support data portability, service interoperability and system portability. From the cloud consumers' perspective, the activities of data portability and service interoperability between public cloud data and external information systems are performed. These use cases are described in Systems integration and Data synchronization refined use cases.

{U.C.17.1} Systems Integration: The platform administrator integrates data and services from multiple cloud providers so that they are accessed through a single management interface for end users.

{U.C.17.2} Data Synchronization: The platform administrator transfers data objects to and from the cloud, or uses bulk data transfer to disks, performing data extractions and backups using standard formats.

{U.C.18} Cloud security and privacy management: performs a set of functionalities that allow the platform administrator to manage backups, monitor activities and configure access controls. In a logic of shared responsibilities between the Cloud Provider and Cloud Consumers, these components are provided by the Provider for privileged access to resource services. The management of application user accounts is the responsibility of the Cloud Consumer. These use cases are detailed in Backup management, Configuration of data access control and Activity monitoring refined use cases.

{U.C.18.1} Backup Management: The platform administrator manages data backups and service recovery. This management includes the functions of periodic data copy, and maintenance and management of code versions. The backups must follow standard formats to ensure correct data portability (for multi-cloud communications) and restoration. Backups are performed on data synchronized in {U.C.17.2} Data Synchronization.

{U.C.18.2} Configuration of data access control: The PaaS and SaaS consumer service models present different types of operation management, exposing different entry points into cloud systems, while also creating different attack surfaces.

From the perspective of the SaaS consumer model, the client account administrator provides access to cloud applications using an internet connection through a web browser, and is responsible for application access controls.

The platform administrator (PaaS consumer), on the other hand, uses access controls to the platform provided by and under the responsibility of the provider. The platform administrator also defines access to services, or manages credentials and

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permissions for access to these services, such as cloud middleware, in {U.C.15.1} Account Management.

{U.C.18.3} Activity Monitoring: The platform administrator monitors activities and audits traces to verify and address threats to the system. These reports can be generated in {U.C.16.2} Platform Monitoring.

6 Conclusions

This deliverable presents the work carried out as part of Task T2.2 of the P3-Maintenance Management Platform Bike4All. The main outcome of the task presented in this report was the Use cases that describe the platform's functionality.

We developed 18 level 0 use cases, 50 level 1 domain application use cases, and 13 level 1 cloud management use cases, ensuring a thorough understanding of the platform's functionality.

This outcome will guide the conception of the platform in the next Task, T2.3, which will be implemented as part of the WP2 of the Bike4All Be Neutral project.

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8 Annexes

8.1 Annex 1–Domain Application Requirements Specification

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.1.1} Real-time Location Update	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.1.1} Track Bike Location in Real-Time

1 Brief Description

The system continuously tracks the real-time location of each bike using GPS technology to monitor the usage and position of the bikes.

2 Actor Brief Descriptions

2.1 System:

Responsible for collecting and processing GPS data..

2.2 Mobility Devices Management Platform:

Provides a GPS Tracking System with real-time location data of the individual bikes.

2.3 Maintenance Manager/Maintenance Operator:

Personnel responsible for maintaining and redistributing bikes.

3 Preconditions

Each bike must be equipped with a functional GPS tracker and connected to the Mobility Devices Management Platform.

The system must have active communication with all GPS trackers provided by the Mobility Devices Management Platform.

The system must be operational and capable of processing location data.

4 Basic Flow of Events

1. The Mobility Devices Management Platform sends the location data of the GPS tracker on each bike to the system at regular intervals.
2. The system receives and processes the GPS data.
3. The system updates the bike's location in the database.
4. The system displays the real-time location on the dashboard for monitoring purposes.

5 Alternative Flows

5.1 <Alternate flow: GPS Signal Loss>

If a GPS signal is lost, the system will use the last known location and flag the bike for signal recovery monitoring.

5.2 <Alternate flow: Data Transmission Failure>

If data transmission fails, the system retries multiple times before flagging the bike for manual inspection.

6 Subflows

6.1 <Subflow: Location Data Processing >

1. Validate incoming GPS data.
2. Convert GPS coordinates into readable format.

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3. Update the database with the latest location data.

7 Key Scenarios

7.1 <Scenario: Location Update Success>

1. A bike's location is updated and accurately reflected on the monitoring dashboard.

7.2 <Scenario: Data Not Received>

1. The system flags a bike if GPS data is not received for a specified period.

8 Post-conditions

8.1 <post-condition 1>

The system accurately reflects the real-time location of each bike.

8.2 <post-condition 2>

The system logs any bikes with signal or data transmission issues for maintenance follow-up.

9 Special Requirements (FURPS+)

Functional Requirements (F):

The system must handle real-time data processing for up to 10,000 bikes simultaneously.

Usability Requirements (U):

The dashboard interface must be intuitive for maintenance staff to use.

Reliability Requirements (R):

The system must have a 99.9% uptime.

Performance Requirements (P):

The system must process location updates within 2 seconds of receiving data.

Supportability Requirements (S):

The system must be easy to maintain and update for new GPS technology.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.1.2} Usage Monitoring	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.1.2} Identify Idle Bikes

1 Brief Description

The system identifies bikes that have been stationary for a prolonged period, indicating potential underutilization or need for maintenance.

2 Actor Brief Descriptions

2.1 System:

Responsible for analyzing bike activity.

2.2 Maintenance Manager/ Maintenance Operator:

Personnel responsible for inspecting and redistributing idle bikes.

3 Preconditions

The system must be operational and have up-to-date location data for all bikes.

Each bike's location data must be recorded continuously.

4 Basic Flow of Events

1. The system continuously monitors the location data of all bikes.
2. The system identifies bikes that have not moved for a predefined duration.
3. The system flags these bikes as idle and sends an alert to the maintenance staff.
4. Maintenance Manager/ Maintenance Operator reviews the alert and takes necessary action (e.g., inspection, redistribution).

5 Alternative Flows

5.1 <Alternate flow: False Positives>

If a bike is flagged as idle but is actually in use, the system allows manual override by the Maintenance Manager/ Maintenance Operator.

5.2 <Alternate flow: Multiple Idle Bikes in Same Location>

If multiple bikes are idle in the same location, the system prioritizes alerts based on the duration of inactivity.

6 Subflows

6.1 <Subflow: Idle Bike Detection>

Compare current and previous location data for movement.

Calculate the duration of inactivity.

Flag and alert Maintenance Manager/ Maintenance Operator if the inactivity exceeds the threshold.

7 Key Scenarios

7.1 <Scenario: Flagging idle bikes>

1. A bike's location is updated and accurately reflected on the monitoring dashboard.

7.2 <Scenario: Redistributing bikes>

1. The system flags a bike if GPS data is not received for a specified period.

8 Post-conditions

8.1 <post-condition 1>

The system accurately identifies and flags idle bikes.

8.2 <post-condition 2>

The maintenance Manager/ Maintenance Operator is alerted promptly for necessary action.

9 Special Requirements (FURPS+)

Functional Requirements (F):

The system must identify idle bikes based on customizable inactivity thresholds.

Usability Requirements (U):

Alerts must be clear and actionable for the Maintenance Manager/ Maintenance Operator.

Reliability Requirements (R):

The system must accurately flag idle bikes with minimal false positives.

Performance Requirements (P):

The system must process and flag idle bikes within 5 minutes of detecting inactivity.

Supportability Requirements (S):

The system must allow easy adjustment of inactivity thresholds and alert settings.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.2.1} Manage Bike Details	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.2.1} Manage Bike Details

1 Brief Description

This use case describes the process by which the system maintains and updates details about each bike in the inventory.

2 Actor Brief Descriptions

2.1 System:

Processes and stores the data.

2.2 Maintenance Manager:

Manages the system and ensures data integrity.

3 Preconditions

The system is operational.

The Maintenance Manager has access to the system.

4 Basic Flow of Events

1. The Maintenance Manager inputs or updates bike details (model, age, condition).
2. The System processes and records the data.

5 Alternative Flows

5.1 <Alternate flow: Data Entry Error:>

If incorrect data is entered, the system prompts the Maintenance Manager to correct it.

6 Subflows

6.1 <Subflow: N/A >

7 Key Scenarios

7.1 <Scenario: Regular updates>

1. Regular updates by the Maintenance Manager to ensure accurate records.

7.2 <Scenario: Error correction>

1. Error correction prompted by the system.

8 Post-conditions

8.1 <post-condition 1>

Bike details are accurately stored and up-to-date in the system database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Ability to input and update bike details..

Usability Requirements (U):

Easy-to-use interface for Maintenance Manager.

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Reliability Requirements (R):

System must ensure data integrity.

Performance Requirements (P):

Updates should be processed quickly.

Supportability Requirements (S):

Easy maintenance and updates to the system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.2.2} Track Bike Availability Status	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.2.2} Track Bike Availability Status

1 Brief Description

This use case describes the process by which the system tracks and updates the current status of each bike.

2 Actor Brief Descriptions

2.1 System:

Processes the data to update bike status.

2.2 Mobility Devices Management Platform:

Sends real-time location data to the System.

3 Preconditions

The system is operational.

Each bike is equipped with a functioning GPS device, and connected to the Mobility Devices Management Platform in an operational status.

4 Basic Flow of Events

1. The Mobility Devices Management Platform sends the bike's location data to the System.
2. The System updates the bike's status (available, in use, under maintenance).

5 Alternative Flows

5.1 <Alternate flow: GPS Device Failure>

If the Mobility Devices Management Platform fails to deliver data from a GPS device, the system flags the bike for inspection.

6 Subflows

6.1 <Subflow: N/A >

7 Key Scenarios

7.1 <Scenario: Regular updates>

1. Regular updates to maintain an accurate status of each bike.

7.2 <Scenario: Handling device failures>

1. Handling device failures by flagging bikes for maintenance.

8 Post-conditions

8.1 <post-condition 1>

Bike status is accurately reflected in the system database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Ability to track and update bike status.

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Usability Requirements (U):

Intuitive interface for viewing status.

Reliability Requirements (R):

High reliability of status updates.

Performance Requirements (P):

Real-time processing of status updates.

Supportability Requirements (S):

Easy troubleshooting and maintenance of tracking system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.2.3} Update Bike Inventory	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.2.3} Update Bike Inventory

1 Brief Description

This use case describes the process by which the system updates the inventory to reflect changes in the fleet.

2 Actor Brief Descriptions

2.1 System:

Processes and records inventory data.

2.2 Maintenance Manager:

Inputs and updates inventory data.

3 Preconditions

The system is operational.

The maintenance Manager has access to the backend.

4 Basic Flow of Events

1. The Maintenance Manager adds or removes bikes from the inventory.
2. The System processes and records these changes..

5 Alternative Flows

5.1 <Alternate flow: Data Entry Error>

If incorrect data is entered, the system prompts the Maintenance Manager to correct it.

6 Subflows

6.1 <Subflow: N/A >

7 Key Scenarios

7.1 <Scenario: Regular updates>

1. Regular updates to reflect the current fleet accurately..

7.2 <Scenario: Error correction>

1. Error correction prompted by the system.

8 Post-conditions

8.1 <post-condition 1>

The inventory is updated accurately in the system database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Ability to add or remove bikes from the inventory..

Usability Requirements (U):

User-friendly interface for Maintenance Manager.

Reliability Requirements (R):

Ensuring data integrity during updates.

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Performance Requirements (P):

Efficient processing of inventory updates.

Supportability Requirements (S):

Easy maintenance and updates to the inventory system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.3.1} Collect Historical Data	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.3.1} Collect Historical Data

1 Brief Description

This use case describes the process by which the system collects historical data on bike usage, including times and locations.

2 Actor Brief Descriptions

2.1 System:

Collects and stores historical data.

2.2 Mobility Devices Management Platform:

Sends real-time location data.

3 Preconditions

The GPS devices attached to the Mobility Devices Management Platform are operational.

The system is operational.

4 Basic Flow of Events

1. The Mobility Devices Management Platform sends real-time GPS device location data to the System.
2. The System stores this data in a historical database.

5 Alternative Flows

5.1 <Alternate flow: GPS Device Failure>

If a GPS device attached to the Mobility Devices Management Platform fails, the system logs an error and continues collecting data from other devices.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Regular collection>

1. Regular collection of location data without interruptions.

7.2 <Scenario: Handling device failures >

1. Handling GPS device failures gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Historical data on bike usage is stored in the system database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

The system shall store real-time GPS data.

The system shall log errors when GPS devices fail.

Usability Requirements (U):

The system shall provide an interface for monitoring data collection status.

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Reliability Requirements (R):

The system shall ensure continuous data collection even if some devices fail.

Performance Requirements (P):

The system shall process incoming data in real-time.

Supportability Requirements (S):

The system shall allow for easy maintenance and updates.

+ (Other):

Security: Data must be securely transmitted and stored.

Scalability: The system should handle an increasing number of devices.

Extensibility: The system should support new types of data sources.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.3.2} Train Predictive Model	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.3.2} Train Predictive Model

1 Brief Description

The system shall train a predictive model using collected historical data to identify patterns in bike demand.

2 Actor Brief Descriptions

2.1 System:

Performs data analysis and model training.

3 Preconditions

Historical data on bike usage is available.

The system is operational.

4 Basic Flow of Events

1. The system retrieves historical data on bike usage.
2. The system processes the data to identify demand patterns.
3. The system trains the predictive model with the processed data.

5 Alternative Flows

5.1 <Alternate flow: Insufficient Data>

If historical data is insufficient, the system logs the issue and notifies the maintenance manager.

6 Subflows

6.1 <Subflow: Data Validation>

The system validates historical data before analysis to ensure accuracy.

7 Key Scenarios

7.1 <Scenario: Regular analysis>

1. Successful model training with sufficient data.

7.2 <Scenario: Handling anomalies >

1. Handling insufficient data gracefully.

8 Post-conditions

8.1 <Post-condition 1>

A trained predictive model is available for generating demand predictions.

9 Special Requirements (FURPS+)

Functional Requirements (F):

The system shall validate and process historical data.

The system shall train the predictive model using validated data.

Usability Requirements (U):

The system shall provide progress and status updates during model training..

Reliability Requirements (R):

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The system shall handle data insufficiencies gracefully and notify the user.

Performance Requirements (P):

The system shall complete model training within a reasonable time frame.

Supportability Requirements (S):

The system shall allow for easy updates and maintenance of the predictive model.

+ (Other):

Security: The model training process must be secure and protected from unauthorized access.

Scalability: The system should handle increasing volumes of historical data.

Extensibility: The system should support integration with new data sources and training algorithms.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.3.3} Predict Bike Demand	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.3.3} Predict Bike Demand

1 Brief Description

The system shall use the trained predictive model to generate predictions of future bike demand at different times and locations.

2 Actor Brief Descriptions

2.1 System:

Uses the predictive model to generate demand predictions.

2.2 Maintenance Manager:

Utilizes the predictions for operational planning.

3 Preconditions

A trained predictive model is available.

The system is operational.

4 Basic Flow of Events

1. The system uses the predictive model to generate predictions of future bike demand.
2. The system presents the demand predictions to the Maintenance Manager.
3. The Maintenance Manager reviews and utilizes the predictions for allocation and distribution decisions.

5 Alternative Flows

5.1 <Alternate flow: Prediction Error>

If significant errors are found in the predictions, the system logs the errors.

The system notifies the Maintenance Manager of the prediction errors.

The system adjusts the prediction model to improve accuracy.

6 Subflows

6.1 <Subflow: Model Adjustment>

The system analyzes prediction errors and adjusts the predictive model accordingly

7 Key Scenarios

7.1 <Scenario: Successful generation predictions >

1. Successful generation of accurate demand predictions.

7.2 <Scenario: Handling prediction errors >

1. Handling prediction errors and improving model accuracy.

8 Post-conditions

8.1 <Post-condition 1>

Accurate bike demand predictions are available for operational planning.

8.2 <Post-condition 2>

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The predictive model is adjusted and improved if prediction errors are detected..

9 Special Requirements (FURPS+)

Functional Requirements (F):

The system shall generate and present bike demand predictions.

The system shall log prediction errors and adjust the model accordingly..

Usability:

The system shall provide an intuitive interface for reviewing predictions.

Reliability:

The system shall ensure high accuracy of demand predictions.

Performance:

The system shall generate predictions within a reasonable time frame.

Supportability:

The system shall allow for easy updates and maintenance of the predictive model.

+ (Other):

Security: Predictions must be securely generated and presented.

Scalability: The system should handle increasing volumes of prediction requests.

Extensibility: The system should support integration with new prediction algorithms.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.3.4} Optimize Bike Allocation	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.3.4} Optimize Bike Allocation

1 Brief Description

This use case describes the process by which the system optimizes the allocation of bikes based on demand predictions.

2 Actor Brief Descriptions

2.1 System:

Provides optimization recommendations.

2.2 Maintenance Manager:

Implements the recommendations.

3 Preconditions

Accurate demand predictions are available.

The system is operational.

4 Basic Flow of Events

1. The System analyzes demand predictions and current inventory.
2. The System generates recommendations for bike allocation.
3. The Maintenance Manager reviews and implements the recommendations.

5 Alternative Flows

5.1 <Alternate flow: Resource Constraints>

If there are constraints (e.g., limited bikes), the system adjusts recommendations accordingly.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Regular optimization of bike allocation>

1. Regular optimization of bike allocation.

7.2 <Scenario: Adjustment of recommendations based on resource constraints>

1. Adjustment of recommendations based on resource constraints.

8 Post-conditions

8.1 <Post-condition 1>

Bikes are allocated efficiently to meet predicted demand.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Optimization of bike allocation.

Usability Requirements (U):

Clear and actionable recommendations.

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Reliability Requirements (R):

Accurate and feasible allocation recommendations.

Performance Requirements (P):

Timely generation of recommendations.

Supportability Requirements (S):

Easy updates to allocation algorithms.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.4.1} Collect Usage Data	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.4.1} Collect Usage Data

1 Brief Description

This use case describes the process by which the system collects data on bike usage, including start and end times, routes taken, and trip durations.

2 Actor Brief Descriptions

2.1 System:

Collects and stores usage data.

2.2 Mobility Devices Management Platform:

Sends real-time location data.

3 Preconditions

GPS devices on bikes connected to the Mobility Devices Management Platform are operational.

The system is operational.

4 Basic Flow of Events

1. Mobility Devices Management Platform sends real-time location and trip data to the System.
2. The System stores this data in a usage database.

5 Alternative Flows

5.1 <Alternate flow: GPS Device Failure>

If the Mobility Devices Management Platform fails to deliver data from a GPS device, the system logs an error and continues collecting data from other devices.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Continuous and accurate collection>

1. Continuous and accurate collection of usage data.

7.2 <Scenario: Handling of GPS device failures gracefully>

1. Handling of GPS device failures gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Comprehensive usage data is stored in the system database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Real-time data collection and storage.

Usability Requirements (U):

Access to usage data for analysis.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Reliability Requirements (R):

Continuous data collection.

Performance Requirements (P):

Efficient storage and retrieval of large data sets.

Supportability Requirements (S):

Easy maintenance and updates to data collection processes.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.4.2}Analyze Peak Usage Times	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.4.2} Analyze Peak Usage Times

1 Brief Description

This use case describes the process by which the system analyzes usage data to identify peak usage times.

2 Actor Brief Descriptions

2.1 System:

Analyzes usage data.

2.2 Data Analyst:

Reviews and validates the analysis.

3 Preconditions

Usage data is available.

The system is operational.

4 Basic Flow of Events

1. The System processes usage data to identify peak usage times.
2. The Data Analyst reviews the analysis for accuracy.

5 Alternative Flows

5.1 <Alternate flow: Data Anomaly>

If anomalies are detected, the system flags them for further review.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Regular analysis>

1. Regular analysis to identify accurate peak usage times.

7.2 <Scenario: Handling anomalies>

1. Handling and flagging data anomalies.

8 Post-conditions

8.1 <Post-condition 1>

Accurate peak usage times are identified and validated.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Data processing and peak time identification.

Usability Requirements (U):

Clear presentation of analysis results.

Reliability Requirements (R):

Accurate analysis of data.

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Performance Requirements (P):

Efficient processing of large data sets.

Supportability Requirements (S):

Easy updates to analysis algorithms.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.4.3} Analyze Popular Routes	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.4.3} Analyze Popular Routes

1 Brief Description

This use case describes the process by which the system analyzes usage data to identify popular routes taken by users.

2 Actor Brief Descriptions

2.1 System:

Analyzes usage data..

2.2 Data Analyst:

Reviews and validates the analysis.

3 Preconditions

Usage data is available.

The system is operational.

4 Basic Flow of Events

1. The System processes usage data to identify popular routes.
2. The Data Analyst reviews the analysis for accuracy.

5 Alternative Flows

5.1 <Alternate flow: Data Anomaly>

If anomalies are detected, the system flags them for further review.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Regular analysis>

1. Regular analysis to identify accurate popular routes.

7.2 <Scenario: Handling anomalies>

1. Handling and flagging data anomalies.

8 Post-conditions

8.1 <Post-condition 1>

Accurate popular routes are identified and validated..

9 Special Requirements (FURPS+)

Functional Requirements (F):

Data processing and route identification.

Usability Requirements (U):

Clear presentation of analysis results.

Reliability Requirements (R):

Accurate analysis of data.

Performance Requirements (P):

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Efficient processing of large data sets.

Supportability Requirements (S):

Easy updates to analysis algorithms.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.4.4} Analyze Average Trip Duration	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.4.4} Analyze Average Trip Duration

1 Brief Description

This use case describes the process by which the system analyzes usage data to calculate average trip durations.

2 Actor Brief Descriptions

2.1 System:

Analyzes usage data.

2.2 Data Analyst:

Reviews and validates the analysis.

3 Preconditions

Usage data is available.

The system backend is operational.

4 Basic Flow of Events

1. The System processes usage data to calculate average trip durations.
2. The Data Analyst reviews the analysis for accuracy.

5 Alternative Flows

5.1 <Alternate flow: Data Anomaly>

If anomalies are detected, the system flags them for further review.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Regular analysis>

1. Regular analysis to calculate accurate average trip durations.

7.2 <Scenario: Handling anomalies>

1. Handling and flagging data anomalies.

8 Post-conditions

8.1 <Post-condition 1>

Accurate average trip durations are calculated and validated.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Data processing and trip duration calculation.

Usability Requirements (U):

Clear presentation of analysis results.

Reliability Requirements (R):

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Accurate analysis of data.

Performance Requirements (P):

Efficient processing of large data sets.

Supportability Requirements (S):

Easy updates to analysis algorithms.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.4.5} Generate Usage Insights Reports	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.4.5} Generate Usage Insights Reports

1 Brief Description

This use case describes the process by which the system generates reports based on the analysis of usage patterns, including peak times, popular routes, and average trip durations.

2 Actor Brief Descriptions

2.1 System:

Generates reports.

2.2 Maintenance Manager:

Reviews and distributes reports.

3 Preconditions

Analysis of usage patterns is complete.

The system is operational.

4 Basic Flow of Events

1. The System compiles analysis results into comprehensive reports.
2. The Maintenance Manager reviews the reports for accuracy.
3. The Maintenance Manager distributes the reports to relevant stakeholders.

5 Alternative Flows

5.1 <Alternate flow: Report Generation Error>

If errors occur during report generation, the system logs the errors and retries the process.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Regular generation and distribution>

1. Regular generation and distribution of usage insights reports.

7.2 <Scenario: Handling report generation errors>

1. Handling and retrying report generation errors.

8 Post-conditions

8.1 <Post-condition 1>

Comprehensive and accurate usage insights reports are available for review and distribution.

9 Special Requirements (FURPS+)

Functional Requirements (F):

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Report generation based on analysis results.

Usability Requirements (U):

Clear and comprehensive reports.

Reliability Requirements (R):

Accurate and timely report generation.

Performance Requirements (P):

Efficient compilation of large data sets into reports.

Supportability Requirements (S):

Easy updates to report generation processes.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.5.1} Receive Issue Report	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.5.1} Receive Issue Report

1 Brief Description

This use case describes the process by which the system receives issues or incidents reported by users related to the bikes.

2 Actor Brief Descriptions

2.1 System:

Receives and logs the issue reports.

2.2 Bike4All Platform:

Sends issue reports to the system.

3 Preconditions

The Bike4All Platform is operational.

The system is operational.

4 Basic Flow of Events

1. The Bike4All Platform sends an issue report to the system.
2. The system receives the issue report.
3. The system logs the issue report in the database.

5 Alternative Flows

5.1 <Alternate flow: Transmission Error>

If there is an error in receiving the report, the system logs the error and requests a retransmission.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Successful receipt and logging>

1. Successful receipt and logging of issue reports.

7.2 <Scenario: Handling transmission errors>

1. Handling transmission errors effectively.

8 Post-conditions

8.1 <Post-condition 1>

The issue report is recorded in the system's database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Receive and log issue reports.

Usability Requirements (U):

Ensure clear logging and access to issue reports.

Reliability Requirements (R):

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High reliability in receiving and logging reports.

Performance Requirements (P):

Efficient handling and logging of incoming reports.

Supportability Requirements (S):

Easy maintenance and updates to the issue reception system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.5.2} Process and Assign Issue	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.5.2} Process and Assign Issue

1 Brief Description

This use case describes the process by which the system processes and assigns received issues for resolution.

2 Actor Brief Descriptions

2.1 System:

Processes and assigns issues to maintenance staff.

2.2 Maintenance Operator:

Receives assigned issues for resolution.

3 Preconditions

Issue reports are logged in the system.

The system and maintenance Operator are operational.

4 Basic Flow of Events

1. The system processes the received issue report.
2. The system assigns the issue to the appropriate maintenance Operator.
3. The maintenance Operator receives the assigned issue.

5 Alternative Flows

5.1 <Alternate flow: Assignment Error>

If there is an error in assignment, the system logs the error and retries the assignment process.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Successful processing and assignment>

1. Successful processing and assignment of issues.

7.2 <Scenario: Handling assignment errors>

1. Handling errors in the assignment process.

8 Post-conditions

8.1 <Post-condition 1>

Issues are assigned to the maintenance Operator for resolution.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Process and assign issues to the maintenance Operator.

Usability Requirements (U):

Clear assignment notifications to the maintenance Operator.

Reliability Requirements (R):

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High reliability in processing and assigning issues.

Performance Requirements (P):

Efficient assignment of issues to relevant Operator.

Supportability Requirements (S):

Easy maintenance and updates to the issue processing system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.5.3} Track and Update Issue Resolution	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.5.3} Track and Update Issue Resolution

1 Brief Description

This use case describes the process by which the system tracks and updates the resolution status of reported issues.

2 Actor Brief Descriptions

2.1 System:

Tracks and records the resolution status.

2.2 Maintenance Operator:

Updates the resolution status of issues.

3 Preconditions

Issues are assigned to the maintenance operator.

The system and maintenance operator are operational.

4 Basic Flow of Events

1. The Maintenance Operator updates the status of the assigned issue as they work on it.
2. The system tracks these updates and records the resolution status.

5 Alternative Flows

5.1 <Alternate flow: Update Error>

If there is an error in updating the status, the system logs the error and requests a status update retry.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Successful tracking and updating>

1. Successful tracking and updating of issue resolution statuses.

7.2 <Scenario: Handling errors in status updates>

1. Handling errors in status updates.

8 Post-conditions

8.1 <Post-condition 1>

The resolution status of issues is accurately recorded in the system.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Track and update issue resolution statuses.

Usability Requirements (U):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Ensure easy and clear status updates by maintenance operator.

Reliability Requirements (R):

High reliability in tracking and recording status updates.

Performance Requirements (P):

Efficient handling and recording of status updates.

Supportability Requirements (S):

Easy maintenance and updates to the status tracking system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.5.4} Send Resolution Feedback	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.5.4} Send Resolution Feedback

1 Brief Description

This use case describes the process by which the system sends feedback on the resolution of issues to the Bike4All Platform.

2 Actor Brief Descriptions

2.1 System:

Sends resolution feedback.

2.2 Bike4All Platform:

Receives feedback on issue resolution.

3 Preconditions

Issues are resolved and their statuses are updated in the system.

The system is operational.

4 Basic Flow of Events

1. The system generates resolution feedback based on updated statuses.
2. The system sends this feedback to the Bike4All Platform.
3. The Bike4All Platform receives and logs the resolution feedback.

5 Alternative Flows

5.1 <Alternate flow: Transmission Error>

If there is an error in sending feedback, the system logs the error and retries the transmission.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Successful sending of feedback>

1. Successful sending of resolution feedback.

7.2 <Scenario: Handling transmission errors>

1. Handling errors in feedback transmission.

8 Post-conditions

8.1 <Post-condition 1>

Resolution feedback is logged in the Bike4All Platform.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Send resolution feedback.

Usability Requirements (U):

Ensure clear and timely feedback to the Bike4All Platform.

Reliability Requirements (R):

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High reliability in sending feedback.

Performance Requirements (P):

Efficient handling and transmission of feedback.

Supportability Requirements (S):

Easy maintenance and updates to the feedback system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.6.1} Activate Remote Lock	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.6.1} Activate Remote Lock

1 Brief Description

This use case describes the process by which the system can remotely lock a bike to prevent unauthorized use.

2 Actor Brief Descriptions

2.1 System:

Sends commands to lock the bike.

2.2 Mobility Devices Management Platform:

Receives commands and locks the bike.

3 Preconditions

Each bike is equipped with a remote lock mechanism connected to the Mobility Devices Management Platform.

The system is operational.

4 Basic Flow of Events

1. The system sends a lock command to the Mobility Devices Management Platform.
2. The Mobility Devices Management Platform receives the command and locks the bike.
3. The Mobility Devices Management Platform sends a confirmation back to the system.

5 Alternative Flows

5.1 <Alternate flow: Lock Mechanism Failure>

If the Mobility Devices Management Platform fails to lock, the system logs an error and retries the command.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Successful remote locking>

1. Successful remote locking of bikes.

7.2 <Scenario: Handling lock mechanism failures>

1. Handling lock mechanism failures and retrying commands.

8 Post-conditions

8.1 <Post-condition 1>

The bike is locked and cannot be used without authorization.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Remote locking of bikes.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Usability Requirements (U):

N/A (backend process).

Reliability Requirements (R):

High reliability of the remote lock mechanism.

Performance Requirements (P):

Quick and efficient remote locking process.

Supportability Requirements (S):

N/A.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.6.2} Trigger Alarm System	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.6.2} Trigger Alarm System

1 Brief Description

This use case describes the process by which the system can trigger an alarm on a bike to deter theft or misuse.

2 Actor Brief Descriptions

2.1 System:

Sends commands to trigger the alarm.

2.2 Mobility Devices Management Platform:

Receives commands and activates the alarm.

3 Preconditions

Each bike is equipped with an alarm system connected to the Mobility Devices Management Platform.

The system is operational.

4 Basic Flow of Events

1. The system sends an alarm command to the Mobility Devices Management Platform.
2. The Mobility Devices Management Platform receives the command and activates the alarm on the bike.
3. The Mobility Devices Management Platform sends a confirmation back to the system.

5 Alternative Flows

5.1 <Alternate flow: Mobility Devices Management Platform Failure>

If the Mobility Devices Management Platform fails to activate, the system logs an error and retries the command.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Successful activation>

1. Successful activation of the Mobility Devices Management Platform to deter theft or misuse.

7.2 <Scenario: Handling platform failures>

1. Handling Mobility Devices Management Platform failures and retrying commands.

8 Post-conditions

8.1 <Post-condition 1>

The bike's alarm system is activated, deterring theft or misuse.

9 Special Requirements (FURPS+)

Functional Requirements (F):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Triggering alarm systems remotely.

Usability Requirements (U):

N/A (system process).

Reliability Requirements (R):

High reliability of the alarm system.

Performance Requirements (P):

Quick and efficient alarm activation process.

Supportability Requirements (S):

N/A.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.6.3} Track Bike Safety Status	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.6.3} Track Bike Safety Status

1 Brief Description

This use case describes the process by which the system continuously monitors the status of bikes for any signs of unauthorized use or tampering.

2 Actor Brief Descriptions

2.1 System:

Monitors and processes bike status data.

2.2 Mobility Devices Management Platform:

Sends GPS device status data, sends lock status data, sends alarm status data.

3 Preconditions

Each bike is equipped with functioning GPS, lock, and alarm systems.

The system is operational.

4 Basic Flow of Events

1. The Mobility Devices Management Platform send status data about the GPS device, remote lock mechanism, and alarm system of each bike to the system at regular intervals.
2. The system monitors and processes the status data.
3. The system flags any signs of unauthorized use or tampering for further inspection.

5 Alternative Flows

5.1 <Alternate flow: System Error>

If there is a system error, the system logs the error and continues monitoring other bikes.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Continuous monitoring>

1. Continuous and accurate monitoring of bike status.

7.2 <Scenario: Handling system errors>

1. Handling system errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The bike's status is continuously monitored and any signs of unauthorized use or tampering are flagged for inspection.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Continuous monitoring of bike status.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Usability Requirements (U):

N/A (system process).

Reliability Requirements (R):

High reliability of status monitoring.

Performance Requirements (P):

Efficient processing of status data.

Supportability Requirements (S):

Easy maintenance and updates to the monitoring system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.6.4} Respond to Security Alerts	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.6.4} Respond to Security Alerts

1 Brief Description

This use case describes the process by which the system responds to security alerts generated by the monitoring system.

2 Actor Brief Descriptions

2.1 System:

Generates and processes security alerts.

2.2 Maintenance Manager:

Reviews and responds to security alerts.

2.3 Maintenance Operator:

Acts on security alerts for physical inspection and repair.

3 Preconditions

The system is operational.

Security alerts are being generated by the monitoring system.

4 Basic Flow of Events

1. The system generates a security alert when signs of unauthorized use or tampering are detected.
2. The system notifies the Maintenance Manager of the security alert.
3. The Maintenance Manager reviews the alert and decides on the necessary action.
4. If a physical inspection or repair is needed, the Maintenance Manager assigns the task to the maintenance operator.
5. The Maintenance Operator acts on the alert and updates the system with the resolution status.

5 Alternative Flows

5.1 <Alternate flow: False Alarm>

If the security alert is found to be a false alarm, the Maintenance Manager marks it as resolved without further action.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Timely response>

1. Timely response to security alerts.

7.2 <Scenario: Handling false alarms>

1. Handling and marking false alarms.

8 Post-conditions

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8.1 <Post-condition 1>

Security alerts are reviewed and acted upon, ensuring bike security and maintenance.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Generate and respond to security alerts.

Usability Requirements (U):

Clear interface for reviewing and responding to alerts.

Reliability Requirements (R):

High reliability of alert generation and response system.

Performance Requirements (P):

Quick generation and processing of alerts.

Supportability Requirements (S):

Easy maintenance and updates to the alert system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.7.1} View Fleet Overview	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.7.1} View Fleet Overview

1 Brief Description

This use case describes the process by which Maintenance Manager can view an overview of the entire bike fleet, including the status of each bike.

2 Actor Brief Descriptions

2.1 System:

Provides data and status updates.

2.2 Maintenance Manager:

Uses the interface to manage the fleet.

3 Preconditions

The system is operational.

The Maintenance Manager is logged into the system.

4 Basic Flow of Events

1. The System retrieves and displays an overview of the fleet, including bike status (available, in use, under maintenance).
2. The Maintenance Manager reviews the overview to manage the fleet.

5 Alternative Flows

5.1 <Alternate flow: Data Retrieval Error>

If the system fails to retrieve data, an error message is displayed and the system retries.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: View bike status>

1. Maintenance Manager can easily view the status of all bikes.

7.2 <Scenario: Handling retrieval errors>

1. Handling data retrieval errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The Maintenance Manager has an up-to-date overview of the fleet.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Display a comprehensive fleet overview.

Usability Requirements (U):

Easy-to-navigate interface.

Reliability Requirements (R):

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Accurate and up-to-date information.

Performance Requirements (P):

Quick data retrieval and display.

Supportability Requirements (S):

Easy updates to the interface.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.7.2} Assign Maintenance Tasks	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.7.2} Assign Maintenance Tasks

1 Brief Description

This use case describes the process by which Maintenance Manager can assign maintenance tasks to the Maintenance Operator.

2 Actor Brief Descriptions

2.1 System:

Manages task assignments and updates.

2.2 Maintenance Manager:

Assigns maintenance tasks.

2.3 Maintenance Operator:

Receives and performs tasks.

3 Preconditions

The system is operational.

The Maintenance Manager is logged into the system.

4 Basic Flow of Events

1. The Maintenance Manager identifies a bike requiring maintenance from the fleet overview.
2. The Maintenance Manager assigns a maintenance task to the Maintenance Operator.
3. The System notifies the Maintenance Operator and logs the task assignment.

5 Alternative Flows

5.1 <Alternate flow: Notification Failure>

If the notification fails, the system logs an error and retries.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Assign maintenance tasks>

1. Maintenance Manager can efficiently assign maintenance tasks.

7.2 <Scenario: Handle notification failures>

1. Handling notification failures.

8 Post-conditions

8.1 <Post-condition 1>

The Maintenance Operator is aware of and can act on assigned tasks.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Assign maintenance tasks.

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Usability Requirements (U):

Easy task assignment interface.

Reliability Requirements (R):

Ensure task assignments are successfully communicated.

Performance Requirements (P):

Quick assignment and notification.

Supportability Requirements (S):

Easy updates to the task assignment process.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.7.3} Monitor Bike Status	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.7.3} Monitor Bike Status

1 Brief Description

This use case describes the process by which Maintenance Manager can monitor the real-time status of bikes in the fleet.

2 Actor Brief Descriptions

2.1 System:

Provides real-time status updates.

2.2 Maintenance Manager:

Monitors bike status.

3 Preconditions

The system is operational.

The Maintenance Manager is logged into the system.

4 Basic Flow of Events

1. The Maintenance Manager accesses the bike status monitoring interface.
2. The Maintenance Manager selects a specific bike to see the details.
3. The System provides real-time updates on that bike status (e.g., location, usage, maintenance).

5 Alternative Flows

5.1 <Alternate flow: Update Failure>

If status updates fail, the system logs an error and retries.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Real-time monitoring>

1. Maintenance Manager can monitor bike status in real-time.

7.2 <Scenario: Handling update failures>

1. Handling update failures gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The Maintenance Manager has accurate, real-time status information.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Real-time status monitoring.

Usability Requirements (U):

Intuitive interface for monitoring.

Reliability Requirements (R):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Ensure continuous and accurate updates.

Performance Requirements (P):

Efficient processing and display of real-time data.

Supportability Requirements (S):

Easy maintenance and updates to the monitoring system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.7.4} Generate Reports	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.7.4} Generate Reports

1 Brief Description

This use case describes the process by which the Maintenance Manager can generate reports on various aspects of the fleet, such as usage statistics, maintenance history, and performance metrics.

2 Actor Brief Descriptions

2.1 System:

Compiles and generates reports.

2.2 Maintenance Manager:

Generates and reviews reports.

3 Preconditions

The system is operational.

The Maintenance Manager is logged into the system.

4 Basic Flow of Events

1. The Maintenance Manager selects the type of report to generate.
2. The System compiles the necessary data, generates the report and sends it to the Maintenance Manager.
3. The Maintenance Manager reviews and analyzes the report.

5 Alternative Flows

5.1 <Alternate flow: Report Generation Error>

If there is an error in generating the report, the system logs the error and retries.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Generating comprehensive reports>

1. Maintenance Manager can generate comprehensive reports.

7.2 <Scenario: Handling report generation errors>

1. Handling report generation errors.

8 Post-conditions

8.1 <Post-condition 1>

The Maintenance Manager has access to detailed reports for analysis.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Generate various types of reports.

Usability Requirements (U):

Simple interface for selecting and generating reports.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Reliability Requirements (R):

Accurate and comprehensive reporting.

Performance Requirements (P):

Quick compilation and generation of reports.

Supportability Requirements (S):

Easy updates to the reporting system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.8.1} Integrate with Bike4All Platform for Customer Support	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.8.1} Integrate with Bike4All Platform

1 Brief Description

This use case describes how the maintenance bike-sharing platform integrates with the Bike4All platform to handle customer support issues.

2 Actor Brief Descriptions

2.1 System:

Handles the integration and data synchronization.

2.2 Bike4All Platform:

Manages user inquiries and issues.

2.3 Maintenance Manager/Maintenance Operator

Responds to inquiries or issues through the system

3 Preconditions

The Bike4All Platform API is available and operational.

The system is configured to communicate with the Bike4All Platform.

4 Basic Flow of Events

1. The Bike4All Platform forwards the inquiry or issue to the system.
2. The system receives and processes the inquiry or issue.
3. The system notifies the maintenance operator/manager of the new inquiry or issue.
4. The maintenance operator/manager resolves the inquiry or issue and responds through the system.
5. The system sends the response back to the Bike4All Platform.

5 Alternative Flows

5.1 <Alternate flow: Support System Unavailable>

If the system is unavailable, log the issue and notify the user of the delay.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Seamless submission and processing>

1. Seamless submission and processing of user inquiries and issues.

7.2 <Scenario: Handling system unavailability>

1. Handling the unavailability of the customer support system gracefully.

8 Post-conditions

8.1 <Post-condition 1>

User inquiries and issues are addressed promptly through the integrated customer support system.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

9 Special Requirements (FURPS+)

Functional Requirements (F):

Ensure reliable communication between the system and the Bike4All Platform.

Usability Requirements (U):

Easy access for maintenance operators/managers to view and respond to issues.

Reliability Requirements (R):

High availability and fault-tolerance for the integration.

Performance Requirements (P):

Quick processing and response times.

Supportability Requirements (S):

Clear logging and notification mechanisms for failures.

+ (Other):

Compliance with relevant data protection regulations.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.8.2} Integrate with Mobility Devices Management Platform for GPS Tracking	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.8.2} Integrate with Mobility Devices Management Platform

1 Brief Description

This use case describes how the maintenance bike-sharing platform integrates with the Mobility Devices Management Platform to enable GPS tracking of bikes.

2 Actor Brief Descriptions

2.1 System:

Manages integration and data synchronization.

2.2 Mobility Devices Management Platform:

Provides GPS tracking data.

2.3 Maintenance Manager/Maintenance Operator

Accesses the GPS tracking data.

3 Preconditions

The Mobility Devices Management Platform API is available and operational.

The system is configured to communicate with the Mobility Devices Management Platform.

4 Basic Flow of Events

1. The system requests GPS tracking data from the Mobility Devices Management Platform.
2. The Mobility Devices Management Platform sends GPS tracking data to the system.
3. The system receives and processes the GPS tracking data.
4. The system updates the bike statuses and locations in its database.
5. Maintenance operators/managers access GPS tracking data through the system to locate and manage bikes.

5 Alternative Flows

5.1 <Alternate flow: GPS Data Unavailable>

If GPS data is unavailable, log the issue and notify the maintenance operator/manager of the delay.

6 Subflows

6.1 <Subflow: N/A>

7 Key Scenarios

7.1 <Scenario: Accurate and real-time tracking>

1. Accurate and real-time GPS tracking of bikes.

7.2 <Scenario: Handling data unavailability>

1. Handling unavailability of GPS data gracefully.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

8 Post-conditions

8.1 <Post-condition 1>

Bikes are tracked accurately, allowing for efficient maintenance and management.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Ensure reliable communication between the system and the Mobility Devices Management Platform.

Usability Requirements (U):

Easy access for maintenance operators/managers to view GPS data.

Reliability Requirements (R):

High availability and fault-tolerance for the integration.

Performance Requirements (P):

Quick processing and real-time update of GPS data.

Supportability Requirements (S):

Clear logging and notification mechanisms for failures.

+ (Other):

Compliance with relevant data protection regulations.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.9.1} Define Maintenance Criteria	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.9.1} Define Maintenance Criteria

1 Brief Description

This use case describes the process by which administrators define the criteria for scheduling regular maintenance checks for bikes, such as usage, time, or distance traveled.

2 Actor Brief Descriptions

2.1 System:

Stores and processes the criteria.

2.2 Maintenance Manager:

Defines and updates maintenance criteria.

3 Preconditions

The system is operational.

The Maintenance Manager is logged into the system.

4 Basic Flow of Events

1. The Maintenance Manager navigates to the maintenance criteria settings.
2. The Maintenance Manager defines or updates the criteria for maintenance checks (e.g., usage hours, time intervals, distance traveled).
3. The System saves the criteria and applies them to the fleet.

5 Alternative Flows

5.1 <Alternate flow: Invalid Criteria Entry>

If the entered criteria are invalid, the system prompts the Maintenance Manager to correct them.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Define and update maintenance criteria>

1. Maintenance Manager can easily define and update maintenance criteria.

7.2 <Scenario: Handling invalid criteria entries>

1. Handling invalid criteria entries.

8 Post-conditions

8.1 <Post-condition 1>

The maintenance criteria are defined and stored in the system.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Allow definition and updating of maintenance criteria.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Usability Requirements (U):

User-friendly interface for defining criteria.

Reliability Requirements (R):

Ensure data integrity and validity of criteria.

Performance Requirements (P):

Efficient processing and application of criteria.

Supportability Requirements (S):

Easy maintenance and updates to the criteria settings.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.9.2} Schedule Maintenance Check	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.9.2} Schedule Maintenance Check

1 Brief Description

This use case describes the process by which the system schedules regular maintenance checks for each bike based on the defined criteria.

2 Actor Brief Descriptions

2.1 System:

Analyzes data and schedules maintenance checks.

3 Preconditions

Maintenance criteria are defined and stored in the system.

The system is operational.

4 Basic Flow of Events

1. The System periodically analyzes each bike's data (e.g., usage, time, distance traveled).
2. The System determines which bikes meet the criteria for a maintenance check.
3. The System schedules maintenance checks for these bikes and logs the schedule.

5 Alternative Flows

5.1 <Alternate flow: Data Analysis Error>

If there is an error in data analysis, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Regular and accurate scheduling>

1. Regular and accurate scheduling of maintenance checks.

7.2 <Scenario: Handling data analysis errors>

1. Handling data analysis errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Maintenance checks are scheduled for bikes that meet the defined criteria.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Schedule maintenance checks based on defined criteria.

Usability Requirements (U):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

N/A (backend process).

Reliability Requirements (R):

High reliability in scheduling checks.

Performance Requirements (P):

Efficient processing and scheduling.

Supportability Requirements (S):

Easy maintenance and updates to the scheduling algorithm.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.9.3} Notify Maintenance Operator	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.9.3} Notify Maintenance Team

1 Brief Description

This use case describes the process by which the system notifies the maintenance manager/ maintenance operator about scheduled maintenance checks.

2 Actor Brief Descriptions

2.1 System:

Generates and sends notifications.

2.2 Maintenance Manager/Maintenance Operator:

Receives and reviews notifications.

3 Preconditions

Maintenance checks are scheduled.

The system is operational.

4 Basic Flow of Events

1. The System generates notifications for the scheduled maintenance checks.
2. The System sends notifications to the maintenance manager/ maintenance operator.
3. The Maintenance Manager/Maintenance Operator receives and reviews the notifications.

5 Alternative Flows

5.1 <Alternate flow: Notification Failure>

If the notification fails to send, the system logs an error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Timely notification>

1. Timely notifications to the maintenance manager/ maintenance operator about scheduled checks.

7.2 <Scenario: Handling notification failures>

1. Handling notification failures gracefully.

8 Post Conditions

8.1 <Post-condition 1>

The maintenance manager/ maintenance operator is aware of the scheduled maintenance checks.

9 Special Requirements (FURPS+)

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Functional Requirements (F):

Generate and send notifications for maintenance checks.

Usability Requirements (U):

Clear and actionable notifications.

Reliability Requirements (R):

High reliability in sending notifications.

Performance Requirements (P):

Quick generation and delivery of notifications.

Supportability Requirements (S):

Easy maintenance and updates to the notification system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.9.4} Perform Maintenance Check	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.9.4} Perform Maintenance Check

1 Brief Description

This use case describes the process by which the Maintenance Manager/ Maintenance Operator performs the scheduled maintenance checks on bikes.

2 Actor Brief Descriptions

2.1 System:

Logs maintenance activities.

2.2 Maintenance Operator:

Performs maintenance checks.

3 Preconditions

The Maintenance Manager/ Maintenance Operator is notified of scheduled checks.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Operator reviews the notifications and prepares for the maintenance checks.
2. The Maintenance Operator performs the maintenance checks on the scheduled bikes.

5 Alternative Flows

5.1 <Alternate flow: Unable to Perform Maintenance>

If the Maintenance Operator is unable to perform the check (e.g., bike not available), the system logs the issue and reschedules.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Performed scheduled maintenance>

1. Maintenance checks are performed as scheduled.

7.2 <Scenario: Handling maintenance issues>

1. Handling situations where maintenance cannot be performed.

8 Post-conditions

8.1 <Post-condition 1>

The maintenance checks are performed, and the system is updated with the results.

9 Special Requirements (FURPS+)

Functional Requirements (F):

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Perform and log maintenance checks.

Usability Requirements (U):

Clear process and interface for logging maintenance activities.

Reliability Requirements (R):

Ensure accurate logging of maintenance results.

Performance Requirements (P):

Efficient performance and logging of maintenance checks.

Supportability Requirements (S):

Easy updates to the maintenance process and logging system.

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Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.9.5} Update Maintenance Records	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.9.5} Update Maintenance Records

1 Brief Description

This use case describes the process by which the system updates maintenance records based on the performed maintenance checks.

2 Actor Brief Descriptions

2.1 System:

Updates and stores maintenance records.

2.2 Maintenance Operator:

Provides status and results of maintenance checks.

3 Preconditions

Maintenance checks are performed and results are available.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Operator provides the status and results of the performed maintenance checks.
2. The System updates the maintenance records in the database.
3. The System confirms the update to the Maintenance Operator.

5 Alternative Flows

5.1 <Alternate flow: Update Error>

If there is an error in updating the records, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Accurate and timely updates>

1. Accurate and timely updating of maintenance records.

7.2 <Scenario: Handling update errors>

1. Handling update errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Maintenance records are updated with the latest information.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Update maintenance records based on checks.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Usability Requirements (U):

N/A (backend process).

Reliability Requirements (R):

High reliability in updating records.

Performance Requirements (P):

Efficient processing and updating of records.

Supportability Requirements (S):

Easy maintenance and updates to the record-keeping system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.10.1} Generate Maintenance Alerts	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.10.1} Generate Maintenance Alerts

1 Brief Description

This use case describes the process by which the system generates alerts for bikes that are identified as due for maintenance.

2 Actor Brief Descriptions

2.1 System:

Generates alerts based on maintenance schedules

3 Preconditions

The System generates maintenance alerts for bikes due for maintenance.

The System logs the alerts in the database.

4 Basic Flow of Events

1. The System generates maintenance alerts for bikes due for maintenance.
2. The System logs the alerts in the database.

5 Alternative Flows

5.1 <Alternate flow: Alert Generation Error>

If there is an error in generating alerts, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Timely and accurate generation>

1. Timely and accurate generation of maintenance alerts.

7.2 <Scenario: Handling alert generation errors>

1. Handling alert generation errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Maintenance alerts are generated and logged in the system.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Generate alerts for bikes due for maintenance.

Usability Requirements (U):

N/A (backend process).

Reliability Requirements (R):

High reliability in generating alerts.

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Performance Requirements (P):

Efficient alert generation process.

Supportability Requirements (S):

Easy maintenance and updates to the alert generation system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.10.2} Send Maintenance Notifications	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.10.2} Send Maintenance Notifications

1 Brief Description

This use case describes the process by which the system sends notifications to the Maintenance Manager/ Maintenance Operator about bikes due for maintenance.

2 Actor Brief Descriptions

2.1 System:

Sends notifications based on generated alerts.

2.2 Maintenance Manager/Maintenance Operator:

Receives and acts on notifications.

3 Preconditions

Maintenance alerts are generated and logged in the system.

The system is operational.

4 Basic Flow of Events

1. The System generates notifications based on maintenance alerts.
2. The System sends notifications to the maintenance team.
3. The Maintenance Manager/Maintenance Operator receives and reviews the notifications.

5 Alternative Flows

5.1 <Alternate flow: Notification Failure>

If a notification fails to send, the system logs an error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Timely and accurate notifications>

1. Timely and accurate notifications to the Maintenance Manager/ Maintenance Operator.

7.2 <Scenario: Handling notification failures>

1. Handling notification failures.

8 Post-conditions

8.1 <Post-condition 1>

The Maintenance Manager/Maintenance Operator is notified about bikes due for maintenance.

9 Special Requirements (FURPS+)

Functional Requirements (F):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Send notifications for maintenance alerts.

Usability Requirements (U):

Clear and actionable notifications.

Reliability Requirements (R):

High reliability in sending notifications.

Performance Requirements (P):

Quick notification generation and delivery.

Supportability Requirements (S):

Easy maintenance and updates to the notification system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.10.3} Acknowledge Maintenance Alerts	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.10.3} Acknowledge Maintenance Alerts

1 Brief Description

This use case describes the process by which the maintenance operator acknowledges the receipt of maintenance alerts and plans for the checks.

2 Actor Brief Descriptions

2.1 System:

Logs acknowledgments and updates status.

2.2 Maintenance Operator:

Acknowledges alerts and plans maintenance.

3 Preconditions

The Maintenance Operator has received notifications about bikes due for maintenance.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Operator reviews the received notifications.
2. The Maintenance Operator acknowledges the alerts in the system.
3. The System logs the acknowledgments and updates the alert status.

5 Alternative Flows

5.1 <Alternate flow: Acknowledgment Failure>

If the acknowledgment fails, the system logs an error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Timely acknowledgment>

1. Timely acknowledgment of maintenance alerts by the Maintenance Operator.

7.2 <Scenario: Handling acknowledgment failures>

1. Handling acknowledgment failures gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Maintenance alerts are acknowledged and planned for action.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Acknowledge maintenance alerts.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Usability Requirements (U):

User-friendly interface for acknowledgments.

Reliability Requirements (R):

High reliability in logging acknowledgments.

Performance Requirements (P):

Efficient acknowledgment process.

Supportability Requirements (S):

Easy maintenance and updates to the acknowledgment system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.10.4} Escalate Unacknowledged Alerts	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.10.4} Escalate Unacknowledged Alerts

1 Brief Description

This use case describes the process by which the system escalates alerts that have not been acknowledged within a certain timeframe.

2 Actor Brief Descriptions

2.1 System:

Monitors acknowledgment status and escalates unacknowledged alerts.

2.2 Maintenance Manager:

Reviews escalated alerts and takes necessary action.

2.3 Maintenance Operator:

Acts on escalated alerts.

3 Preconditions

Maintenance alerts have been generated and notifications sent.

The system is operational.

4 Basic Flow of Events

1. The System monitors the acknowledgment status of maintenance alerts.
2. If an alert is not acknowledged within a predefined timeframe, the System Backend escalates the alert.
3. The System notifies the Maintenance Manager of the escalated alert.
4. The Maintenance Manager reviews the escalated alert and takes necessary action.
5. If necessary, the Maintenance Manager assigns tasks to the Maintenance Operator.
6. The Maintenance Operator updates the system with the actions taken.

5 Alternative Flows

5.1 <Alternate flow: Escalation Error:>

If there is an error in the escalation process, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Timely escalation of unacknowledged alerts>

1. Timely escalation of unacknowledged alerts.

7.2 <Scenario: Handling escalation errors>

1. Handling escalation errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

Unacknowledged alerts are escalated and reviewed for further action.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Escalate unacknowledged alerts.

Usability Requirements (U):

Clear notification and escalation process.

Reliability Requirements (R):

High reliability in monitoring and escalating alerts.

Performance Requirements (P):

Efficient escalation process.

Supportability Requirements (S):

Easy maintenance and updates to the escalation system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.11.1} Retrieve Maintenance History	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.11.1} Retrieve Maintenance History

1 Brief Description

This use case describes the process by which the Maintenance Manager can retrieve the maintenance history of a specific bike.

2 Actor Brief Descriptions

2.1 System:

Provides access to stored maintenance records.

2.2 Maintenance Manager:

Retrieves maintenance history.

3 Preconditions

Maintenance records are stored in the system.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Manager accesses the management interface.
2. The Maintenance Manager selects a specific bike and requests its maintenance history.
3. The System retrieves and displays the maintenance history for the selected bike.

5 Alternative Flows

5.1 <Alternate flow: Data Retrieval Error>

If there is an error in retrieving the data, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful retrieval>

1. Successful retrieval of maintenance history.

7.2 <Scenario: Handling retrieval errors>

1. Handling errors in data retrieval.

8 Post-conditions

8.1 <Post-condition 1>

The maintenance history of the bike is displayed to the Maintenance Manager.

9 Special Requirements (FURPS+)

Functional Requirements (F):

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Retrieve and display maintenance history.

Usability Requirements (U):

Clear and accessible interface for viewing history.

Reliability Requirements (R):

High reliability in retrieving accurate records.

Performance Requirements (P):

Efficient data retrieval process.

Supportability Requirements (S):

Easy maintenance and updates to the retrieval system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.11.2} Evaluate Repair Effectiveness	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.11.2} Evaluate Repair Effectiveness

1 Brief Description

This use case describes the process by which the system evaluates the effectiveness of previous repairs based on subsequent maintenance records.

2 Actor Brief Descriptions

2.1 System:

Analyzes maintenance records to evaluate repair effectiveness.

2.2 Maintenance manager:

Reviews evaluation results.

3 Preconditions

Maintenance records are stored in the system.

The system is operational.

4 Basic Flow of Events

1. The System analyzes maintenance records to evaluate the effectiveness of previous repairs.
2. The System generates a report on repair effectiveness.
3. The Maintenance Manager reviews the report and takes necessary action.

5 Alternative Flows

5.1 <Alternate flow: Evaluation Error>

If there is an error in the evaluation process, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful evaluation>

1. Successful evaluation of repair effectiveness.

7.2 <Scenario: Handling evaluation errors>

1. Handling errors in the evaluation process.

8 Post-conditions

8.1 <Post-condition 1>

A report on repair effectiveness is generated and reviewed.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Evaluate the effectiveness of repairs.

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Usability Requirements (U):

Clear and actionable reports.

Reliability Requirements (R):

High reliability in the evaluation process.

Performance Requirements (P):

Efficient data analysis.

Supportability Requirements (S):

Easy maintenance and updates to the evaluation system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.11.3} Plan Future Maintenance	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.11.3} Plan Future Maintenance

1 Brief Description

This use case describes the process by which the system uses maintenance records and analysis to plan future maintenance activities.

2 Actor Brief Descriptions

2.1 System

Uses data to plan maintenance activities.

2.2 Maintenance Manager:

Reviews and approves maintenance plans.

3 Preconditions

Maintenance records and analysis reports are available.

The system is operational.

4 Basic Flow of Events

1. The System analyzes maintenance records and reports to plan future maintenance activities.
2. The System generates a maintenance plan.
3. The Maintenance Manager reviews and approves the maintenance plan.

5 Alternative Flows

5.1 <Alternate flow: Planning Error:>

If there is an error in the planning process, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful planning>

1. Successful planning of future maintenance activities.

7.2 <Scenario: Handling planning errors>

1. Handling errors in the planning process.

8 Post-conditions

8.1 <Post-condition 1>

A maintenance plan is generated, reviewed, and approved.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Plan future maintenance activities.

Usability Requirements (U):

Clear and actionable maintenance plans.

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Reliability Requirements (R):

High reliability in planning.

Performance Requirements (P):

Efficient data analysis and planning process.

Supportability Requirements (S):

Easy maintenance and updates to the planning system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.12.1} Track Issues Identified During Maintenance	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.12.1} Track Issues Identified During Maintenance

1 Brief Description

This use case describes the process by which the system tracks issues identified during maintenance checks on bikes.

2 Actor Brief Descriptions

2.1 System:

Stores and processes issue data.

2.2 Maintenance Operator:

Identifies and logs issues.

3 Preconditions

1. Maintenance checks are being performed.
2. The system is operational.

4 Basic Flow of Events

1. The Maintenance Operator performs a maintenance check on a bike.
2. The Maintenance Operator identifies an issue and logs it into the system.
3. The System records the issue in the maintenance database.

5 Alternative Flows

5.1 <Alternate flow: Logging Error>

If there is an error in logging the issue, the system logs the error and prompts the Maintenance Operator to retry.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful logging>

2. Successful logging of issues identified during maintenance.

7.2 <Scenario: Handling logging errors>

3. Handling errors in logging issues.

8 Post-conditions

8.1 <Post-condition 1>

Issues identified during maintenance are recorded in the system's database.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Log issues identified during maintenance.

Usability Requirements (U):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

User-friendly interface for logging issues.

Reliability Requirements (R):

High reliability in storing issue data.

Performance Requirements (P):

Efficient logging and storage of issue records.

Supportability Requirements (S):

Easy updates and maintenance of the logging system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.12.2} Flag Bikes Due for Maintenance or Repair	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.12.2} Flag Bikes Due for Maintenance or Repair

1 Brief Description

This use case describes the process by which the system flags bikes that are due for maintenance or repair based on identified issues.

2 Actor Brief Descriptions

2.1 System:

Flags bikes based on logged issues and maintenance criteria.

3 Preconditions

Issues are categorized and logged in the system.

The system is operational.

4 Basic Flow of Events

1. The System reviews categorized issues.
2. The System flags bikes that are due for maintenance or repair.
3. The System updates the status of flagged bikes in the database.

5 Alternative Flows

5.1 <Alternate flow: Flagging Error>

If there is an error in flagging a bike, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful flagging>

1. Successful flagging of bikes due for maintenance or repair.

7.2 <Scenario: Handling flagging errors>

1. Handling errors in the flagging process.

8 Post-conditions

8.1 <Post-condition 1>

Bikes due for maintenance or repair are flagged in the system.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Flag bikes due for maintenance or repair.

Usability Requirements (U):

Clear status indicators for flagged bikes.

Reliability Requirements (R):

High reliability in the flagging process.

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Performance Requirements (P):

Efficient flagging process.

Supportability Requirements (S):

Easy updates and maintenance of the flagging system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.12.3} Prioritize Repairs	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.12.3} Prioritize Repairs

1 Brief Description

This use case describes the process by which the system prioritizes repairs based on the severity and frequency of identified issues.

2 Actor Brief Descriptions

2.1 System:

Analyzes issues and prioritizes repairs.

2.2 Maintenance Manager:

Reviews and adjusts repair priorities if necessary.

3 Preconditions

Bikes are flagged for maintenance or repair.

The system is operational.

4 Basic Flow of Events

1. The System analyzes the flagged bikes and the severity of their issues.
2. The System prioritizes the repairs based on predefined criteria (e.g., safety, frequency of issues).
3. The Maintenance Manager reviews and adjusts the priorities if necessary.
4. The System updates the priority list in the database.

5 Alternative Flows

5.1 <Alternate flow: Prioritization Error>

If there is an error in prioritizing repairs, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful prioritization>

1. Successful prioritization of repairs.

7.2 <Scenario: Handling prioritization errors.>

1. Handling errors in the prioritization process.

8 Post-conditions

8.1 <Post-condition 1>

Repairs are prioritized and listed in the system.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Prioritize repairs based on severity and frequency.

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Usability Requirements (U):

Clear prioritization interface for administrators.

Reliability Requirements (R):

High reliability in the prioritization process.

Performance Requirements (P):

Efficient prioritization process.

Supportability Requirements (S):

Easy updates and maintenance of the prioritization system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.13.1} Manage Spare Parts Inventory	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.13.1} Manage Spare Parts Inventory

1 Brief Description

This use case describes the process by which the system manages an inventory of spare parts and other materials needed for bike maintenance.

2 Actor Brief Descriptions

2.1 System:

Stores and processes inventory data.

2.2 Maintenance Manager/Maintenance Operator

Manages inventory records.

3 Preconditions

The system is operational.

The Maintenance Manager is logged into the system.

4 Basic Flow of Events

1. The Maintenance Manager/Maintenance Operator logs into the inventory management interface.
2. The Maintenance Manager/Maintenance Operator adds, updates, or removes spare parts and materials in the inventory.
3. The System processes and stores the updated inventory data.

5 Alternative Flows

5.1 <Alternate flow: Data Entry Error>

If there is an error in data entry, the system prompts the Maintenance Manager/Maintenance Operator to correct it.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful inventory management>

1. Successful management of spare parts inventory.

7.2 <Scenario: Handling data entry errors>

1. Handling data entry errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The inventory of spare parts and materials is accurately updated in the system.

9 Special Requirements (FURPS+)

Functional Requirements (F):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Manage spare parts inventory.

Usability Requirements (U):

User-friendly interface for inventory management.

Reliability Requirements (R):

High reliability in storing inventory data.

Performance Requirements (P):

Efficient processing and updating of inventory records.

Supportability Requirements (S):

Easy updates and maintenance of the inventory management system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.13.2} Track Usage of Spare Parts	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.13.2} Track Usage of Spare Parts

1 Brief Description

This use case describes the process by which the system tracks the usage of spare parts during maintenance activities.

2 Actor Brief Descriptions

2.1 System:

Records and processes usage data.

2.3 Maintenance Operator:

Logs the usage of spare parts.

3 Preconditions

Maintenance activities are being performed.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Operator logs the usage of spare parts during maintenance activities.
2. The System records the usage data in the inventory database.
3. The System updates the inventory count for the used spare parts.

5 Alternative Flows

5.1 <Alternate flow: Logging Error>

If there is an error in logging usage, the system logs the error and prompts the team to retry.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful tracking of spare parts usage>

1. Successful tracking of spare parts usage.

7.2 <Scenario: Handling errors in logging usage data>

1. Handling errors in logging usage data..

8 Post-conditions

8.1 <Post-condition 1>

The usage of spare parts is accurately recorded and inventory counts are updated.

9 Special Requirements (FURPS+)

Functional Requirements (F):

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Track usage of spare parts.

Usability Requirements (U):

Easy-to-use interface for logging usage.

Reliability Requirements (R):

High reliability in recording usage data.

Performance Requirements (P):

Efficient processing and updating of usage records.

Supportability Requirements (S):

Easy updates and maintenance of the usage tracking system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.13.3} Trigger Low Stock Alerts	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.13.3} Trigger Low Stock Alerts

1 Brief Description

This use case describes the process by which the system triggers alerts when the stock of spare parts is low.

2 Actor Brief Descriptions

2.1 System:

Monitors inventory levels and triggers alerts.

2.2 Maintenance Manager:

Receives and acts on low stock alerts.

3 Preconditions

Inventory levels are being tracked in the system.

The system is operational.

4 Basic Flow of Events

1. The System continuously monitors the inventory levels of spare parts.
2. When the stock of a spare part falls below a predefined threshold, the System triggers a low stock alert.
3. The Maintenance Manager receives the low stock alert and reviews the inventory.

5 Alternative Flows

5.1 <Alternate flow: Alert Failure>

If there is an error in triggering the alert, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful triggering>

1. Successful triggering of low stock alerts.

7.2 <Scenario: Handling alert errors>

1. Handling errors in the alert process..

8 Post-conditions

8.1 <Post-condition 1>

The Maintenance Manager is notified of low stock levels and can take appropriate action.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Trigger low stock alerts.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Usability Requirements (U):

Clear and actionable alerts.

Reliability Requirements (R):

High reliability in monitoring inventory levels and triggering alerts.

Performance Requirements (P):

Efficient alert triggering process.

Supportability Requirements (S):

Easy updates and maintenance of the alert system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.13.4} Generate Inventory Reports	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.13.4} Generate Inventory Reports

1 Brief Description

This use case describes the process by which the system generates reports on inventory levels, usage, and replenishment activities.

2 Actor Brief Descriptions

2.1 System:

Compiles and generates reports.

2.3 Maintenance Manager:

Generates and reviews inventory reports.

3 Preconditions

Inventory data is being tracked and stored in the system.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Manager selects the type of inventory report to generate.
2. The System compiles the necessary data and generates the report.
3. The Maintenance Manager reviews the generated report and takes necessary action.

5 Alternative Flows

5.1 <Alternate flow: Report Generation Error>

Report Generation Error: If there is an error in generating the report, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful report generation>

1. Successful generation of inventory reports.

7.2 <Scenario: Handling report generation errors>

1. Handling errors in the report generation process.

8 Post-conditions

8.1 <Post-condition 1>

Inventory reports are generated and reviewed by the Maintenance Manager.

9 Special Requirements (FURPS+)

Functional Requirements (F):

Generate various types of inventory reports.

Usability Requirements (U):

Clear and comprehensive reports.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Reliability Requirements (R):

High reliability in report generation.

Performance Requirements (P):

Efficient compilation and generation of reports.

Supportability Requirements (S):

Easy updates and maintenance of the reporting system.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.14.1} Generate Common Issues Identified Report	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.14.1} Generate Common Issues Identified Report

1 Brief Description

This use case describes the process by which the system generates a report detailing common issues identified during maintenance.

2 Actor Brief Descriptions

2.1 System:

Compiles and generates the report.

2.3 Maintenance Manager:

Reviews the report.

3 Preconditions

Maintenance activity data is stored in the system.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Manager selects the reporting period.
2. The System retrieves maintenance activity data for the specified period.
3. The System analyzes the data to identify common issues.
4. The System compiles the data to generate the report.
5. The System presents the report to the Maintenance Manager.
6. The Maintenance Manager reviews the report.

5 Alternative Flows

5.1 <Alternate flow: Data Analysis Error>

If there is an error in analyzing data, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful report generation>

1. Successful generation of the common issues identified report.

7.2 <Scenario: Handling report generation errors>

1. Handling data analysis errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The report detailing common issues identified is generated and reviewed.

10 Special Requirements (FURPS+)

Functional Requirements (F):

The system must analyze and compile maintenance data accurately.

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Usability Requirements (U):

The report must be easy to generate and interpret.

Reliability Requirements (R):

The system should ensure data accuracy and handle errors gracefully.

Performance Requirements (P):

The report should be generated in a timely manner.

Supportability Requirements (S):

The system should allow easy updates for reporting requirements.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.14.2} Generate Parts Used Report	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.14.2} Generate Parts Used Report

1 Brief Description

This use case describes the process by which the system generates a report detailing the spare parts used during maintenance activities.

2 Actor Brief Descriptions

2.1 System:

Compiles and generates the report.

2.3 Maintenance Manager:

Reviews the report.

3 Preconditions

Maintenance activity data is stored in the system.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Manager selects the reporting period.
2. The System retrieves maintenance activity data for the specified period.
3. The System analyzes the data to identify parts used.
4. The System compiles the data to generate the report.
5. The System presents the report to the Maintenance Manager.
6. The Maintenance Manager reviews the report.

5 Alternative Flows

5.1 <Alternate flow: Data Analysis Error>

If there is an error in analyzing data, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful report generation>

1. Successful generation of the parts used report..

7.2 <Scenario: Handling data analysis errors>

1. Handling data analysis errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The report detailing the parts used is generated and reviewed.

11 Special Requirements (FURPS+)

Functional Requirements (F):

The system must analyze and compile maintenance data accurately.

Usability Requirements (U):

The report must be easy to generate and interpret.

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

Reliability Requirements (R):

The system should ensure data accuracy and handle errors gracefully.

Performance Requirements (P):

The report should be generated in a timely manner.

Supportability Requirements (S):

The system should allow easy updates for reporting requirements.

Be.Neutral-Project#3[PPS4]: Maintenance Management Platform	
Use-case Specification: {U.C.14.3} Generate Comprehensive Maintenance Report	Date: 17-06-2024

Project#3[PPS4]: Maintenance Management Platform

Use-Case: {U.C.14.3} Generate Comprehensive Maintenance Report

1 Brief Description

This use case describes the process by which the system generates a comprehensive report detailing maintenance activities, including the number of bikes serviced, common issues identified, and parts used.

2 Actor Brief Descriptions

2.1 System:

Compiles and generates the report.

2.3 Maintenance Manager:

Reviews the report.

3 Preconditions

Maintenance activity data is stored in the system.

The system is operational.

4 Basic Flow of Events

1. The Maintenance Manager selects the reporting period.
2. The System retrieves maintenance activity data for the specified period.
3. The System compiles the data to generate the comprehensive report.
4. The System presents the report to the Maintenance Manager.
5. The Maintenance Manager reviews the report.

5 Alternative Flows

5.1 <Alternate flow: Report Generation Error>

If there is an error in generating the report, the system logs the error and retries.

6 Subflows

6.1 N/A

7 Key Scenarios

7.1 <Scenario: Successful report generation>

1. Successful generation of the comprehensive maintenance report.

7.2 <Scenario: Handling report generation errors>

1. Handling report generation errors gracefully.

8 Post-conditions

8.1 <Post-condition 1>

The comprehensive maintenance report is generated and reviewed.

12Special Requirements (FURPS+)

Functional Requirements (F):

P3_E2- Requirement's analysis of the Bike4All Maintenance platform

The system must compile and generate detailed maintenance reports accurately.

Usability Requirements (U):

The report must be easy to generate and interpret.

Reliability Requirements (R):

The system should ensure data accuracy and handle errors gracefully.

Performance Requirements (P):

The report should be generated in a timely manner.

Supportability Requirements (S):

The system should allow easy updates for reporting requirements.

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